An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields

Bernhard Mueller[1,2,*] Mario Poneder Jinwook Kim[3] Daniel Ari Friedman[4,5]

November 7, 2026

Paper release: r2062 **Released:** November 7, 2026

[1]Pragma Research Inc. [2]Information Physics Institute [3]Oraclizer Labs [4]Active Inference Institute [5]FractiAI
[*]Corresponding author: bernhard@floatingpragma.ai

Abstract

An executable model of Observer Patch Holography specifies a twelve-port icosahedral carrier, local C/E seam repair, quantum record instruments, memory and observer access. We classify the admissible record laws, derive the exact repair gap $(5-\sqrt{5})/60$, and construct continuing observers, complete-record recovery and event clocks on this source. Specified interacting actions admit source-compiled detectors with controlled occupation limits and complete-output error bounds. A refining spherical source has a spectral-dimension-two continuum operator; a declared radial extension gives a smooth $1+3$ free-scalar description for fixed finite observer menus. These results distinguish native dynamics from supplied coherent control and physical interpretation. The accompanying particle construction joins gauge currents, localized modes and a pixel-dependent mass operator on specified carriers. A common interacting chiral continuum, a spectrum matched to measured particles and responsive Einstein geometry require additional realization and comparison theorems.

1 The executable model and its interpretation

Observer Patch Holography proposes that observers recover the specification of their world from shared records and realize that specification within the same world. An implementation therefore accounts for its carrier, initialization, repair, controls, memory, readout and resource costs. The construction studied here specifies these objects and proves what its observers can record. A consistent implementation is not by itself an identification with nature: that identification compares complete preparation, intervention and detection laws in common physical units.

The carrier and repair law are given in Section 2; the finite observer, recovery and local clock are in Section 3. Overlapping observers read one process in Section 4. Sections 5–7 derive detector and continuum results on explicitly stated actions and refinement domains. Section 9 gives operational comparisons, and Sections 10–11 state their physical boundaries.

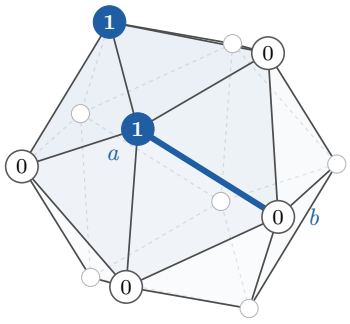
The mathematical roles are distinct. A finite theorem proves an implication on its stated domain; a certificate verifies a finite computation; an analytic limit theorem requires its displayed normalization and order of limits. A supplied action, state, instrument or decoder is a design input. Numerical agreement with data used to choose that input is a conditional postdiction. Physical validation requires further independent observations. Host runtime, seam-attempt counts, event-clock readings, modular parameters and calibrated duration are related only by explicit maps.

The consensus paper proves quotient repair and fixed-point results [5]; the observer paper supplies the algebraic framework [6]; the carrier, gauge and spacetime papers establish their respective conditional theorems [9, 7, 8]. The flagship gives the combined mathematical argument [3]. Closure alone entails neither least-informative state selection nor inner implementation of connected response: those are separate premises in the basis registry [4].

2 The specification

The carrier, repair and record laws, observer algebra, services and time parameters below define the source. Each extension states its additional action, preparation and control inputs.

2.1 The carrier and its load configurations



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 1: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

The specification has one carrier with twelve primitive ports. Its incidence is the oriented boundary of an icosahedron, with thirty seams and twenty faces; five seams meet at every port, the neighbours of each port form a five-cycle, and every seam lies on two oppositely oriented faces [9, 3]. Write A for the adjacency matrix of the seam graph and $L = 5I - A$ for its Laplacian. The spectrum of L is 0, $5 - \sqrt{5}$ with multiplicity three, 6 with multiplicity five and $5 + \sqrt{5}$ with multiplicity three, so its characteristic polynomial is $x(x-6)^5(x^2 - 10x + 20)^3$ [48, 89, 82].

A load configuration assigns a nonnegative integer x_p to each port p, with total load $\sum_p x_p = 2$. There are 78 configurations. The 66 with two distinct occupied ports form the binary shell; the other twelve are collisions $|2u\rangle$ with load two at one port u. We write $\mathcal{H}_{78} = \mathbb{C}^{78}$ for the quantum load space with basis $|x\rangle$, $\mathcal{H}_{66}$ for the span of the binary shell, and M_{78}, M_{66} for the corresponding full matrix algebras. For a binary configuration occupied at ports u and w we also write $|uw\rangle$ or $|\{u, w\}\rangle$.

Ports are numbered 0 to 11 by the frame vectors u_p: $(-1, \varphi, 0)$, $(1, \varphi, 0)$, $(-1, -\varphi, 0)$, $(1, -\varphi, 0)$, $(0, -1, \varphi)$, $(0, 1, \varphi)$, $(0, -1, -\varphi)$, $(0, 1, -\varphi)$, $(\varphi, 0, -1)$, $(\varphi, 0, 1)$, $(-\varphi, 0, -1)$, $(-\varphi, 0, 1)$, with $\varphi = (1 + \sqrt{5})/2$. The thirty seams are the pairs with $|u_a - u_b|^2 = 4$, each written (a, b) with $a < b$, and that order is the orientation used by the repair law below [82, 83].

The construction takes this incidence as a design input. The carrier paper gives a conditional selector of the same unoriented incidence from a supplied three-dimensional quadratic readout [9].

Premise A1 of the theorem library is the layer of carriers, their federation through typed seams and coherent triple overlaps, the oriented spherical support and the complete reversible port response [4]. The construction realizes one carrier of that layer: its ports, seams and faces, its load configurations and the repair events on its seams. Coherent quantum operations run through a six-level processor whose twelve source generators, one per port, exponentiate to every executed gate on two- and four-dimensional proper codes through declared load and export services [69, 60]. Spherical-screen executions join many carriers under explicit access and refinement contracts [11]; they identify no physical current for the complete response algebra and do not discharge the full A1 attachment.

2.2 Native repair and its quantum record laws

Fix an oriented seam $e = (a, b)$ and a fair placement coin $c \in \{0, 1\}$. The native repair law is

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(\mathbf{e}_b - \mathbf{e}_a), \tag{1}$$

where $\mathbf{e}_p$ is the unit load vector of port p and y is the repaired configuration. The two endpoint loads become the floor and the ceiling of their mean; the coin assigns the ceiling to a when $c = 1$; all other loads are unchanged.

Proposition 2.1 (The classical seam rule)**.** *Let a stochastic rule on one seam (i) conserve $x_a + x_b$, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be invariant under exchange of the two endpoints. Then it is* (1) *with a fair coin.*

The carrier paper proves this characterization [9]. Without (iii), the rule that leaves the extra unit with the endpoint holding more satisfies the other three requirements. It fixes every binary configuration, so the binary shell carries no motion and the records carry no geometry. Requirement (iii) is an independent restriction on the admitted tie-breaker: it excludes retaining a preference for the endpoint holding more at the input. Internal realization alone does not impose that restriction.

The record is $r = (c, j)$, the coin and the signed current. Compatibility on a seam is specified independently of the update; it requires the two endpoint loads to differ by at most one. On the quantum load space the record defines the current instrument C with Kraus operators and record-discarded channel

$$L^e_{(c,j)} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \qquad \mathsf{C}_e(\rho) = \sum_r L^e_r\, \rho\, L^{e\dagger}_r. \tag{2}$$

Both maps are written in Kraus form [109]. The contrast-record variant E keeps $r = (c, d)$ instead, with Kraus operators $M^e_{(c,d)} = 2^{-1/2} \sum_{x:\, x_a - x_b = d} |y(x,c)\rangle\langle x|$ and record-discarded channel E_e. Uniform independent seam attempts form the baseline schedule; on diagonal binary states the repair is then the lazy symmetric exclusion process with two particles, with generator $\frac{1}{60}\sum_e (S_e - I)$ per attempt, S_e exchanging the endpoint loads [42, 48].

Proposition 2.2 (Native repair instrument)**.** *Let $e = (a, b)$ be an oriented seam of the twelve-port carrier, let the placement coin be fair, and let C_e and E_e act on $\mathcal{H}_{78}$. Then:*

(i) For every record value the branch map $x \mapsto y(x, c)$ is an injective translation of its support. Consequently $\sum_r L^{e\dagger}_r L^e_r = I$, and C_e is completely positive and trace preserving (CPTP), also when tensored with an untouched reference system. The same holds for E_e.

(ii) *Every branch conserves the total load, leaves spectator ports unchanged, never increases* $\sum_p x_p^2$*, and maps the binary shell into itself.*

(iii) *The current labels satisfy* $c \in \{0,1\}$ *and* $j \in \{-1,0,1\}$*; all six labels occur on* $\mathcal{H}_{78}$ *and four occur on the binary shell.*

(iv) C *and* E *induce the same classical transition on every configuration, and*

$$\mathsf{E}_e^2 = \mathsf{E}_e, \qquad \mathsf{E}_e\mathsf{C}_e = \mathsf{C}_e\mathsf{E}_e = \mathsf{E}_e, \qquad \mathsf{C}_e^n = \mathsf{E}_e + 2^{1-n}(\mathsf{C}_e - \mathsf{E}_e) \quad (n \geq 1).$$

On the binary shell E_e *is a trace-preserving conditional expectation.*

(v) *With the seam-total parity* $p = (x_a + x_b) \bmod 2$ *one has* $d = 2j + (2c-1)p$*, and* E *equals* C *followed by a binary record of* p *and this relabelling, as an identity of complete unnormalized branches.*

Proof sketch. The inverse of a fixed-current branch adds j at a and subtracts it at b; every basis vector has one branch for each coin, so each column of the joint record/source isometry has two entries $2^{-1/2}$ with disjoint supports [46]. For fixed contrast, the unchanged spectators and the endpoint sum reconstruct the input, which gives (i) for E. Items (ii) and (iv) are the single-seam relations of the candidate register [42]. Item (v) follows from $j = \lceil d/2 \rceil$ for $c = 0$ and $j = \lfloor d/2 \rfloor$ for $c = 1$; the parity-refined Kraus operator $\Pi_p L_{(c,j)}$ has exactly the contrast group as support [63]. The executable checks all 312 outputs of the observer's two seams against an independent floor and ceiling rule [46]. □

The current instrument is no classical coarse-graining of the contrast instrument: grouping the labels of E sums density operators and cannot restore the coherent Kraus sums of C. Both belong to the family of quantum record laws classified in Section 2.3. One further law of the candidate register, the reciprocal exchange N4 that couples the native current to private six-level response factors, enters only the local-tick certificate of Section 3.8 [42].

2.3 Admissible record laws

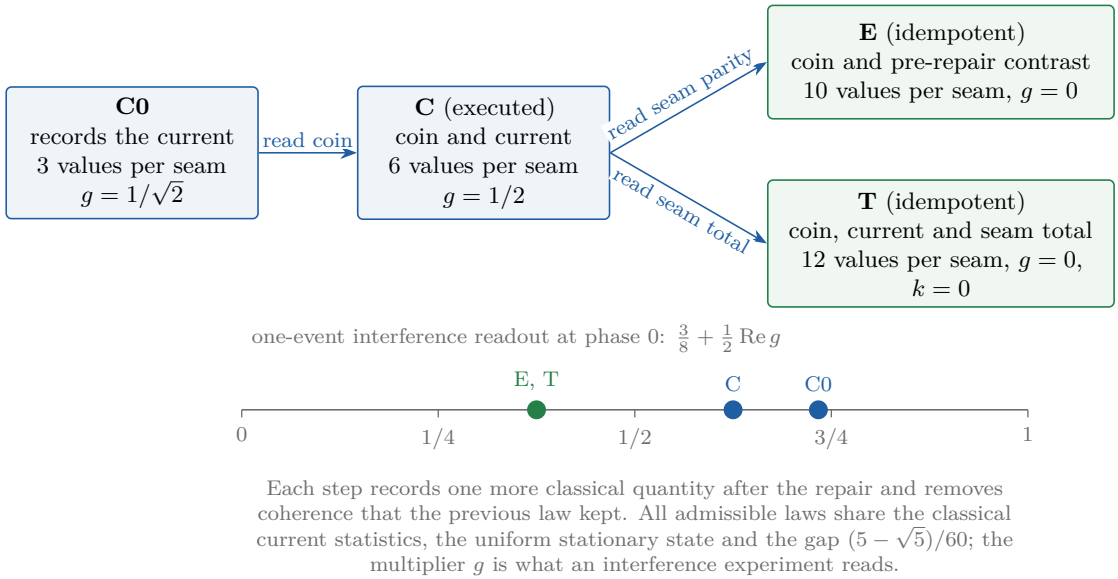
The classical rule fixes the diagonal of a record law and leaves its coherences free. For the seam $e = (a,b)$ write $\kappa(x) = (x_a, x_b)$ for the seam key and $D = \mathbf{e}_b - \mathbf{e}_a$. A channel Φ on $\mathcal{H}_{78}$ is an admissible record law for e if it acts as the identity on the ten spectator ports, reproduces the classical repair distribution on every $|x\rangle\langle x|$, commutes with $U_t = \exp(i\sum_p t_p N_p)$ for every $t \in \mathbb{R}^{12}$, and is invariant under the carrier rotation that exchanges a and b with the current reversed. The third condition makes every local load phase a symmetry of the unread channel. It is a premise of the classification, a local charge-superselection rule for the repair channel, and it is what excludes a recorder of a pre-repair endpoint load. It guarantees that a definite-current Kraus representation exists; it does not force every recorded implementation of the channel to commit a definite current, since a change of record basis before commitment mixes the current labels and leaves the channel unchanged [41]. The classification below concerns unread channels, and statements about current histories use the compatible definite-current instrument.

Theorem 2.3 (Coherence tables)**.** *A channel* Φ *is an admissible record law if and only if*

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j(\kappa(x), \kappa(x'))\, |x + jD\rangle\langle x' + jD|$$

for endpoint-symmetric positive semidefinite matrices G_j on the seam keys that can carry the current j, with $G_j(\kappa,\kappa) = \Pr(j \mid \kappa)$. On $\mathcal{H}_{78}$ the free entries are five multipliers: k between the empty and the full seam, g between the empty seam and a waiting single load, h between the full seam and a waiting single load, a real m between the two waiting orientations, and n between a moving single load and a moving collision. Positivity gives $|k| \le 1$, $|g|, |h|, |n| \le 2^{-1/2}$ and $|m| \le \frac{1}{2}$, and $k = 1$ forces $g = h$. Repairing one seam with (k, g, h, m, n) and then with (k', g', h', m', n') gives $(kk', gg', hh', mm', n\bar{h}')$. Exactly two admissible laws are idempotent, those with $g = h = m = n = 0$ and $k \in \{0, 1\}$, and every law with $k = 1$ converges under repetition to the idempotent law with $k = 1$.

The carrier paper proves this classification [9], and the exact composition checks are retained with the record-law receipt [40].



Each step records one more classical quantity after the repair and removes coherence that the previous law kept. All admissible laws share the classical current statistics, the uniform stationary state and the gap $(5 - \sqrt{5})/60$; the multiplier g is what an interference experiment reads.

Figure 2: The record-law ladder of Theorem 2.3 and Table 1, and the one-event interference readout $\frac{3}{8} + \frac{1}{2}\,\mathrm{Re}\,g$ at phase zero that separates its members.

Four members form a ladder (Table 1 and Figure 2): C is C_0 followed by a diagonal readout of the coin, E is C followed by a readout of the seam parity as in Proposition 2.2(v), and T is C followed by a readout of the seam total. The completion of C in Proposition 2.2(iv) is the case $g = h = n = \frac{1}{2}$, $m = 0$, and E is the idempotent law with $k = 1$. A recorder of an endpoint's pre-repair load keeps the classical law and fails covariance, because one of its branches carries two currents. C_0 records only what conservation requires, C records the tie-break as an event, E is the coarsest law for which one repair completes the seam, and T records the full seam occupation; none of these principles follows from the theorem premises. The construction executes C as its convention and compares it with E. Under a common definite-current record interface all members assign the same probabilities to every classical current observation; Section 9 reads g in one coherent event, and Section 4.3 identifies all nine real coordinates. Every admissible table, including complex phases and different laws on different seams, has the uniform binary density as its unique stationary source state and the same spectral gap $(5 - \sqrt{5})/60$, so equilibrium selects none of the coordinates; a finite unread warm-up of 221 attempts on the binary shell, or 318 on the full carrier, brings an arbitrary initial density within 1/1000 of that state, uniformly over untouched references [40].

Table 1: Four distinguished admissible record laws [40].

Law	Each event records	Values per seam	k	$g = h$	Idempotent
C_0	the current	3	1	$2^{-1/2}$	no
C	coin and current	6	1	$1/2$	no
E	coin and pre-repair contrast	10	1	0	yes
T	coin and both endpoint loads	12	0	0	yes

2.4 The finite observer algebra

An observer of length n on the carrier consists of a preparation on $\mathcal{H}_{78}$, a record tape $R^{\otimes n}$, seam policies s_k that map committed records and classical randomness to seams or to supplied detector outcomes, the coherent writes W_{s_k}, optional supplied instruments whose committed bits may enter s_k, and a halting rule after n writes; a policy reads only committed records. Definition 2.4 is the instance $n = 2$, and Sections 3.4, 5 and 6 add declared instruments to it.

Definition 2.4 (Two-event native observer)**.** Let $R = \mathbb{C}^6$ carry one basis vector per current label (c, j). The observer carrier is

$$\mathcal{H}_{\mathrm{cap}} = \mathcal{H}_{78} \oplus (R \otimes \mathcal{H}_{78}) \oplus (R \otimes R \otimes \mathcal{H}_{78}),$$

with sectors of dimensions 78, 468 and 2808, total 3354; the sector index is the phase, the number of filled record slots.

(a) *Preparation.* The scored preparation is uniform on the 66 binary configurations; the full input family uses all 78.

(b) *Measurement and record.* At phase zero the coherent write $W_e = \sum_r |r\rangle \otimes L_r^e$ acts on the seam $e = (0, 1)$.

(c) *Record-only feedback.* At phase one the seam is $s(r) = (0, 5)$ if the stored current is positive and $(0, 1)$ otherwise, and $V_1(|r\rangle \otimes \psi) = |r\rangle \otimes W_{s(r)}\psi$.

(d) *Forecast.* Before the second event the controller reads its stored letter and issues a supplied forecast $q(r)$ of a positive second current.

(e) *Halting.* At phase two the observer acts as the identity.

The controller reads only the committed first record; it has no access to source loads or future outcomes.

Proposition 2.5 (One CPTP map with classical-prefix preservation)**.** *Let $V_0 = W_{(0,1)}$ and V_1 be extended by zero to the other sectors, giving A_0 and A_1, and let P_2 project onto the full sector. Then:*

(i) V_0 and V_1 are isometries on every nonfull sector, including unreachable combinations of letters and loads, and $T(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger + P_2\rho P_2$ is CPTP;

(ii) on the inhabited subspace $K = \mathcal{H}_1 \oplus \mathcal{H}_2$ every first-record projector Q_r obeys $T^*(Q_r) = Q_r$, so a committed first record is reread with probability one;

(iii) *for the blank input with occupations* $(1, 1, 0, \ldots, 0)$*, the first-record projector of the letter* $(0, 0)$*, extended by zero to phase zero, has expectation* 0 *before the tick and* $1/2$ *after it, so writing a record is a transition distinct from preserving it;*

(iv) *for the equal superposition of* $\{0, 2\}$ *and* $\{1, 2\}$*, the reduced-density entry of the first-record pair* $((0, 1), (0, 0))$ *equals* $1/4$ *after the first write and* $1/8$ *after the second while the classical diagonals are unchanged, so coherence between different record values need not survive feedback.*

Proof sketch. Orthogonal retained labels and $W_s^\dagger W_s = I$ give $V_0^\dagger V_0 = V_1^\dagger V_1 = I$, hence $A_0^\dagger A_0 + A_1^\dagger A_1 + P_2 = I$. Every subsequent map intertwines Q_r on K. In (iv) the working state attached to the entry has occupations $\{1, 2\}$, on which the isometries of $(0, 5)$ and $(0, 1)$ overlap by $1/2$ [46]. The checkpoint adapter verifies all $78 + 468 + 2808$ columns and both the preservation identity on the 3276-dimensional checkpoint space and the blank-sector control [69, 95]. □

Lemma 2.6 (Feedback requires classical record preservation)**.** *Let* Λ *be a CPTP map from* $R \otimes W$ *to* $R \otimes W'$ *with* $\Lambda^*(B \otimes I) = B \otimes I$ *for every* B *in the full matrix algebra of* R*. Then* $\Lambda^*(I \otimes O) = I \otimes \Phi^*(O)$ *for a unital completely positive map* Φ^**, so changing only the record preparation of a product input changes no output statistic on* W'*.*

Proof sketch. The operators $B \otimes I$ lie in the multiplicative domain of the unital completely positive map Λ^*, because the Schwarz inequality is an equality for them [102]. Then $\Lambda^*(I \otimes O)$ commutes with every $B \otimes I$ and has the stated form [46]. □

Lemma 2.6 fixes the shape of the observer contract. A record that steers the working state is preserved as a classical record, with its projectors fixed, while coherence between different record values may change.

2.5 Declared services

Each component names the services it assumes; they belong to the specification and enter the resource counts where a component counts resources. The services are coherent proper-code load and export, with complementary failure outcomes retained by a complete channel definition; blank initialization, reset, storage and routing of quantum registers and classical records; Born readout of pointers and fair coins; and analogue precision of source pulses. Particular components add a calibrated external flux and probe (Section 3.4), incident-event recognition and bare-occupation readout (Section 3.8), coherent phase preparation and readout (Section 9), and, for the native process results, independent uniform seam scheduling, uniform fixed-number preparation, full initial and current readback, coherent wave processing, the return-length sampler and the response controller (Sections 7, 4 and 9). The coherent services are not products of the native grammar: committed C/E repair with diagonal accessible inputs and fresh independent diagonal ancillas cannot prepare the declared coherent reference to arbitrary accuracy, even with adaptive heralding. This coherence floor of the native interface makes the coherent services a necessary resource of the specification. The sharp joint instrument of Section 6.1 uses native integer repair as a declared primitive, and Section 6.5 attaches the compiled 78-dimensional C/E channels, routing and encoded recovery to the continuing work history. The processor algebra is M_6; its twelve generators are the exact source generators of the compiler, one per port [60, 69]; a proper code is a two- or four-dimensional subspace of $\mathbb{C}^6$ with its load and export channels and the complementary failure outcome; and the service error model is the compositional contract of Section 9.2. Exact certificates exclude numerical execution error and physical service error unless a component states otherwise.

Direct native acceptance effects are diagonal, so complex tomography of the binary shell, with real and imaginary analyzers on all 4,356 Hermitian coordinates, uses these coherent services [32]. A finite native warm-up replaces exact stationary input for one complete readout (Section 9.5), and Table 5 separates what repair and its counts supply from what each result declares.

2.6 The closure inputs P and N

The pixel ratio P is the area of one screen cell in units of $\ell_\star^2$, and N is a record capacity in nats [6, 54]. The construction accepts both as closure-derived inputs and fits neither to an output. P enters through a declared coupling map, the source-only structured-running branch of the fine-structure closure (one-loop running and matching conventions with the tree-level m_Z closure), defined in the P-derivation code of the repository [2] and consumed unchanged; the map is a declared selection, not a result of this construction. At the pinned closure value $P = 1.630972095858897\ldots$, the midpoint of the pinned forward-closure interval, the map gives $\alpha^{-1}(P) \approx 136.9948351774$ and $g(P) = \sqrt{4\pi\alpha(P)} \approx 0.302867613117$ in Heaviside–Lorentz natural units, numerical evaluations of the executed map [52, 91]. N enters only as a bound, and no component consumes its closure value. For a declared capacity carrier of dimension D, the record capacity $N_{\mathrm{record}} = \ln D$ bounds every correctable classical record code by $\ln M_0 \le N_{\mathrm{record}}$ [39, 47]. The cosmological relation $N = 3\pi/(\Lambda\ell_\star^2)$ additionally requires a physical identification of capacity with horizon area, which the construction does not make.

2.7 Time parameters

The construction keeps the time parameters of Table 2 distinct. Conversions between them need explicit maps. The construction uses four: the fixed-reference relation $t = -\beta s$ of Section 6.6 between a model coordinate and a modular parameter, the closed-history relation $\tau = -Ls$ of the observers paper [6], the instrument-valued conversion between two local event clocks of Section 4.2, which keeps their whole count law, and the rescaling of native attempt count along the refining screens of Section 7. A calibration of native calls against a monotonic host counter is an engineered software correspondence. No preferred universal physical clock is required.

3 The finite observer

This section is layer three on one carrier: an observer that prepares, records, retains and acts on native records, with its capacity, memory and recovery limits.

3.1 The native observer

The native observer executes Definition 2.4 on the scored preparation [45, 46]. A trial is one fresh preparation followed by two repair events and a halt; running many trials repeats the preparation and does not extend one observer's memory. Event order is the time parameter. The tape, prior, forecast table, seam selector and halting rule are designed apparatus, and the forecast is a fixed table without learning.

Proposition 3.1 (Statistics of the native observer)**.** *Prepare the 66 binary configurations uniformly, use fair coins and the controller of Definition 2.4, and let Y indicate a positive second current. Then:*

(i) the 264 histories, two coins for each of the 66 configurations at each of the two events, each have weight $1/264$*;*

Table 2: Time parameters and where they occur.

Parameter	**Meaning**	**Occurrence**
Host time	Runtime of a software execution; a resource cost	A software calibration of native calls; never a physical quantity
Solver counter	Position in a search or repair schedule; the attempt count of a declared schedule	History repair (Section 6.7); attempt counts in the observer theorems
Causal order	Precedence of recorded events; counts of native operations	Observers, detectors, phase protocol, local ticks
Local event count	Attempted repairs incident to an observer region, or occupation changes at a port	Overlapping observers, local and active event clocks, matched screens (Section 4)
Closed-history length	Integer attempt count L of a closed native return, the inverse-temperature multiplier of the closed-history functional	Closed-history functional and modular correlations (Section 4.5)
Scaled attempt count	Native attempt count n at level l rescaled to $\tau = n/(1260 \cdot 16^l)$ on the refining screens	Native density limits and the controlled repair response (Sections 7.3, 9)
Coherent wave parameter	Parameter of the admitted coherent continuation $\cos(t\sqrt{A})$	Native waves and the return reference (Sections 7.2, 7.7)
Model coordinate	Parameter of a declared Hamiltonian or action	Scalar detectors, two-site apparatus, common source, compact interacting core
Modular parameter	Parameter of the modular automorphisms of an algebra-state pair	Modular clock and fixed-reference relation (Section 6.6), closed-history and regional references (Sections 4.5, 6.10)
Operational clock	Observer-accessible reading with calibrated duration	Supplied by no component

(ii) the first record has masses $14/33$, $14/33$, $5/66$, $5/66$ *for* $(c, j) = (0, 0), (1, 0), (0, 1), (1, -1)$, *the conditional probabilities* $q(r) = \Pr(Y = 1 \mid r)$ *are* 0, $5/56$, 0, $1/2$ *in the same order, and* $\Pr(Y = 1) = 5/66$;

(iii) the Brier risk of forecasting Y *is* $395/7392$ *with* $q(r)$, $140/2013$ *with the best forecast based on the chosen seam alone, and* $12875/150304$ *when* q *is evaluated on an independent record resampled within the same seam;*

(iv) the mean final load at port 0 *is* $35/264$ *under record feedback and* $1/6$ *when the selector is disconnected or replaced by an independent selector with the same seam frequencies; the feedback effect is* $-3/88$.

Proof sketch. For (iii), $\mathbb{E}[(Y - q(R))^2] = \mathbb{E}[\mathrm{Var}(Y \mid R)]$. The chosen seam S is a function of R; the best S-only forecast has excess risk $\mathbb{E}[\mathrm{Var}(q(R) \mid S)]$, and an independent resample within the seam adds twice that amount. For (iv), a positive first current occurs on ten binary inputs with coin zero, mass $5/66$. On that branch $x_0 = 0$, $x_1 = 1$ and $\mathbb{E}[x_5] = 1/10$; a second repair on $(0, 1)$ gives port-0 mean $1/2$ and one on $(0, 5)$ gives $1/20$. Hence the change is $(5/66)(1/20 - 1/2) = -3/88$ and the mean is $1/6 - 3/88 = 35/264$. An independent selector leaves the uniform shell uniform under either seam [46]. □

The other two current letters occur only on the full carrier, where the forecast abstains. The exact receipt retains all 264 histories [86], and a seeded execution of 20,000 fresh trials matches every trial against its exact path reference, with sample final port-0 loads 0.1327 with feedback, 0.16645 disconnected and 0.1668 with the independent selector [87]. Since the selector is a design choice and the forecast table is the exact conditional law, (iii) and (iv) are consistency checks: the Bayes forecast beats coarser forecasts and the record-wired selector moves the occupation.

3.2 Continuing memory and the actual stationary reference

A continuing observer [2] extends the two-event construction with an explicitly declared memory law. It keeps the same source and complete C/E instruments. A classical tag chooses C with probability λ and E otherwise; a repair occurs with probability r, with an identity hold otherwise. A controller chooses a preferred seam from its retained records. The actual seam distribution is

$$p(e \mid m) = \frac{\epsilon}{30} + (1 - \epsilon)\pi(e \mid m), \qquad r, \epsilon \geq \frac{1}{16}, \quad 0 \leq \lambda \leq 1.$$

The rational parameter bound and permitted controller tables are part of the specified design, not measured physical constants. The executed controller proposes $(0, 5)$ after a positive current and $(0, 1)$ otherwise. It does not read the hidden source load. A proposal intervention preserves the uniform exploration floor. Holds are recorded with current zero.

For each classical memory input, every source branch is $K_b = \sqrt{w_b} M_b$, with M_b a native partial injection or identity and $\sum_b K_b^* K_b = I_{78}$ exactly. Tensoring with the identity preserves an arbitrary untouched reference. The tests independently reconstruct the branch supports by endpoint floor/ceiling arithmetic, including their off-diagonal action. Composition therefore defines every finite history with its unnormalized quantum output and actual probability. Exact prefix replay and retained six-attempt source/reference/last-record marginals verify C, E and the delayed tagged mixture; the complete history tree remains implicit in that channel kernel.

One to eight complete records occupy the declared buffer. Under the executed one-slot export policy, n attempts leave one record and export $n - 1$ records for $n \geq 1$; expected repair and hold counts are nr and $n(1-r)$. Concatenating exports and the remaining buffer reconstructs the history.

The halt-on-full control instead preserves its source and records with an explicit timeout outcome. These are different resource contracts. Bounded working memory with growing exports is not closed finite-resource operation for unlimited duration. No new native source-word implementation of the supplied initialization, selection, write, export or delivery services is claimed.

The stationary reference must include this controller. For $r = 1$ and $\epsilon = 1/2$, a uniform binary source with blank memory becomes nonuniform after two attempts: 30 source probabilities change, with maximum deviation 1/960 from 1/66. Exact rational solution of the actual source/current-sign kernel instead gives one stationary law on a 129-state recurrent subset of the 132 source/bit states. Applying the complete C/E observer instrument to that law constructs the full-record reference. Its projection recovers the bit law, and one further actual observer step preserves the entire full-record density exactly. The resulting 128 C or 192 E record blocks have 4,218 positive diagonal entries overall. Positivity establishes faithfulness only on the explicitly compressed support of each block. The growing exported tape is not a stationary object, and the delayed member's hold updates require a separate stationary solve.

This reference also makes a clock obstruction precise. In an actual nontracial block choose positive weights with ratio $q = p_i/p_j > 1$ and the dimensionless modular parameter $s_* = \pi/(2 \log q)$. The state-dual modular action takes the real pair probe $(|i\rangle + |j\rangle)/\sqrt{2}$ to a density with off-diagonal entry $-i/2$. Every admitted history, with the declared real joint ancillas and classical memory, preserves real density matrices. A norm-one imaginary pair observable separates the target from every such normalized output, giving trace-distance error at least 1/2. This witness is not a native detector, and the probe is an all-input map test, not an asserted native preparation. On the accessible commutative record algebra the modular action is trivial. Other algebras, encodings or coherent services have different contracts; no general physical-clock exclusion follows. The exact reference, implementation bound and full qualifications are retained with the executable model [2].

3.3 Record capacity

The retained records are evaluated with the correctable-public-record construction of the theorem library [6, 47]. There M_0 is the largest correctable set of reachable public records, the independence number of the compound checkpoint confusability graph; the record capacity is $N_{\mathrm{record}} = \ln M_0$ nats; and $M_0 \le D$ for a declared carrier of dimension D. The counts below describe the reachable record interface; program, source, probe and service memory are additional.

Proposition 3.2 (Record capacity of the native observer)**.** *Take halt and exact classical reread as the declared terminal continuations of the observer of Definition 2.4. Then the compound graph has no edges, and:*

(i) $M_0 = 11$ *on the binary preparation (four reachable first records, 264 histories) and* $M_0 = 17$ *on the full family of 78 inputs (six first records, 312 histories);*

(ii) each terminal word projector has rank 78 in the 3354-dimensional carrier, distinct projectors are orthogonal, and the eleven reachable ones have total rank 858, so the bound is unsaturated with slack $\ln(3354/11)$*;*

(iii) the binary preparation reaches $1+4+11 = 16$ *classical prefixes, so four bits encode the complete controller history; a compact controller that reads only this code reproduces the phase, action, forecast, source loads, record and halt of all 264 histories; the full family reaches 24 prefixes and needs five bits;*

(iv) coarser retained readouts give correctable codes of size 4 for the first letter alone, 4 for the second letter alone, 3 for the forecast alone, 2 for the chosen seam alone, 4 for forecast and seam, and 1 after complete erasure.

Proof sketch. Halt and reread leave the tape unchanged, so the joint kernel is the identity and the existing evaluator returns the number of reachable terminal words. The terminal projector of word w occupies the basis coordinates $546 + 78w$ to $546 + 78w + 77$ of the carrier [47, 88]. □

The terminal capacity is $\ln 11$ nats, about 3.459 bits, on the binary preparation and $\ln 17$ nats on the full family; the Shannon entropy of the benchmark record law, about 2.023 nats, is a different quantity. The library's global proposal adds closure across the complete terminal fiber, $N_\star = \ln M_0(U_{N_\star})$, and a de Sitter horizon identification giving $\Lambda\ell_\star^2 = 3\pi/N_\star$ [6, 47]; the native records supply an actual finite input to that evaluator, and the two further steps are not taken here.

3.4 The P-dependent detector

The detector carries the closure input P through the electromagnetic coupling into a recorded binary observation that controls the next repair [52, 53]. The charge assignment, probe, calibrated flux, readout and controller are specified apparatus (Section 9.3).

Choose the charged realization $Q_v = g(P)N_v$ of the native occupation N_v. A current record j on the seam (a, b) then transfers the charge $\Delta Q = g(P)\, j\, (\mathbf{e}_b - \mathbf{e}_a)$, and total charge is conserved. For port gauge parameters λ_v and $U_\lambda|x\rangle = e^{ig\sum_v \lambda_v x_v}|x\rangle$, every current-grouped Kraus operator obeys $U_\lambda L_{(c,j)} U_\lambda^\dagger = e^{igj(\lambda_b - \lambda_a)} L_{(c,j)}$. The probe is a unit-charge path qubit prepared in $|{+X}\rangle$. The first native current routes it around a calibrated external loop with signed exposure $j\chi$, where $\chi = \oint A \cdot dx$ is prepared independently of g, so the relative phase is $\varphi_j = g(P)\, j\, \chi$, the Aharonov-Bohm phase [98]. A Y measurement commits $y = \pm 1$ with amplitudes $m_y(j) = \frac{1}{2}[e^{-i\varphi_j/2} - iy\, e^{i\varphi_j/2}]$ and effects $E_y = \sum_r \frac{1}{2}[1 + y \sin\varphi_{j(r)}]|r\rangle\langle r|$ on the old current register. The actuator receives only y. It repairs $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, keeps y unchanged, and the protocol halts.

Proposition 3.3 (P-dependent response)**.** *Prepare the 66 binary configurations uniformly, run the protocol above, and put $s = \sin(g(P)\chi)$. Then:*

(i) $\{E_y\}$ is a binary instrument on every input, including coherent records; each $M_y = \sum_r m_y(j(r))|r\rangle\langle r|$ commutes with the old record projectors, and the composed experiment is one CPTP map that preserves the committed bit;

(ii) $\Pr(y \mid j_1, P, \chi) = \frac{1}{2}[1 + y \sin(g(P) j_1 \chi)]$*;*

(iii) at $\chi = 1$ all 528 state, coin, detector and coin histories have positive probability, and

$$\mathbb{E}[y] = 0, \quad \mathbb{E}[y j_1] = \tfrac{5}{33} s, \quad \mathbb{E}[j_2] = \tfrac{5}{264} s, \quad \mathbb{E}[g(P) j_2] = \tfrac{5 g(P)}{264} s;$$

(iv) for weak exposure, $\mathbb{E}[g\, j_2] = \frac{5\pi}{66}\alpha(P)\chi + O(\chi^3)$;

(v) disconnecting the probe, or replacing y by an independent fair bit, gives $\mathbb{E}[j_2] = 0$ with the same detector and seam frequencies.

Proof sketch. The eigenvalues of E_y lie in $[0, 1]$ and sum to one, which gives (i). For (iii), after the first repair let the source be z; averaging the second coin gives $\mathbb{E}[j_2 \mid z, j] = \frac{1}{4}(2z_0 - z_1 - z_5) + \frac{sj}{4}(z_1 - z_5)$. The first term averages to zero because the first repair preserves the uniform shell.

Currents ± 1 each have mass $5/66$, and on either branch the other load is uniform over ten ports, so $\mathbb{E}[j(z_1 - z_5)] = 5/66$. An independent floor and ceiling enumerator reproduces every coefficient over all $66 \cdot 2 \cdot 2 \cdot 2 = 528$ paths [53, 55]. □

At the computed P and $\chi = 1$ the plus probability is 0.649129261647 for $j = +1$, 0.350870738353 for $j = -1$ and $1/2$ for $j = 0$; the mean second current is 0.00564883566843, the mean transferred charge 0.00171084937579 in natural units, and the final port-0 load 0.161017830998235 against $1/6$ for the controls. Off-closure controls at $0.999P$ and $1.001P$ change the detector and feedback probabilities at fixed χ, and a seeded run of 200,000 trials gives mean second current 0.006390 with standard error 0.00085715 [52, 91]. The audit record (r_1, y, r_2) has 24 terminal words and $M_0 = 24$ [54]. If a cut enclosing a retained record obeys the area-capacity law with area K chart cells, then $N_{\mathrm{cut}} = KP/4$ and $\ln M_0 \le N_{\mathrm{cut}}$ give $K \ge 4 \ln M_0 / P$; with the pinned P one detector bit needs $K \ge 2$ ($4 \ln 2 / P \approx 1.6999608573$) and the full audit $K \ge 8$ ($4 \ln 24 / P \approx 7.7942567832$).

3.5 Quantum mixing and memory

Discarding every seam, coin and repair record after each event turns the repairs into channels Φ_{C} and Φ_{E} on the source. Under independent seam attempts these channels have an exact spectral theory [48]. Theorems 3.4 and 3.5 concern this record-discarded channel under independent uniform attempts; they do not describe the retained tape or the feedback controller.

Theorem 3.4 (Signed-difference reduction and quantum gap)**.** *Let G be a finite connected graph with $n \ge 2$ vertices, positive edge weights w_{ab} and $W = \sum w_{ab}$. Let each event choose the edge ab with probability w_{ab}/W and a fresh fair coin, and discard its record. On the binary space $\mathcal{H}_k$ of load k, $1 \le k \le n-1$, with $D = \binom{n}{k}$:*

(i) for a matrix unit $|x\rangle\langle y|$ the signed difference $z = x - y$ is invariant; if z has r positive entries and $Z = \{v : z_v = 0\}$, then on the sector of z

$$\Phi_{\mathsf{L}}\big|_z = I - \frac{L_{\mathrm{excl}}(G[Z], k - r) + K_{\mathsf{L}}(z)\, I}{2W}, \qquad \mathsf{L} \in \{\mathsf{C}, \mathsf{E}\},$$

with the exclusion Laplacian L_{excl} of the induced graph, $K_{\mathsf{C}}(z) = \sum w_{ab}|z_a - z_b|$ and $K_{\mathsf{E}}(z) = 2 \sum w_{ab} \mathbf{1}[z_a \ne z_b]$;

(ii) both channels are unital and Hilbert-Schmidt self-adjoint with spectrum in $[0, 1]$, their fixed algebra consists of the scalars, and both have gap $\gamma = \lambda_1(G)/(2W)$, where $\lambda_1(G)$ is the first positive Laplacian eigenvalue; and $\|\Phi_{\mathsf{L}}^t - \mathrm{Dep}_D\|_{2\to 2} = (1 - \gamma)^t$;

(iii) for every input and every untouched reference R, $\mathrm{dist}\big((\Phi_{\mathsf{L}}^t \otimes \mathrm{id}_R)(\rho), \tau \otimes \rho_R\big) \le \min\{1, \frac{D}{2}(1 - \gamma)^t\}$, with $\tau = I_D/D$ and $\mathrm{dist}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

For the twelve-port carrier at load two ($n = 12$, $W = 30$, $\lambda_1 = 5 - \sqrt{5}$, $D = 66$) the gap is $\gamma = (5 - \sqrt{5})/60 \approx 0.046066$, and the slowest off-diagonal sectors contract by exactly $r_{\mathsf{C}} = 5/6$ and $r_{\mathsf{E}} = 7/10$ per attempt. On the full 78-state carrier the gap is the same, the full channel is block triangular with a diagonal collision block whose entries lie in $[0, 5/6]$, and the unique stationary density is the uniform binary density, of rank 66 on dimension 78.

Proof sketch. A matrix unit survives a coin branch exactly when its two records agree, and then both configurations translate by the same current; comparing the coin branches gives (i). The diagonal sector is the symmetric k-particle exclusion chain, whose gap equals $\lambda_1(G)$ by the theorem of Caputo,

Liggett and Richthammer [101]. For a nonzero sector, $K_{\mathsf{E}} \geq K_{\mathsf{C}} \geq \frac{1}{2}\sum w_{ab}(z_a - z_b)^2 \geq r\lambda_1 \geq \lambda_1$, so no off-diagonal mode is slower. Item (iii) uses Schmidt rank at most D and $\|Y\|_1 \leq D\|Y\|_2$. On the carrier, $r = 1$ gives $K_{\mathsf{C}} = 10$ and $K_{\mathsf{E}} = 2(10 - \mathbf{1}[uv \text{ edge}])$, while $r = 2$ gives $K_{\mathsf{C}} \geq 16$ and $K_{\mathsf{E}} \geq 24$. The collision block has largest entry $1 - d_{\min}/W = 5/6 \leq 1 - \gamma$ [48]. The executable checks the reduction against the native primitive on all 4,356 binary matrix units and the transient structure on all 6,084 full matrix units [89]. □

Choose adjacent ports u, v and the two mixtures $\rho_\pm = \frac{1}{20}\sum_{w\notin\{u,v\}}(|uw\rangle \pm |vw\rangle)(\langle uw| \pm \langle vw|)$. They have identical load populations and orthogonal supports, and

$$\operatorname{dist}(\Phi_{\mathsf{L}}^t(\rho_+), \Phi_{\mathsf{L}}^t(\rho_-)) = r_{\mathsf{L}}^t, \qquad \operatorname{dist}(\Phi_{\mathsf{C}}^t(\rho_+), \Phi_{\mathsf{E}}^t(\rho_+)) = \tfrac{1}{2}(r_{\mathsf{C}}^t - r_{\mathsf{E}}^t).$$

The second distance is $1/15$ after one attempt and peaks at four attempts at $12259/101250$; the binary effect $\frac{1}{2}(I + A_{uv} + A_{uv}^\dagger)$ with $A_{uv} = \sum_{w\notin\{u,v\}}|uw\rangle\langle vw|$ attains it. Load populations, classical load histories and diagonal geometric observables obey the same exclusion process under both laws, so only coherent preparations and readouts distinguish them under this schedule.

Theorem 3.5 (Finite quantum memory)**.** *Under the schedule of Theorem 3.4 on the twelve-port carrier:*

(i) on the binary shell both record-discarded channels are entanglement breaking after at most 134 attempts;

(ii) on the full carrier let u, v be adjacent and $|\Psi\rangle = (|2u\rangle|0\rangle + |uv\rangle|1\rangle)/\sqrt{2}$ with a reference qubit; after t attempts the coefficient of $|2u\rangle\langle uv|$ is a_{L}^t, with $a_{\mathsf{C}} = 23/30$ and $a_{\mathsf{E}} = 7/10$, and the partial transpose on the reference has a principal 2×2 minor equal to $-a_{\mathsf{L}}^{2t}/4 < 0$; no finite power of either full channel is entanglement breaking;

(iii) every input converges to $\tau_P \otimes \rho_R$, where τ_P is the uniform binary density, with dist $\leq \min\{1, \mu^{s/2} + \mu^s + \frac{D}{2}(1-\gamma)^{t-s}\}$ *for all $0 \leq s \leq t$, $\mu = 5/6$ and $D = 66$.*

Proof sketch. For the normalized Choi state J_t, $\|J_t - I/D^2\|_2^2 = D^{-2}\sum_{i>0}\lambda_i^{2t}$. The separable ball of Gurvits and Barnum [105] and the equivalence of separable Choi states with entanglement-breaking channels [107] reduce (i) to $65\, r_{\text{mix}}^{2t} + 4290\, r_{\mathsf{L}}^{2t} \leq 1/4355$, where $r_{\text{mix}} = 1 - \gamma$. The rational enclosure $559/250 < \sqrt{5} < 2237/1000$ certifies this at $t = 134$ and its failure at $t = 133$; treating all modes with r_{mix} would certify only 178. For (ii), 25 edges avoid u and four of them touch v; C keeps one coin on each of those four and E keeps neither, and the triangular structure prevents any return from the binary space. Separable states have positive partial transpose [111]; the complete argument is in [48, 89]. □

Replacing the full carrier by its attracting binary shell therefore changes an exact finite-time quantum property, although both converge to the same stationary density. The number 134 is a sufficient certificate time; the exact index of entanglement breaking on the shell is not determined.

3.6 Recovery from complete records

Preserving committed records and recovering the unknown input from them are different questions [49]. Here a seam policy may depend on committed records and classical randomness and may not inspect hidden loads or insert quantum corrections; a final decoder may act on the full load register and tape, keep the tape, use blank ancillas and return a state on the original carrier. That decoder access is stronger than the record-only access of the observer's controller.

Theorem 3.6 (Optimal recovery from complete native records)**.** *Let X be the 78 configurations or the 66-state shell and $D = |X|$. For a complete admitted history h let X_h be its compatible inputs, P_h the projector onto them, $d_h = \operatorname{rank} P_h$, and w_h the product of its seam and coin factors. Then:*

(i) the history Kraus operator is $K_h = \sqrt{w_h}\, U_h P_h$ with $U_h^\dagger U_h = P_h$ and $\sum_h w_h P_h = I$; a tape-controlled inverse outputs $\sum_h w_h |h\rangle\langle h| \otimes P_h \rho P_h$, so every initial load population is recovered;

(ii) among all final CPTP decoders the optimal entanglement fidelity for the maximally mixed input is $F_{\mathrm{opt}} = D^{-2} \sum_h w_h d_h^2 = D^{-2} \sum_{x,y} H_{xy}$, with $H_{xy} = \sum_{h:\, x,y \in X_h} w_h$;

(iii) for the two-event observer, F_{opt} equals 5543/8712 (binary) and 1301/2028 (full) under C, and 769/1452 and 3329/6084 under E with the same current-sign selector applied to its decoded current;

(iv) the maximal exactly correctable algebra on the whole input is the commutant of $\{P_h\}$, and one event with full-support random seam choice reduces it to the diagonal load algebra; nevertheless the qubit code $|0_L\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $|1_L\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ satisfies the Knill-Laflamme condition $QP_hQ = \alpha_h Q$ for every seam, coin and record of either law and is recovered exactly after one event;

(v) under independent seam attempts $H_{xy}(t) = q_{\mathsf{L}}(x-y)^t$, with largest off-diagonal rates 5/6 for C and 7/10 for E on both carriers; $F_{\mathrm{opt}}(t) \to 1/D$, and neither retained instrument is entanglement breaking at any finite horizon;

(vi) a fixed E seam word identifies every full-carrier input exactly when at most one visited component has size one or two, and every binary input exactly when at most one component is a singleton and at most one has size two; the minimal such words have eight and seven distinct edges;

(vii) no bounded record-only adaptive policy makes C entanglement breaking; a C word that doubles a minimal forest, 16 or 14 events, excludes every nontrivial exact code; the policy that repeats each forest edge until its coin first differs identifies the input almost surely, with stopping time $T = m + \sum_{i=1}^{m} G_i$ for independent geometric G_i of success probability one half, $\Pr(T = t) = \binom{t-m-1}{m-1} 2^{-(t-m)}$ for $t \geq 2m$, and $\mathbb{E}[T] = 3m$, namely 24 on the full carrier and 21 on the binary shell.

Proof sketch. Every retained current determines a translation, so reversing them gives the input uniquely from the terminal configuration and h. Optimality follows from Cauchy-Schwarz on each classical branch: $\sum_a |\mathrm{Tr}(R_{ha} U_h P_h)|^2 \leq d_h^2$. For (iv), commutation is sufficient by (i), and necessity follows from equality in the Schwarz inequalities in a Stinespring dilation; the code has nonadjacent occupied pairs, equal one-port means and vanishing products $N_a N_b$ on every edge, which gives the Knill-Laflamme identity [108]. For (vi), matching E histories are those for which z is constant on every visited component. For (vii), the inputs $\{u, w\}$ and $\{v, w\}$ with u, v nonadjacent keep equal currents for at least one coin on every edge, so a history of length at most T shares a rank-two fiber with probability at least 2^{-T}; repeating an edge with the opposite coin reveals the initial parity through $d = 2j_1 + j_*$ [49, 90]. □

The minimal fixed E words are the forests $(0,1),(0,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the full carrier and $(1,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the binary shell, with component sizes $(3,3,3,3)$ and $(1,2,3,3,3)$. The full-input correctable blocks of the two-event observer

have sizes $46, 10, 9, 1$ on the binary shell and $56, 10, 9, 1, 1, 1$ on the full carrier, for both laws. Existing source inverse words implement the single-seam branch reversal of (i) while keeping the classical archive [49, 90]. Native current records thus suffice to reverse classical load transport, without restoring arbitrary coherence lost across record groups. The source encoder and active recovery of Section 6.5 instead preserve the stated logical qubit between events; they do not invert the whole bare-load channel.

3.7 A coherent continuation and its selection boundary

An exact finite experiment attaches a reflection walk to the binary native branches, with coherent event access in place of a committed tape [50]. For each binary configuration x, seam e and coin c, let (y, j) be the native output and current. On the $66 \cdot 60 = 3960$ arc basis vectors define

$$S|x,e,c\rangle = |y,e,c'\rangle, \qquad c' = \begin{cases} c, & j = 0, \\ 1-c, & j \neq 0, \end{cases} \qquad A|x\rangle = \frac{1}{\sqrt{60}} \sum_{e,c} |x,e,c\rangle.$$

Every arc is checked: $S^2 = I$, with 300 exchanged pairs and 3360 fixed arcs, and the outgoing native record (y, e, c, j) is recovered bijectively from the reversed arc. Thus $A^\dagger A = I$ and $A^\dagger S A = K$, the actual symmetric native transition kernel, and $W = S(2AA^\dagger - I)$ satisfies the hypotheses of the quantum-walk spectral mapping theorem with discriminant K [119, 120]. The identity $A^\dagger W^n A = T_n(K)$, with T_n the Chebyshev polynomial, gives for the centered loads $X_{x,p} = x_p - 1/6$

$$KX = X(I - L/60), \qquad U_{n+1} - 2U_n + U_{n-1} = -U_n L/30, \qquad U_n = A^\dagger W^n A X.$$

These are compressed amplitudes, and n counts the supplied coherent continuation with no identification as duration. The binary domain is essential: on the full carrier $(2, 0, \ldots) \mapsto (1, 1, \ldots)$ has no native reverse event, so its asymmetric kernel equals $A^\dagger S A$ for no involutive unitary on the same domain. Twelve port addresses and sixty seam/coin events give a 720-arc source-compiled walk with discriminant $K_{12} = I - L/60$; the map $F_{x,p} = (x_p - 1/6)/\sqrt{10}$ satisfies $F^* F = I - \mathbf{1}\mathbf{1}^T/12$ and $K_{66} F = F K_{12}$, so it realizes the shared mean-zero density sector. Among the sixty proper carrier rotations a unique inverse-symmetric order-five class $\mathcal{C}$ of size twelve gives $L = 6I - \frac{1}{2}\sum_{g \in \mathcal{C}} U_g$, with multiplicity-free density representation $1 \oplus 3 \oplus 5 \oplus 3$. The other order-five class is isospectral but assigns probability zero to a native seam whose actual one-attempt probability is $1/60$, so the seam records select the class.

The emitted walk word uses ten address qubits, five blank scratch qubits and 114,910 source pulses per walk with 229,820 load/export events; uniform event preparation costs a separate 36,288 pulses. The reversible circuit is checked on all 32,768 basis inputs, every clean-scratch address returns clean scratch, including 304 padding addresses, and the fifteen-qubit execution reuses raw pulse factors without target-matrix insertion or output normalization. Floating routine guards sum to $s = 829/(5 \cdot 10^9)$, giving complete output trace-norm error at most $2\delta + \delta^2$ with $\delta \leq s/(1-s)$; these are numerical guards, not physical precision certificates.

A finite reference and its modular clock. Biased native event amplitudes, the compiled reversible seam shift, inverse event preparation and a herald realize $F_p = I - pL/60$ on the protected site code. From the supplied $I_{12}/12$, n successive successes give $F_p^{2n}/\operatorname{Tr} F_p^{2n}$ with success probability at least $1/12$, every failure recorded; for $p = 30t/n \leq 1$ this tends to $e^{-tL}/\operatorname{Tr} e^{-tL}$ with relative-entropy and total-variation errors at most $16t^2/[n(1 - 4t/n)]$. The execution at $p = 1/2$, $n = 2$, $t = 1/30$ tests all twelve input columns with success $4382551/5184000$ and bound $1/105$, and

a source interference readout has contrast 260343/43825510 against zero after dephasing. This state is central on the 44-dimensional represented group algebra, where its modular flow is trivial; nontrivial modular gaps require the full protected matrix algebra [20].

Positive subordination gives square-root weights on the same register. With $F = I - hL/2$, $h = p/30$, a Poisson number of Sibuya jumps gives positive weights with

$$\sum_n w_n F^{2n} = e^{-R\sqrt{I-F^2}} = e^{-b\sqrt{L-hL^2/4}}, \qquad b = R\sqrt{h}.$$

Finite jump and count caps and dyadic rounding retain missing mass as sampler rejection. The source instance $p = R = 1/2$ uses 64 native coin records and at most four raw filters, retains each first failure and all sixteen successful interference outcomes, and has Gibbs trace-distance bound below 0.217 and spectral diagnostic 0.01344. Its finite polynomial $M = \sum_n w_n F^{2n}$ fixes $H = -\log(M/m_0)$, $m_0 = \sum_n w_n$. Raw source words execute that clock, a current-controlled intersector kick and complete spectral-pair detectors on the same register, retaining all 3,600 two-attempt native tapes, sampler and herald failures, sixteen detector addresses and twelve reference columns. The modular identity $\sigma_s = \alpha_{-s}$ holds for the ideal successful marginal $M/\operatorname{Tr} M$ on $\mathcal{B}(\mathbb{C}^{12})$; it does not hold for every retained request block.

This density supplies eleven pairs $Q_j = X_{0j}/\sqrt{2\Delta_j}$, $P_j = Y_{0j}/\sqrt{2\Delta_j}$, $\Delta_j = p_0 - p_j > 0$, with $\langle [Q_j, P_k]\rangle = i\delta_{jk}$, symmetric covariance $\delta_{jk}\coth(g_j/2)/2$, $g_j = \log(p_0/p_j)$, and frequency g_j under H. For chronological factors $U_N = e^{iA_m/\sqrt{N}}\cdots e^{iA_1/\sqrt{N}}$ built from these pair observables on independent identical copies, a parity commuting with the state and anticommuting with every A_k gives

$$\left| z_N^N - e^{-C} \right| \le \frac{D^4/24 + |C|^2/2}{N}, \qquad z_N = \operatorname{Tr}(\rho U_N),\ D = \sum_k \|A_k\|,$$

with $C = \frac{1}{2}\sum_k \langle A_k^2\rangle + \sum_{k>j}\langle A_k A_j\rangle$. Source pointer reads execute the one-copy factors for three band representatives with all 32 work/pointer addresses retained; a growing copy register is not executed. With physical normalization $q_j = Q_j/\sqrt{\omega_j}$, $\pi_j = \sqrt{\omega_j}P_j$ and $g_j = b\omega_j$, the clock H/b gives thermal covariance $\coth(b\omega_j/2)\cos(\omega_j t)/(2\omega_j) - i\sin(\omega_j t)/(2\omega_j)$. For m smeared factors with $R = \sum_k R_k$, $R_k^2 = \sum_j (f_{kj}^2/\omega_j + \omega_j h_{kj}^2)$, gaps $g_j, b\omega_j \ge \gamma > 0$, $\vartheta = \gamma/(1+\gamma)$ and $Z = 1/p_0$,

$$|z_N^N - e^{-C}| \le \frac{R^4}{\vartheta^2 N}\left(\frac{(m+1)Z}{96} + \frac{1}{8}\right),$$

so $Z/N \to 0$ suffices on a fixed positive band; the family $\rho = \operatorname{diag}(2,1,\dots,1)/(d+1)$ with $N = d$ has instead the limit $e^{3(\cos(1/\sqrt{2})-1)} \ne e^{-3/4}$. On the native family $D_l = 4^l L_l$ one has $\lambda_n(D_l) \ge n/81940$, hence $\operatorname{Tr} e^{-b\sqrt{D_l}} \le 1 + 163880/b^2$, and with the lazy filter $p = 630h16^l \le 1$,

$$Z \le \frac{1 + 327760/b^2}{m_0}, \qquad P_{\text{success}} \ge \frac{m_0}{240\,4^l};$$

both prepared and target gaps exceed $b/182$ when $(1-m_0)/m_0 \le b/182$. Preparation error E with success floor $p > 2E$ gives conditional trace error at most $4E/p$, and with one-copy word error ϵ the N-copy amplification is at most $N\zeta/(1-N\zeta)$, $\zeta = 4E/p + \epsilon$. On a fixed band $0 \le D_0 \le \Lambda$ with $h\Lambda < 2$, retained positive weights with missing mass $\delta < a \le e^{-b\sqrt{\Lambda}}$ give

$$\|\log(m_0)I - \log M - b\sqrt{D_0}\| \le \frac{\delta}{a-\delta} + \frac{bh\Lambda^{3/2}}{4(1+\sqrt{1-h\Lambda/4})}.$$

A constructive growing word family sets $k = 2l + c$ with $4^c \geq b^2 \max(630, \Lambda)$, $R = 2^k$, $h = b^2 4^{-k}$, $a = 4^{-k}$, jump cap 2^{6k}, count cap $6R + 2k$ and Taylor order $4k + 3$; the zero-request branch keeps weight at least $f = 2^{-2R}$. For $d = 240\,4^l$, request cap K and word length P, primitive and load/export error at most $fa/(64dKP)$ gives complete stopped preparation error $E \leq fa/(32d)$ and

$$\|dS_{\mathrm{raw}} - M\| \leq 2dE, \qquad \|\log(dS_{\mathrm{raw}}) - \log M\| \leq \frac{a}{16 - a}$$

for the actual successful marginal S_{raw}, without a reducing-band premise. The derived partition bound permits $N = 2^l$ copies with copy, preparation and detector errors tending to zero on a fixed positive band. Growing execution, spectral tails, accurate source services and a zero-energy anchor for the radial product lie outside these bounds, and fixed caps with $N_c h \to 0$ force the bounded-band generator to zero; the uncapped sampler has infinite expected work because the constant sector survives every filter [20].

Selection controls. Two exact controls bound what the committed records select. For $|z| = 1$, $W_z = S[AA^\dagger + z(I - AA^\dagger)]$ has the same first event $W_z A = SA$ but $A^\dagger W_z^2 A = K^2 + z(I - K^2)$; a complete added Ramsey read gives two-step plus probability 1 for $z = 1$ and $113/150$ for $z = -1$. Multiplying each coherent record branch on seam $(0, 1)$ with current $+1$ by -1 preserves every committed C and E branch on arbitrary references, yet the twisted embedding gives $\mathrm{Tr}[(A_\eta^\dagger S A_\eta)^3] - \mathrm{Tr}(K^3) = -1/900$. Both real continuations conserve load, respect all sixty proper rotations, fix the uniform coherent state and admit the time reversal $\Theta = S$ conjugation; under full event-permutation covariance, real orthogonal coins fixing the uniform vector are identity or reflection. A sign change on every moving arc preserves proper covariance and the committed instruments but gives $\mathrm{Tr}(D^3) - \mathrm{Tr}(K^3) = -1/90$ for $D = 2\,\mathrm{diag}(K) - K$. Scalar record phases cancel throughout committed-record histories. A source-executed interface test that reads the low unread record bit in the X basis gives complete X-plus probabilities $3/4$ and $1/4$ for one preparation, against $1/2$ for both after commitment, using 5380 C or 5383 E source pulses per attempt [51]. These are restrictions on selection: coherent access is a supplied interface, and the test selects no phase.

3.8 The local event clock

N4 is a reciprocal extension of the binary repair law in the construction's candidate register [42]. Each port carries a private six-level response factor, and one coherent exchange $W_e = \exp[\theta(Y \otimes \Gamma_e + X \otimes \Delta_e)]$, with $X = S_e(N_a - N_b)^2$, $Y = iS_e(N_a - N_b)$ and response generators Γ_e, Δ_e built from the endpoint response matrices, conjugates the binary repair, $K_r = W_e(L_r \otimes I)W_e^\dagger$, with supplied coupling θ. Private preparations change bare load probabilities: at fixed bare load and $\theta = 1/8$ an endpoint occupation probability moves from 0.482020673 to 0.517979327. An interval certificate proves an edge-Gram margin $\mathcal{G} \geq mI$ at $\theta = 1/16$ and $\theta = 1/8$ [42, 66, 2].

Fix an observer port v. A tick is any attempted seam incident to v, including attempts that leave its load unchanged. Under independent uniform seam selection a tick has probability $p = 1/6$ per attempt, and remote attempts continue between ticks. In independent trials the full tracial state is prepared, the binary occupation N_v is measured, the unmodified source runs until one or two ticks have occurred, and N_v is measured again, with no intermediate measurement, reset or postselection. With $f = N_v - 1/6$ and $\|f\|_\tau^2 = 5/36$, write $c_k = \mathbb{E}[(X_0 - \frac{1}{6})(X_k - \frac{1}{6})]/(5/36)$ for the endpoint correlation after k ticks [66, 67]. The numbers of Theorem 3.7(iii) are certified lower bounds; the full N4 two-tick correlation is not evaluated, and preparation, incident-event detection

and bare-occupation readout are supplied operations that the detector of Section 3.1 does not implement.

Theorem 3.7 (Memory at observer-local ticks)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be the averages over the thirty seams of the remote and incident seam channels, so that the first-hit channel is $\mathcal{K} = \mathcal{B}(I - \mathcal{A})^{-1}$. Then:*

(i) $c_k = \langle f, (\mathcal{K}^*)^k f\rangle / \|f\|^2$, *and* $c_1 = 1 - 6\alpha_{\mathrm{occ}}/11$ *for the occupation compression factor* α_{occ}*;*

(ii) the statistic

$$b_{\mathrm{tick}} = c_2 - c_1^2 - \frac{r^2(209r + 180)}{1331r^2 + 3300r + 1800}, \qquad r = 1 - c_1,$$

vanishes for the native law and for the one-term exchange controls, which have $c_1 = 5/11$ *and* $c_2 = 93/407$*;*

(iii) for both N4 completions, built on C *and on* E, $b_{\mathrm{tick}} \geq 3m/50$, *and the certified lower bounds are*

$$b_{\mathrm{tick}} \geq \frac{139996257}{12500000000000} \approx 0.00001119970056 \quad (\theta = 1/16),$$
$$b_{\mathrm{tick}} \geq \frac{11335290321}{50000000000000} \approx 0.00022670580642 \quad (\theta = 1/8);$$

(iv) $c_{2,\mathsf{C}} \geq c_{2,\mathsf{E}}$ *and* $b_{\mathrm{tick},\mathsf{C}} \geq b_{\mathrm{tick},\mathsf{E}}$.

Proof sketch. Each seam channel is a positive semidefinite contraction of Hilbert-Schmidt space, with spectrum in $\{0, 1/2, 1\}$ for C and $\{0, 1\}$ for E; N4 conjugation preserves this. Hence $0 \leq \mathcal{A} \leq (1-p)I$, the first-hit series converges, and the stopped maps are unital and trace preserving. Remote seams fix f, which gives (i). The closed comparator with seam mean maps $I - \frac{\alpha_{\mathrm{occ}}}{2} w_e w_e^\top$ yields the correction term in (ii) after substituting $\alpha_{\mathrm{occ}} = 11(1-c_1)/6$. With the Schur complement S of $I - \mathcal{A}$ on the centered occupations, $S \leq I$ and $b_{\mathrm{tick}} = \langle z, S^{-1} z\rangle / (p\|f\|^2) \geq \|z\|^2/(p\|f\|^2)$. The hidden defect is $z = -\frac{1}{60}\sum_e t_e S_e$ with five incident coefficients ± 1, so $\|z\|^2 \geq 5m/3600$ and $b_{\mathrm{tick}} \geq 3m/50$. On each seam $\mathsf{C} - \mathsf{E}$ is positive semidefinite, which gives (iv) [66]. □

The correction used for global attempts, $(1 - c_1)^2/10$, assigns $-16/2035$ to the native null at local ticks; the correction of Theorem 3.7(ii) is the one adapted to this stopping instrument. The receipt and eight independent checks bind the result, including a comparison with the full 66-state native stopping kernel and a complete dimension-six quantum model of the resolvent and Schur identities [66, 92].

4 Observers, clocks and screens on one process

This section completes layer three on one native process. Its components run the native C or E instrument itself, with the uniform independent seam schedule and fair placement coin as declared scheduling inputs. Their time parameters are attempt counts and local event counts. Each result states its preparation, observer access and services; none identifies a count with laboratory duration [14].

4.1 Two overlapping observers

On the 66 binary total-two configurations, prepare the uniform state and disclose the bits at ports $(0, 1, 5)$; all eight labels stay in the output menu. Observer A accesses $(0, 1)$ and observer B accesses $(0, 5)$, so the regions overlap at port 0. Each receives its initial bits and every incident local current,

including zero, and ticks when an attempted seam meets its region. The protocol requests two incidents on the union with at most two native attempts per interval; after the first it applies one native repair on $(0,1)$ if the reconstructed n_0 is one and on $(0,5)$ otherwise, and that repair ticks both observers.

Proposition 4.1 (Exact overlapping-observer correspondence)**.** *Normalize the native remote and incident branch counts by* 60*, write* $P = A + B$ *with* $B = \sum_m B_m$ *over incident marks* m*, and put* $D = I - A$*,* $p = 2/5$ *and* $G = D/p$*. With* $K_m = B_m D^{-1}$ *and* $F_m = D^{-1} B_m$*,*

(i) $K_m = G F_m G^{-1}$ *and* $\mathbf{1}^{\mathsf{T}} G = \mathbf{1}^{\mathsf{T}}$*; remote projectors, the selective feedback maps and the disclosed uniform preparation commute with or are fixed by* G*;*

(ii) for attempt cap h *the marked and timeout maps are* $K_{m,h} = B_m(I + A + \cdots + A^{h-1})$ *and* $Q_h = A^h$*, their effective conjugates are* $F_{m,h} = G^{-1} K_{m,h} G$ *and* A^h*, and the terminal decoder* G *returns the complete unnormalized source state on every output word of every finite adaptive protocol in this commuting intervention family;*

(iii) the direct endpoint execution and the effective recursion agree exactly on all 5,768 *output words,* 689 *of positive probability, including all terminal populations, and the timeout probability is* $369/625$*; and*

(iv) the unread completed transfer $F = D^{-1} B$ *is self-adjoint in the metric* G *with* $1/20 \le F \le 1$*, so* $H = -\log F$ *is an effective Euclidean generator per completed union incident tick.*

Proof sketch. Remote repairs leave the union fields fixed and every remote seam is disjoint from the feedback ports, so the conjugation by G cancels between successive reads and allowed feedback maps; induction over the protocol gives (ii). Every native seam count equals $I + \mathrm{SWAP}$, which makes $GF = B/p$ symmetric positive semidefinite, and the native triangle $(0,1,5)$ contributes at least $3I/60$ to B [15]. □

At every positive-weight prefix each observer's decoder, which sees only its local initial pair and incident increments, returns fields that obey native continuity, and the two port-zero values agree. The transformed finite-cap maps preserve the transported cone $G^{-1}\mathbb{R}^{66}_{+}$; they are not entrywise-positive stochastic maps, and the comparison is of decoded full states. The quantum extension checks the native remote resolvent on all 4,356 matrix units with coherent preparations, untouched references and complete local clock records, and an ordinary completely positive effective instrument for local outputs is a positive mixture of native repairs; its one-particle coherent experiment retains all 81 outputs [16]. The compiled clock history replaces the ideal preparation, intervention and pointer of that experiment by words of the six-level source compiler. Exact rational propagation of the binary source arrays keeps all 512 code addresses and 1,656 clock, timeout and complement-abort outputs within 10^{-11} of the ideal history, at most 51 pulses (54 with the compiled Bell preparation), two code services per pulse and five native repair attempts [17]. The native instrument, the scheduler and the code services are specified inputs.

4.2 Joint and active event clocks

A local event clock ticks on actual occupation changes at an observer port, or on attempted repairs incident to a region, while remote repairs continue; its stopped instrument sums the unresolved wait and retains finite timeouts and conditional quantum outputs. Two such clocks on ports 0 and 72 of joined carriers read one continuing history with an observer-chosen field algebra. Exact elimination of the hidden factor gives a record-conditioned quantum memory recurrence valid for

arbitrary untouched references. At completed events the experiment admits a four-mark measure-and-prepare description, while unresolved waits retain coherence that the completed instrument erases. C and E share the completed clock law and predict field-plus probabilities 35/54 and 14/27 after three native attempts. Without feedback each two-tick cycle of clock A carries B-count mean two and variance 16; instrument-valued clock conversion preserves that count law, and direct native elimination and elimination through the joint event instrument agree, including B-triggered field feedback [18].

An active event clock lets each native event incident to port 0, including a zero-current one, trigger one fixed compiled modular word $U(1) = \exp(iH)$ generated by the native three-state attempt transfer; the carrier pauses during probe services, which are charged separately. For caps $c \in \{1, 2\}$ the execution retains all 3,661 joint output labels, the first timeout has probability $(5/6)^c$ and both ticks occur with probability $[1 - (5/6)^c]^2$. Closed native histories derive a faithful thermal reference for that generator, and passive repair itself cannot act as a nontrivial modular automorphism on an invariant observer algebra [19]. On the complete movement-bit tape $B_t = |j_t|$, including zero currents, four consecutive covariances $c_n = \mathbb{E}[B_t B_{t+n+1}] - p^2$ invert to the two transfer roots $\lambda_\pm = (25 \pm \sqrt{3})/30$ with $p = 5/33$, and the same roots predict the coherent quadratures $q_X = (1 + \cos(sE_j))/2$ and $q_Y = (1 + \sin(sE_j))/2$, $E_\pm = -\log \lambda_\pm$, with no frequency fit. The one-exponential approximation matching c_0, c_1 misses c_2 by exactly $1/2673000$. The inference does not distinguish C from E, whose classical tapes agree [20].

4.3 Identification of the record law

Nine source-executed scalar measurements identify all nine real coordinates (k, g, h, m, n) of the record-law table of Theorem 2.3, and with them the coarse definite-current instrument on all 78 total-two states with arbitrary spectator and reference coherences. The execution comprises 36 compiled probes and ten complete two-event histories, five with untouched quantum references, run through the canonical observer runtime with serialized checkpoint validation and continuation. An admissible collision law agrees with E on the whole binary shell, and this experiment separates it. A reference-uniform bound carries measurement uncertainty through the finite feedback histories, so narrowing the admissible probability tubes narrows the complete-history bound [21]. The probes are synthetic; they identify an instrument within the classified family and select no law for nature.

4.4 Federations and matched screens

Federations join 20, 80 or 320 identical twelve-port carriers through the declared seam wiring, and native Kraus maps preserve signed-difference operator blocks with individual outcomes and reference correlations. The endpoint instruments fix the local Liouville action $H_{\text{native}} = \sum_e (I - \Phi_e)$, with $h_e = P_1 + \frac{1}{2}P_{1/2}$ for C and $h_e = P_1$ for E on orthogonal projectors; the discrete uniform transfer is $I - H_{\text{native}}/M$ for M seams. For an observed region R, maximum degree Δ and retained radius r, truncating distant ports changes the complete local record, quantum and reference output by at most $\min(1, |R|(\Delta s)^{r+1}/(r+1)!)$ at auxiliary per-seam Poisson parameter s [22]. Actual source neighborhoods are identified at levels 1/3, 2/4 and 3/5. On the 15,360- and 245,760-port screens matched radius-nine initial restrictions give complete root-observer output distance below 0.000382 at $s = 1/4$; after four local incident ticks, with no auxiliary time coordinate, the pair allowance is below 6.656×10^{-9}, and a density-1/3 preparation keeps a certified two-time field covariance above 0.15332581621. Full-screen executions retain the actual seam scheduler, current records, feedback repair costs, timeouts and serialized continuation. Conditional on a conserved total, native connected exchange derives the unique diagonal equilibrium, so the screens at totals

5,120 and 81,920 keep density 1/3, the feedback reversal of the connected four-field statistic and their finite-number correlations without product preparation [23].

A coherent observer runs on the same matched screens. A logical qubit on ports 0 and 1, possibly entangled with an untouched reference, passes through two native incident ticks and a local phase Z_a after first mark 1, while every other port is occupied with probability 1/3. A degree-two operator dual evaluates the common 222-port ball with 561 seams and all 21 capped record labels, and each full-screen completed output lies within 7.779×10^{-10} of it, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [24]. The native continuum operator of the refining screens and its density, wave and response limits are stated in Section 7.

4.5 What the native counts supply

Let C_{xy} count the sixty seam and coin branches taking a binary configuration x to y and put $P = C/60$. Every attempt counts, including zero currents, so this complete-attempt transfer differs from the incident transfer F of Proposition 4.1. Its closed histories give the positive functional $\rho_L = P^L / \operatorname{Tr} P^L$ on the 66-state matrix algebra through a chosen linear extension, and the same path counts give ordered modular correlations with generator $H = -\log P$ and conventional Heisenberg time $\tau = -Ls$. The entries of ρ_L are classical endpoint probabilities divided by the return probability, native C and E evolution converges to $I/66$ without preparing it, and the current-only adapter uses at most 11,398 attempts per experiment; the observers paper proves these statements [6].

Binary repair is exact conditional resampling at constant energy under the declared fair coin. A stopped coin prefix with stored cost $E = n \log 2$ gives a bounded register refresh that inhabits the finite four-law core in internal record-cost units; that core holds for every full-support law with $E = -\log(\text{ref})$ and formalizes no contact relation [6]. The transfer spectrum and compact cores of the same kernel are in the particle paper [116], and regional entropy on the reference $C^2/171720$ is in the Einstein paper [8].

5 Field detectors on supplied actions

Layer four begins with detectors that couple native records to fields whose actions are supplied. They provide complete-output and refinement machinery that a source-derived field description must reuse; they derive no field.

5.1 The scalar detector and its refinement transfer

The scalar detector joins the native observer to the free-scalar continuum family of the theorem library [56, 57]. The scalar action, quantization, vacuum, preparation, Born readout and model clock are supplied. The field is the massive scalar on the golden tensor grids $q \in \{13, 34, 89, 233\}$, with Dirichlet boundary and mass one; after mass whitening its one-particle operator is $\mathsf{A} = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B$, with $\Omega = \sqrt{\mathsf{A}}$ and $H = d\Gamma(\Omega)$. A preparation profile f and a detector profile g lie in compact left and right slabs.

One trial runs as follows. (1) Prepare one of the 66 binary configurations uniformly, repair $(0, 1)$ and retain (c_1, j). (2) Prepare the field vacuum and apply the momentum displacement of amplitude $\gamma j f$, $\gamma = 4$. (3) Evolve to model time t and read the centered binary Weyl instrument

$$K_y = \tfrac{1}{2}\left[e^{-iX/2} - iy\, e^{iX/2}\right], \qquad X = a^\dagger(v) + a(v), \qquad v = \Omega^{-1/2} g/\sqrt{2},$$

with $K_y^\dagger K_y = \frac{1}{2}[I + y \sin X]$. (4) Commit y, repair $(0,5)$ for $y = +1$ and $(0,1)$ for $y = -1$, and halt. Model times lie in $[19/20, 1]$, with primary setting $39/40$, and carry no laboratory calibration.

Proposition 5.1 (Scalar feedback and refinement transfer)**.** *For the process above at model time t, let $r(t) = \Pr(y = +1 \mid j = 1) - \frac{1}{2}$. Then:*

(i) the branch operators $M_{(r,y,s)} = (L_s L_r) \otimes (K_y e^{-iHt} V_j)$, with V_j the controlled displacement, form one CPTP instrument on arbitrary joint source and field inputs, and each branch retains its conditional field state as a superposition of two coherent components with their relative phase;

(ii) $p(y \mid j) = \frac{1}{2} + yj\, r(t)$ with $r(t) = \frac{1}{2} e^{-\|v\|^2/2} \sin \mu(t)$ and $\mu(t) = \gamma \langle g, \Omega^{-1} \sin(\Omega t) f \rangle$;

(iii) over the 528 histories, $\mathbb{E}[j_2] = 5r/132$, $\mathbb{E}[N_{0,\mathrm{final}}] = \frac{1}{6} - 5r/132$, $\mathbb{E}[yj] = 10r/33$ and $\Pr(y = \pm 1) = \frac{1}{2}$; disconnecting the preparation or replacing y by an independent fair bit gives $\mathbb{E}[j_2] = 0$ with the same seam frequencies, and reversing the kick reverses the sign;

(iv) if $|r_q - r_\infty| \le \varepsilon_q$ on the window, then the complete labelled histories satisfy the total-variation distance $\mathrm{TV} = \frac{5}{33}|r_q - r_\infty| \le 5\varepsilon_q/33$, and $|\Delta \mathbb{E}[j_2]| \le 5\varepsilon_q/132$.

Proof sketch. The Weyl operators $e^{\pm iX/2}$ are displacements by $\pm iv/2$, which gives the branch states and (ii). Conditioned on the initial configuration and first coin, the two detector laws differ in total variation by $|j|\,|r_q - r_\infty|$, and the subsequent native update is common [57]. □

At $q = 233$ the parent certifies, uniformly for $19/20 \le t \le 1$,

$$\varepsilon \le \frac{12513633763}{250000000000}, \qquad -\frac{111322580247}{500000000000} \le r \le -\frac{34438518047}{250000000000},$$

so the finite-grid mean second current satisfies

$$-\frac{37107526749}{4400000000000} \le \mathbb{E}[j_2] \le -\frac{34438518047}{6600000000000},$$

negative throughout the window, with the continuum mean within $\frac{5}{132}\varepsilon \approx 0.001896$ [93, 58]. At $t = 39/40$ the plus probabilities for $j = +1$ are about 0.334515, 0.319421, 0.317093 and 0.316771 on $q = 13, 34, 89, 233$, and the finest-grid prediction is $\mathbb{E}[j_2] \approx -0.00694050$. A seeded execution with 200,000 fresh trials per grid tests Born outcomes against predictions computed before sampling, with reversal, donor and fixed-seam controls; at $q = 233$ all 12,487,168 field modes contribute. An independent diagonalization gives largest relative frequency errors 2.137%, 0.2875% and 0.04116% at $q = 13$, 34 and 89 against the supplied dispersion $\omega^2 = 1 + \pi^2(n_x^2 + n_y^2 + n_z^2)$ [85]. A local kick-drift-kick extension through the six-neighbour stencil runs at $q = 13$ and 34 and transfers the combined spatial and temporal bound through the same factors 5/33 and 5/132; at $q = 233$ the temporal term is at most 0.000008071512 and the combined probability error at most 0.050062606564 [56].

5.2 The retained two-read continuation

The continued observer keeps the disturbed field after its first read [59]. The field is read at model times 19/20 and 1; the first bit y_1 selects the second native seam, the conditional field evolves to time 1, the second bit y_2 selects the third seam, and the observer halts with five records readable. This continuation uses exact continuous spectral evolution and carries its own error bounds.

Proposition 5.2 (Two reads of one field)**.** *In the continued protocol:*

(i) all 2,112 labelled histories are retained; the record-prefix counts at the six phases are $1, 4, 8, 24, 48, 124$*, so the 209 reached prefixes have an eight-bit encoding and the 124 terminal words need seven bits;*

(ii) with the Gram kernel G_{ab} *of the two detector vectors,* $V = G_{11} = G_{22}$*,* $C = \operatorname{Re} G_{12}$*,* $c = 2 \operatorname{Im} G_{12}$ *(so that* $[X_1, X_2] = ic$*),* $S_a(j) = e^{-V/2} \sin(j\mu_a)$*,* $D = \cos(c/2)$ *and* $R_j = \frac{1}{2}[e^{-(V-C)} \cos(j(\mu_1 - \mu_2)) - e^{-(V+C)} \cos(j(\mu_1 + \mu_2))]$*, the joint law is*

$$p_j(y_1, y_2) = \tfrac{1}{4}[1 + y_1 S_1(j) + y_2 D\, S_2(j) + y_1 y_2 R_j];$$

(iii) on the positive-current preparation, performing the first read changes the second plus probability by $(-r(t_2))[1 - \cos(c/2)]$*, certified greater than* 0.000022281162 *at* $q = 233$ *and greater than* 0.0000252366 *in the declared compact continuum; a commuting Gaussian control with the same means and real covariance has* $D = 1$ *and no such disturbance; the zero-current vacuum correlation exceeds* 0.045619388905 *at* $q = 233$*;*

(iv) the total variation distance of the complete labelled history from the declared continuum is at most 1*,* 0.451806*,* 0.052189606637 *and* 0.025709913643 *on* $q = 13, 34, 89, 233$*, including the zero-current histories.*

Proof sketch. A state is stored as $\sum_n c_n D(i \sum_a h_{na} v_a)|\alpha\rangle$ with exact half-integer shifts, its Gram matrix is explicit in G, and both terms of each Kraus operator are applied before the branch norm is taken. For four atoms with Walsh-moment differences a, b, d, $\mathrm{TV} = \frac{1}{2} \max(|a|, |b|, |d|, \frac{1}{2}(|a| + |b| + |d|))$, bounded in directed rational arithmetic [59]. □

At $q = 233$ the execution gives $\mathbb{E}[j_3] \approx -0.00130126$ and $\mathbb{E}[y_1 y_2 \mid j = 0] \approx 0.0514838$; deleting the first field record from the forecast raises its Brier risk by about 0.00064677. The 50,000-trial runs per grid do not resolve the small disturbance statistically.

5.3 The source-compiled two-site apparatus

The two-site apparatus executes preparation, evolution and pointer as emitted twelve-port control words [60, 61]. Its action lies outside the golden refinement family, so its occupation limit carries no spatial refinement. The Hamiltonian is

$$H = \tfrac{1}{2}(P_0^2 + P_1^2 + Q_0^2 + Q_1^2) + \tfrac{\kappa}{2}(Q_0 - Q_1)^2, \qquad \kappa = \tfrac{9}{32},$$

with normal-mode frequencies 1 and $5/4$ and $d \in \{4, 8, 16\}$ number states per site. The executed Hamiltonian uses the compressed squares $Q_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$ and $P_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$, so that $H_d = \Pi_d H \Pi_d$. The prepared state is the normalized projection of the coupled vacuum onto total occupation at most $d-2$; at $d = 16$ its binary amplitude tree costs 5,379 native pulses. The current impulse is $e^{ijQ_{0,d}}$, 64 or 128 symmetric split steps reach $t = 1$, and the pointer coupled by $e^{-iZQ_{1,d}/2}$ and read in the Y basis gives $\Pr(y = +1 \mid j) = \frac{1}{2} + j r_d$ and $\mathbb{E}[j_2] = 5r_d/132$.

Theorem 5.3 (Complete ideal error budget of the two-site instrument)**.** *For the declared two-site action, model time* $0 \le t \le 1$ *and native current* $j \in \{-1, 0, 1\}$*:*

(i) with A *the onsite and* B *the bond part of* H_d *and* $S(h) = e^{-ihA/2} e^{-ihB} e^{-ihA/2}$*,*

$$\|S(t/n)^n - e^{-it(A+B)}\| \le \frac{t^3}{n^2}\Big(\frac{\|[A,[A,B]]\|}{24} + \frac{\|[B,[A,B]]\|}{12}\Big),$$

uniformly over inputs and references;

(ii) the prepared state differs from the infinite coupled vacuum in trace distance by less than 0.007576, 0.000079899 *and* 1.0324×10^{-8} *at* $d = 4, 8, 16$*;*

(iii) the trace distance between the complete ideal finite instrument, including its classical outcome and unnormalized terminal field branches, and the infinite two-oscillator instrument is below 0.799447, 0.077171 *and* 0.000196 *at* $d = 4, 8, 16$*, and the bound tends to zero as* d *grows;*

(iv) at $d = 16$ *and* $n = 128$ *the combined occupation and splitting error is below* 0.001464436*, and the ideal source-word instrument satisfies* $0.514957066204 \leq \Pr(y = +1 \mid j = 1) \leq 0.517885937293$*, an interval that excludes* $1/2$*.*

Proof sketch. Item (i) is a defect bound for the symmetric product formula [114] with nested commutators bounded by exact rational Frobenius norms. The infinite state is a displaced squeezed vacuum throughout; the Gaussian generating function bounds the number tails along impulse, evolution and both pointer arms, and projected Duhamel comparison with the orthogonal leakage gives (iii) without dividing by a Born probability. Item (iv) adds a rational enclosure of the infinite reference probability [61, 94]. □

The executed plus probabilities for a positive unit current, with their rational splitting bounds, are 0.516680958 (0.000103957) and 0.516679563 (0.000025990) at $d = 4$ with 64 and 128 steps, 0.516424174 (0.000693013) and 0.516422803 (0.000173254) at $d = 8$, and 0.516423330 (0.005076875) and 0.516421959 (0.001269219) at $d = 16$. The infinite reference has detector variance $9/20$ and mean $\mu(t) = \frac{1}{2}(\sin t - \frac{4}{5}\sin(5t/4))$, so $\Pr(y = +1 \mid j) = \frac{1}{2}(1 + e^{-9/40}\sin(j\mu(t)))$, equal to 0.516421501748 at $t = 1$ and $j = 1$ [60]. The $d = 16$, 128-step protocol uses nine work qubits, 18,612,125 native pulses and 37,224,250 proper-code load and export events, and retained 200,000-trial runs record the reversal, disconnected and donor controls.

6 The common interacting source

6.1 Construction

The common source places quantum electric flux, a charged fermion pair and a neutral scalar on one evolving work register. The sharp protocol measures it twice and feeds its records into the register and the declared native integer primitive [68, 69]; its four-read smooth extension joins compiled native quantum channels and encoded recovery (Section 6.5). The action, preparation, coupling and model-coordinate schedule are construction inputs.

Order four fermion modes as L_0, R_0, L_1, R_1. The two channels carry charges $q_0 = +1$ and $q_1 = -1$ and one fermion each; $b_j = 0$ means left and $b_j = 1$ right occupation of channel j. The electric link has basis $|\ell\rangle$, $\ell \in \mathbb{Z}$, with $\mathcal{E}|\ell\rangle = \ell|\ell\rangle$ and $U|\ell\rangle = |\ell + 1\rangle$. The isometry

$$W|b_0, b_1\rangle = |\text{modes } b_0 \text{ and } 2 + b_1 \text{ occupied}\rangle \otimes |-b_0 + b_1\rangle$$

maps onto four orthonormal occupation and flux vectors.

Proposition 6.1 (Flux-dressed matter code)**.** *On the image of* W*:*

(i) both operator Gauss constraints hold, $G_L W = G_R W = 0$ *with* $G_L = \mathcal{E} - Q_L$ *and* $G_R = -\mathcal{E} - Q_R$*, the supported flux set is* $\{-1, 0, 1\}$ *inside the infinite link, and fermion parity is even;*

(ii) the dressed hopping terms $h_j = c_{Lj}^\dagger U^{q_j} c_{Rj} + c_{Rj}^\dagger U^{-q_j} c_{Lj}$ *preserve both ambient Gauss values and the code, with* $W^*(h_0 + h_1)W = X_0 + X_1 = H_{\rm hop}$ *and* $W^*\mathcal{E}^2 W = \frac{1}{2}(I - Z_0 Z_1)$*, and* $\|[H_{\rm hop}, \mathcal{E}^2]\| = 2$*;*

(iii) with $\theta = \operatorname{atan2}(4,3)$*,* $u = (3I - 4iX)/5$*,* $U_\theta = u \otimes u = e^{-i\theta H_{\rm hop}}$ *and* $V_\theta = e^{-i\theta\mathcal{E}^2} = \operatorname{diag}(1, z, z, 1)$*,* $z = (3 - 4i)/5$*, the echo* $U_\theta^* V_\theta U_\theta$ *applied to* $|00\rangle$ *gives right occupation of channel* 0 *with probability exactly* 576/3125*, and without* V_θ *the probability is zero.*

Proof sketch. On every image column $Q_L = -b_0 + b_1 = \mathcal{E}$ and $Q_R = -\mathcal{E}$. The four-mode algebra of canonical anticommutation relations (CAR) is checked on all 16 occupation states with independent Jordan-Wigner signs, and all entries of the echo unitaries are Gaussian rationals [69]. □

The echo runs on the source controls with 22 native pulses and 44 coherent service events. Two neutral scalar sites join the same register, and with onsite quadratics compressed from the infinite oscillator the declared dimensionless action is

$$\begin{aligned} H &= H_{\rm hop} + \mathcal{E}^2 + H_{\rm scalar} + H_{\rm int}, \\ H_{\rm scalar} &= \sum_{j=0,1} \tfrac{1}{2}\left[P_j^2 + (1+\kappa)Q_j^2\right] - \kappa Q_0 Q_1, \qquad H_{\rm int} = g\sqrt{2}\, Q_0 (n_{L0} - n_{R0}), \end{aligned} \tag{3}$$

with $\kappa = 9/32$ and $g = 1/4$. The interaction commutes with both Gauss generators and with fermion parity and couples the sectors, $[H_{\rm int}, H_{\rm hop}] = 2ig\sqrt{2}\, Q_0 Y_0$. The experiment proceeds in four steps. (1) Native repair on $(0,1)$ records a current j, which controls the gauge pulse $V_j = e^{-ij\theta X_0}$. (2) The register evolves for interval 1 (U_1), and a source pointer commits the right occupation a of the first charged species (P_a). (3) Only a selects the hopping pulse $F_1 = (3I - 4iX_0)/5$, with $F_0 = I$, and the register evolves for interval 1/2 (U_2). (4) A second pointer commits the sign b of Q_0 (R_b), and only b selects the second native seam, $(0,5)$ for $b = 1$ and $(0,1)$ otherwise.

Proposition 6.2 (One joint instrument with retained histories)**.** *The operators* $K_{(a,b)} = R_b U_2 F_a P_a U_1$ *satisfy* $\sum_{a,b} K_{(a,b)}^\dagger K_{(a,b)} = I$*, and* $M_{(r,a,b,s)} = (L_s^{e(b)} L_r^{(0,1)}) \otimes (K_{(a,b)} V_j)$ *form one CPTP instrument on arbitrary joint native and quantum inputs. Under the uniform binary preparation and the declared quantum vector* ψ*, the* $66 \cdot 2 \cdot 4 \cdot 2 = 1{,}056$ *histories have probabilities* $p(a,b \mid j)/264$ *with* $p(a,b \mid j) = \|K_{(a,b)} V_j \psi\|^2$*, each bound to its terminal density* $|v_{j,a,b}\rangle\langle v_{j,a,b}|/264$*, and the first native record keeps the masses* 14/33*,* 14/33*,* 5/66*,* 5/66*. For a complete record-controlled instrument* $B_y = \sum_r |r\rangle\langle r| \otimes K_{(r,y)}$ *with channel* $\Lambda(X) = \sum_y B_y X B_y^\dagger$*, every old record projector* Q_r *satisfies* $\Lambda^*(Q_r) = Q_r$*, and the block readback* $E_{\rm block}(X) = \sum_r Q_r X Q_r$ *and the public readback* $E_{\rm public}(X) = \sum_r \operatorname{Tr}(Q_r X Q_r) Q_r / \dim W$ *are distinct conditional expectations.*

Proof sketch. Summing $M^\dagger M$ over s, then (a,b), then r gives the identity. Each B_y commutes with Q_r, so $\Lambda^*(Q_r) = Q_r \sum_y B_y^\dagger B_y = Q_r$; every reported probability is checked against the norm of the retained unnormalized vector [69, 95]. □

At 512 steps per interval the emitted words use 4 working qubits, 176,182 native pulses and 352,364 coherent load and export events at $d = 2$, and 6 qubits, 1,879,628 pulses and 3,759,256 events at $d = 4$. Disconnected, feedback-disabled, fixed-seam and independent-bit controls isolate the causal effect of the records [68].

6.2 The feedback contrast at finite and infinite occupation

Theorem 6.3 (Finite feedback contrast)**.** *For the common source with intervals* 1 *and* 1/2*:*

(i) every palindromic product $F(h)$ of the component exponentials whose generators sum to H satisfies $\|F(t/n)^n - e^{-itH}\| \le S^3|t|^3/(3n^2)$ with $S \le 3 + (1+2\kappa)(2d-1) + |g|\sqrt{4d-2}$, on all inputs and references; at $n = 512$ and $t = 1$ the half-diamond bounds are below 0.000728 for $d = 2$ and 0.004184 for $d = 4$;

(ii) at $d = 2$ and zero native preparation current, exact rational Taylor arithmetic of degree 64 with $\|H\| \le 109/16$ encloses the unconditional probability of a positive scalar coordinate and the feedback contrast for the continuous finite Hamiltonian:

$$\begin{aligned} &\textit{feedback enabled:} && [0.370697726608684,\ 0.370697726608685],\\ &\textit{feedback disabled:} && [0.402359206973491,\ 0.402359206973492],\\ &\textit{contrast:} && [-0.031661480364808,\ -0.031661480364807]; \end{aligned}$$

(iii) for the ideal source word at 512 steps per interval the contrast lies in

$$[-0.033297300667839,\ -0.030025660061775],$$

which excludes zero.

Proof sketch. The multinomial Leibniz expansion bounds the third derivative of a product of unitaries by S^3, palindromic symmetry gives second-order agreement, and unitary telescoping gives (i). For (ii) the integral Taylor remainder gives $\|T_m(t) - e^{-itH}\| \le (St)^{m+1}/(m+1)!$, and at $m = 64$ the probability radius is below 7.091×10^{-37}; measurements and feedback contract trace distance, which gives (iii) [69]. □

With $g = 0$ the gauge-only feedback cannot change the scalar marginal, and a coarse splitting or inadequate Taylor degree yields an interval containing zero, which the evaluator reports as no resolved signal. The finite pointer measures the sign projector of the truncated coordinate, which differs from the infinite half-line effect; at $d = 2$ the two differ in norm by $\frac{1}{2} - 1/\sqrt{2\pi} \approx 0.10105772$ [73], so the infinite-occupation statement needs the infinite detector.

Theorem 6.4 (Infinite-occupation feedback contrast)**.** *Take the action (3) on two infinite oscillator Fock spaces, the infinite free coupled vacuum, proportional to $\exp[-(a_0^\dagger - a_1^\dagger)^2/36]|0,0\rangle$ with squared normalization $4\sqrt{5}/9$, the blank gauge code, zero native preparation current, the matter record and feedback of Section 6.1, and the half-line readout of Q_0. Then:*

(i) the unconditional positive-coordinate probability with feedback minus that without lies in $[-0.024045670143,\ -0.024043450955]$, and without the interaction the contrast is exactly zero;

(ii) both matter records and both half-line outcomes retain their quantum states, including the post-read infinite occupation tail;

(iii) along $d = 2^k$, with exact palindromic factors and $n = d^2$ steps per interval, the complete finite sign instruments converge in trace distance to this infinite instrument, terminal states included, and the finite-family contrast is strictly negative for all sufficiently large d.

Proof sketch. The linear force is relatively bounded by the coupled oscillator Hamiltonian, so H is self-adjoint and finite Hermite polynomials lie in every $\mathcal{D}(H^k)$. Each of 24 steps of length 1/16 applies a degree-16 Taylor polynomial of H, charges its remainder exactly, truncates above total occupation 16 and rounds to a 2^{-112} grid, charging both differences. Half-line probabilities are $r + s/\sqrt{2\pi}$ with exact rationals, and π is enclosed by Machin's formula. For (iii), $P_d Q_0 P_d \to Q_0$ in the strong resolvent sense and the spectral measure of Q_0 has no atom at zero [70, 115, 112]. □

The $d = 2$ finite experiment has contrast about -0.03166148; replacing only its detector by the half-line effect gives about -0.02526221, and removing the cutoff gives the interval of Theorem 6.4. The signal survives both limits, which change its size; the interval is specific to zero native preparation current and the declared two-site action [70].

6.3 Occupation refinement and complete-output comparisons

Let P_d project both oscillators onto occupations below d and let $N = n_0 + n_1$ be the total oscillator occupation, a different object from the closure input N.

Theorem 6.5 (Uniform pre-readout occupation bound)**.** *Let the matter start in an arbitrary state entangled with an untouched reference, let the scalar start in the fixed coupled vacuum, and let the finite preparation retain total occupations* $0, 2, \dots, d-2$*. Then:*

(i) $M(s,t) = \langle e^{sN} \rangle$ *satisfies* $\partial_t M \le \frac{11}{16} \sinh(s)\, \partial_s M + \frac{25}{32} \sinh(s)\, M$ *through the matter measurement and feedback;*

(ii) at total model time $T = 3/2$ *the resulting moment bound holds with base* $\varrho = 113/64 \ge 7/4$ *and* $M < 13$*, where* $\varrho \le e^{s(T)}$*;*

(iii) $\|P_d H (1 - P_d)\| \le C_d = \frac{\kappa}{2}\sqrt{d(d+1)} + \frac{3\kappa}{2} d + \frac{1}{4}\sqrt{d}$*, and the retained-history trace distance up to the final scalar read is at most* $\left[(\sqrt{2 v_d} + T C_d \sqrt{\tau_d})^2 + \tau_d \right]^{1/2}$ *with* $\tau_d = M/\varrho^d$ *and* v_d *the omitted preparation probability, an* $O(d\, \varrho^{-d/2})$ *rate;*

(iv) this bound is below 0.011408*,* 0.000002498 *and* 6.175×10^{-14} *at* $d = 32, 64, 128$*.*

Proof sketch. In common and difference modes only the squeezing term and the force change the number; weighted commutator, Cauchy and Young inequalities that hold for matter-entangled vectors give (i), and integration along $\tanh(s(t)/2) = e^{-11t/16} \tanh(s_0/2)$ from the initial value $\sqrt{80/17}$ at $\varrho_0 = 8$ gives (ii). The record dilation and feedback commute with N, and projected Duhamel comparison gives (iii) [71]. □

Theorem 6.6 (Fourth moment and strong convergence of the sharp instrument)**.** *Let* $w(N) = (N+2)^4$ *and let* $M(t) = \langle w(N) \rangle$ *be summed over the unnormalized record branches. Then:*

(i) $|\dot{M}| \le CM$ *with* $C = 8\kappa + 4|g| = 13/4$*, and every finite preparation has* $M(0) \le M_0 = 143748821/8192000$*;*

(ii) the record-dilation vectors immediately before the final read satisfy

$$\|\Psi_d - \Psi\| \le \left[(e_0 + I_T L_d)^2 + \frac{M_0 e^{CT}}{(d+2)^4} \right]^{1/2} = O(1/d),$$

with $I_T = \sqrt{M_0}\, (e^{CT/2} - 1)/(C/2)$*,* $L_d = \kappa/(d+2) + |g|/(d+2)^{3/2}$ *and* e_0 *the preparation error;*

(iii) the sign projectors of $Q_{0,d}$ *converge strongly to those of* Q_0*, and the complete unnormalized classical and quantum history, together with its native continuation, converges in trace norm for all three native currents and both feedback settings.*

Proof sketch. On occupation at most m, $\|H\psi\| \le 4(m+2)\|\psi\|$ and H raises occupation by at most two, so finite-particle vectors are analytic and core convergence gives strong resolvent convergence [113, 115]. The moment coefficients are bounded by 8κ and, through a positive polynomial certificate, $4|g|$; the gauge measurement and rotations preserve $w(N)$ summed over branches [72]. □

Strong convergence supplies no finite-d rate for the sharp readout. The complete-output certificate supplies fixed-state enclosures at $d \in \{2, 32, 64, 128\}$ for the zero-current preparation and the named rectangular occupation codes [73, 96].

Theorem 6.7 (Complete outputs of the sharp detector)**.** *For zero native preparation current, with both matter records and both sign outcomes retained:*

(i) for two complete post-read outputs with rank-one branches, the half trace norm of their difference is $\frac{1}{2}\sum_b \sqrt{(p_b+q_b)^2 - 4|z_b|^2}$*, summed over earlier records, where* p_b*,* q_b *are the branch weights and* z_b *their overlap;*

(ii) the complete continuous-output distances between the finite and the infinite sharp histories lie in the intervals of Table 3; at $d = 2$ *the ideal 512-step source budget widens them to* $[0.454336, 0.455973]$ *with feedback and* $[0.456824, 0.458461]$ *without;*

(iii) every output state supported in the rectangular occupation code of $d = 256$ *lies at distance above* 0.122 *from the true output of either prepared history;*

(iv) for the vacuum control, whose two sign probabilities are both $1/2$*, the best approximation by a state supported in the code with* $d = 2m$ *has distance between* $\sqrt{3/(8\pi)}\, m^{-1/4}$ *and* $\sqrt{4/(3\pi\sqrt{3})}\, m^{-1/4}$*, hence of order* $d^{-1/4}$*, and distance at most* ε *requires* $d \geq 9/(32\pi^2\varepsilon^4)$*.*

Proof sketch. Item (i) is the trace norm of a difference of rank-one operators. For (iii), the operator exchanging the normalized inside and outside directions of each branch is a norm-one witness, giving distance at least $\sum_{a,b}\sqrt{a_{ab}t_{ab}} - \epsilon_{\mathrm{poly}}$; half-line Hermite matrix elements are rational up to $1/\sqrt{2\pi}$ [104]. For (iv) the omitted vacuum mass per sign branch obeys $1/(4\pi\sqrt{m}) \leq T_{2m} \leq 2/(3\pi\sqrt{3m})$ [73]. □

Table 3: Certified complete continuous-output distances between the finite sharp history and the infinite sharp history, zero native preparation current, rounded outward [68, 96].

Onsite dimension d	Feedback enabled	Feedback disabled
2	[0.455154, 0.455155]	[0.457642, 0.457643]
32	[0.203713, 0.226530]	[0.203935, 0.226752]
64	[0.180917, 0.180924]	[0.181090, 0.181097]
128	[0.152201, 0.152204]	[0.152341, 0.152344]

At $d = 64$ the pre-readout error is below 0.000002498, so the sharp detector dominates the final error: accurate sign probabilities do not certify an accurate post-measurement state.

6.4 The smooth Ramsey instrument

A detector variant keeps the preparation, evolution, first record and feedback and replaces the final sharp read by a source Ramsey pointer with work-space Kraus maps

$$M_0 = e^{-i\pi/4}\cos\left(\tfrac{Q_0}{2} - \tfrac{\pi}{4}\right), \qquad M_1 = e^{i\pi/4}\cos\left(\tfrac{Q_0}{2} + \tfrac{\pi}{4}\right),$$

and effects $\frac{1}{2}(I \pm \sin Q_0)$ [74]. Bit zero denotes the plus effect, a detector meaning different from the sharp sign record, so the sharp-sign certificates do not carry over.

Theorem 6.8 (Smooth instrument)**.** *For the declared initial vacuum family, uniformly over the three native current kicks and both feedback settings, with $T = 3/2$ and the constants of Theorem 6.6:*

(i) the complete recorded instrument, including the pointer interaction and both unnormalized outcomes, lies within trace distance $\min(1, E_d)$ *of its infinite counterpart, where* $A_T = \sqrt{M_0}e^{CT/2}$ *and*

$$E_d = \Big[\Big(e_0 + I_T L_d + \frac{A_T(e-1)}{2(d+2)^{3/2}}\Big)^2 + \frac{M_0 e^{CT+2}}{(d+2)^4}\Big]^{1/2} = O(1/d);$$

(ii) with $n(d) = d^2$ symmetric steps per interval the combined ideal temporal and occupation error is $O(1/d)$;

(iii) on normalized inputs in the form domain of the infinite Hamiltonian the unconditional increment of the full energy is exactly $1/8$ *in the declared dimensionless convention.*

Proof sketch. Along the pointer parameter the generator $\frac{1}{2}Z_{\mathrm{anc}}Q_0$ obeys $dM/ds \le 2M$, which gives the pointer term. For (iii), $f_0(q) = \cos(q/2 - \pi/4)$ and $f_1(q) = \cos(q/2 + \pi/4)$ satisfy $\sum_b f_b^2 = 1$, $\sum_b f_b f_b' = 0$ and $\sum_b (f_b')^2 = \frac{1}{4}$, so the kinetic term gains exactly $\frac{1}{8}\|\psi\|^2$ [74]. □

The bound is about 0.2063 at $d = 64$, 0.04096 at $d = 256$ and 0.008840 at $d = 1024$, and vacuous at the executed $d = 2$ and $d = 4$ [68]; the energy identity belongs to the infinite oscillator and fails at a finite cutoff [74, 85].

6.5 Continued instruments, compiled native repair and encoded memory

One execution engine carries the sharp and smooth protocols and the extension

$$U(1),\ P \to r_0,\ F(r_0),\ U(1/2),\ M \to r_1,\ U(1/2),\ M \to r_2,\ F(r_2),\ U(1/2),\ P \to r_3,$$

with P the complete gauge instrument, M the smooth instrument and F the gauge feedback, model time $5/2$ and all sixteen unnormalized work/reference branches retained, with no intermediate work preparation [75]. Complete Kraus families preserve every old-prefix effect, and imported checkpoints are validated by prefix reexecution; this supplies neither physical quantum serialization nor cloning. Carrying a moment bound A and an error e from stage to stage, with $L_H = 9/[32(d+2)] + 1/[4(d+2)^{3/2}]$ and $L_S = 1/[2(d+2)^{3/2}]$, each unitary interval of length t adds $A(e^{at} - 1)L_H/a$ $(a = 13/8)$, each smooth read $A(\mathrm{e} - 1)L_S$, and the final tail gives $E_d = [e^2 + A^2/(d+2)^4]^{1/2}$; with $n(d) = d^2$ the complete output distance is $O(1/d)$ for the scalar vacuum family and vacuous at the executed $d = 2, 4$; the classical native attachment enumerates 4,224 joint outcomes at each executed cutoff. A separate state-specific certificate retains all sixteen outputs for the blank-gauge and coupled vacuum preparation and each $j \in \{-1, 0, 1\}$ [76]; the rank-one trace-norm identity, complete-channel contractivity and directed rational evaluation at 512 splits per interval give

$$\max_{j\in\{-1,0,1\}} D_{d,j}^{\text{ideal product},\infty} < \begin{cases} 0.264092, & d = 2, \\ 0.053956, & d = 4. \end{cases}$$

This bound transfers through the native and protected-memory attachments for arbitrary admitted native and reference inputs with this independently prepared work state; it is not an arbitrary work-input bound and excludes floating execution and physical service errors.

Coherent native attachment. Let $L_h^{A,e}$ be the compiled native Kraus operator for law $A \in \{\mathsf{C}, \mathsf{E}\}$, including its fair coin, and $K_r^{j(h)}$ the work branch including the first record's current-controlled kick. Source routing implements seam $(0,5)$ by exchanging port labels 1 and 5. With $e(0) = (0,5)$ and $e(1) = (0,1)$,

$$M_{h,r,s} = (L_s^{A,e(r_2)} L_h^{A,(0,1)}) \otimes K_r^{j(h)}, \qquad \sum_{h,r,s} M_{h,r,s}^\dagger M_{h,r,s} = I_{78} \otimes I_W,$$

complete on arbitrary entangled native, work and reference inputs [77]. The execution keeps all 512 native amplitudes, padding and abort outcomes, with 295,106 terminal basis outcomes for C and 820,166 for E. For input $(|\{2,3\}\rangle + e^{i\theta}|\{1,2\}\rangle)/\sqrt{2}$ and independent work, the native interference effect gives $p_C(\theta) = B + \frac{1+q}{8}\cos\theta$ and $p_E(\theta) = B$, with $q = \Pr(r_2 = 0 \mid j = 0)$; the C phase contrast is about 0.349153278 at $d = 2$ and 0.351351028 at $d = 4$, against zero for E. Cyclic Gray ordering of the shared nine-qubit controlled-flip macro reduces its source pulse count from 11,782 to 2,044 while preserving the full 512-dimensional operator, including global phase, inverse and padding [122, 77, 78].

Active encoded memory. The code $V|0\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $V|1\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ has the flagged Knill–Laflamme identity $V^\dagger L_h^\dagger L_h V = p_h I_2$, $p_h = k_h/4$, $k_h \in \{0,1,2\}$. A source encoder and record-controlled full-register unitary R_h realize $R_h L_h V = \sqrt{p_h} V$ [78]. Correcting after each native event, with final seam $(2,3)$ for $r_2 = 0$ and $(0,1)$ otherwise,

$$M_{h,r,s}^{\text{corrected}}(V \otimes I_W) = \sqrt{p_h p_s^{e(r_2)}}(V \otimes K_r^{j(h)}),$$

which preserves the logical and reference marginal after summing all work and record outputs, including initially entangled inputs. The passive control without recovery has complete plus/minus output distance $1 - \bar{q}$, with $q_j = \Pr(r_2 = 0 \mid j)$ and $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, so every later decoder errs by at least $\bar{q}/2$ on one input, while the corrected distance is one; for fixed seams $(0,1), (2,3)$ passive Bell fidelity is $1/2$ and corrected fidelity one. Encoding, blank registers, source precision, storage and record services are supplied. This is active protection of the stated qubit before readout, not arbitrary-carrier recovery, fault tolerance or an autonomous observer.

6.6 Observer readouts and the fixed-reference clock

The readouts below extract observer representations, modular flow and regional structure from the same executed history [79]. The quantum algebra, state pairing and access choices are premises, and the Born readout of these instruments is a declared service (Section 2.5).

Proposition 6.9 (Observer representations of the executed history)**.** *Let v_{ab} be the four unnormalized terminal vectors of the history at $d \in \{2,4\}$ and $\rho_{\text{hist}} = \bigoplus_{a,b} |v_{ab}\rangle\langle v_{ab}|$ on $\bigoplus_{a,b} M_{4d^2}$.*

(i) For a positive functional ω the Gelfand-Naimark-Segal (GNS) construction $\langle [A], [B] \rangle = \omega(A^*B)$, $\pi(X)[A] = [XA]$, $\Omega = [I]$ has dimension dr on a block M_d whose density has rank r. The record and work algebra has GNS dimension $4(4d^2)$, that is 64 and 256; the four committed record projectors give a four-dimensional classical GNS space; the site-0 observer that forgets both records has GNS dimension d^2, that is 4 and 16; and the regional maps $J_R[A] = [\text{embed}_R(A)]$ are isometric and intertwine the regional actions.

(ii) For faithful ρ_0 the modular group $\sigma_s(A) = \rho_0^{is} A \rho_0^{-is}$ moves number observables; the commutative record algebra has identity flow; states conditioned on the sharp terminal bit are nonfaithful on the full scalar algebra, with rank at most $d/2$.

(iii) *At $d = 2$, $\rho_0 = \frac{1}{2}(I - \delta X)$ with $\delta = P(b = 0) - P(b = 1)$ and modular frequency $\nu = \log(P(b = 0)/P(b = 1))$. A source-compiled continuation of every terminal branch by $Z(\pi/4)$, $X(s\nu/2)$ and a fresh number pointer preserves the old two-bit record marginal and has readout law $P(1 \mid s) = \frac{1}{2}(m + \delta \sin(\nu s))$, where m is the total branch mass, equal to one for the normalized history.*

(iv) *Scalar regional restrictions are completely positive conditional expectations, normalized partial trace followed by identity embedding; they are unital and idempotent and in general not multiplicative. The centered components of the actual Hamiltonian give an interaction graph that is a path through scalar site 1, scalar site 0 and the shared charged-matter and link sector.*

Proof sketch. For $\rho = V \operatorname{diag}(\lambda) V^*$ of rank r the map $[A] \mapsto AV\sqrt{\operatorname{diag}(\lambda)}$ is the GNS isometry [99]; the modular identities, including the finite Kubo-Martin-Schwinger (KMS) condition, are checked with this sign convention [100], and at $d = 2$ the probe implements the Schrödinger dual of σ_s [79]. □

In the executed history $\nu \approx 0.5292248981159262$ and $\delta \approx 0.2586046272446501$ [97]; ensemble frequencies read a modular interval on a declared monotone window, and full-cycle aliases are indistinguishable. The post-measurement site-0 state taken as its own modular reference fails to reproduce the interacting dynamics: its modular flow preserves it while its density changes under H with derivative norms about 0.2086 at $d = 2$ and 0.6603 at $d = 4$ [85]. A fixed reference on the full work algebra removes this obstruction [80].

Proposition 6.10 (Fixed-reference modular relation)**.** *Let H be the finite interacting Hamiltonian on the work space of dimension $4d^2$, $\beta > 0$ and $\sigma_\beta = e^{-\beta H} / \operatorname{Tr} e^{-\beta H}$. Then*

$$\sigma_\beta^{is} A \, \sigma_\beta^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \alpha_{-\beta s}(A), \qquad \alpha_t(A) = e^{itH} A e^{-itH},$$

so continuous Hamiltonian segments correspond to modular flow at $t = -\beta s$. For the full matrix algebra this identification of the two automorphism groups forces the Gibbs form of the reference. With record blocks $\Sigma_\beta = \bigoplus_r q_r \sigma_\beta$, $q_r > 0$, the weights cancel and every record projector is fixed. The actual state $\omega(t) = e^{-itH} \omega e^{itH}$ can change while σ_β is stationary, and $\operatorname{Tr}(\omega(t)E) = \operatorname{Tr}(\omega \, \sigma_{-t/\beta}(E))$ for every effect E.

Proof sketch. $\log \sigma_\beta = -\beta H - \log Z_\beta \, I$, and equality of the generators requires $[\log \sigma + \beta H, A] = 0$ for every matrix A, whose commutant is scalar [80, 103]. □

The reference, β and full-work access are declared, and measurements and feedback act outside this one-parameter group. The receipt uses $\beta = 1$ and reference weights $(1/10, 2/10, 3/10, 4/10)$, with source continuations at model times 0, 1/2 and 1 appending a gauge-number pointer to every post-sharp branch and retaining all eight outcomes [80].

6.7 History repair of the completed record

Theorem 6.11 (Local repair of the completed history)**.** *Fix $d \in \{2, 4\}$, the native current j and the feedback setting. Let twelve event nodes carry unnormalized work vectors subject to the compatibility equations $x_{\text{root}} = \text{prepared}(j)$, $x_1 = U_1 x_{\text{root}}$, $x_{(a)} = P_a x_1$, $y_{(a)} = F_a x_{(a)}$, $z_{(a)} = U_2 y_{(a)}$ and $w_{(a,b)} = B_b z_{(a)}$, of depths 0 to 5. Then:*

(i) *exactly one assignment satisfies all equations;*

(ii) *the common fixed set of the local updates $x_v \leftarrow A_v x_{\text{parent}(v)}$ equals that compatibility set;*

(iii) *every node-fair schedule reaches the compatible assignment from arbitrary initial vectors after finitely many updates and stays there; five covering rounds suffice, a forward topological pass needs eleven updates, and a reverse order may need 55;*

(iv) *with subtree weights w_v, equal to* 11, 5, 4, 3 *and* 1 *at depths one to five, the potential $\Phi = \sum_{v \neq \mathrm{root}} w_v \|x_v - A_v x_{\mathrm{parent}(v)}\|$ decreases by at least the repaired residual at every active ideal repair;*

(v) *every deterministic extraction from the completed assignment, including branch probabilities, terminal densities, GNS representations and modular parameters, is independent of the schedule.*

Proof sketch. Agreement at depth zero forces agreement at each further depth. A node whose ancestors are correct becomes correct permanently at its next selection, and after the kth covering round every node of depth at most k is correct. Updating v zeroes its residual and changes each child's residual by at most $\|r_v\|$, since all maps are contractions [81]. □

The solved object is the complete branching instrument law for given apparatus inputs; its uniqueness selects no realized record pair, and the auxiliary solver is a host computation distinct from a native hardware process.

6.8 A same-action particle corner and regional interfaces

A finite attachment uses the supplied Weyl/contact CAR action $H = d\Gamma[-(1/8)\sigma\cdot\sin k] + \sum_x n_{x0} n_{x1}$ on its complete two-fermion sector. An isolated bound-pair band has an independently checked position-space wavepacket, number and energy currents, and complete contact/scattering spectral records. Its topology is trivial; it supplies neither a nontrivial native defect invariant nor a physical mass assignment. Its exact rational dispersion enclosure gives a kinetic coordinate mass different from the value inferred from its energy difference and the same speed parameter. All of these tests use this one action.

At total momentum (π, π, π), an on-site singlet d and a zero-contact singlet r have energies one and zero, respectively. They have the same particle number and total momentum. A supplied isometry maps the native configurations $\{1, 2\}$ and $\{7, 2\}$ to (r, d). The native product reference restricts to $(264/7505)\,\mathrm{diag}(25/33, 8/33)$, furnishing an exactly faithful Gibbs reference on this invariant M_2 corner, with $\beta = \log(25/8)$ and Heisenberg time $t = -\beta s$.

The executed native protocol retains an initial coarse corner herald, a balanced probe, every uniformly chosen C/E seam outcome, a second herald, the record-derived clock, an analyzer and the final current/coin energy label. Rejection, leakage, zero slots and export aborts remain outputs. Each coarse herald is a 2,068-pulse source word checked on all 512 original register inputs. Preparation error is charged before the first herald, and the complete average half-trace bound is below 0.018762 without success conditioning. The selected two levels do not prepare or encode the full spatial field, and this corner supplies no clock units, many-site isometry or full-field native reference.

The same local field action also has signed CAR restrictions retaining all regional number sectors zero, one and two. Their Kraus maps prove complete positivity, trace preservation, untouched-reference compatibility and composition. Number-preserving local observables have true algebra inclusions; compressed odd operators do not inherit a CAR homomorphism. The corresponding ambient conditional expectations satisfy tower laws but are not multiplicative. This instantiates the CP restriction interface; a scalar-bottom homomorphic retraction is impossible on this algebra.

Increasing open boxes admit exterior-square inclusions J satisfying $J^* H_{\mathrm{large}} J = H_{\mathrm{small}}$. Boundary hopping makes $H_{\mathrm{large}} J - J H_{\mathrm{small}} \neq 0$, so this equality alone does not intertwine dynamics. The complete two-read feedback histories instead use the existing finite-boundary and product-formula bounds, followed by contractive regional restriction. Every quantum record and discarded local particle sector remains in that comparison. The growing boxes hold spacing fixed and do not reconstruct Lorentzian geometry.

On the same full finite action, local energy continuity and quantum Gibbs/entropy identities supply further finite premises. The regional modular generator fails the proposed affine identification with bare regional energy even after number and number-squared terms are allowed. The full interacting Weyl action also couples every proper momentum pocket to its complement. Consequently neither a regional clock nor a chiral sector follows by merely deleting terms or modes. The source-bound construction is recorded with the executable model [10].

The separate continuing native memory. The endpoint-biased 60-code process also continues through every forbidden antipodal flag by a supplied coherent decode/re-encode service. It preserves the unknown logical state and keeps boundary, zero and abort outcomes. Positive endpoint coins leave the exact flagged-code conditions unchanged. Every all-seam correctable sharp-$q = 1$ source qubit, without persistent ancillary storage of its logical state, has both core occupation compressions equal to $I_2/2$, so its core kernel is independent of the history-selected code. For this entire class, the exact absorption potential $h(c) = 1 + \sum_d Q(c,d)h(d)$ has a unique maximum $H \simeq 203.025378$ attempted repairs at core $(0,1)$. Every counted reset raises h by at most H, so telescoping gives $\mathbb{E}N_n \geq \max\{0, (n - h(c_0))/H\}$. Resetting only at the boundary to $(0,1)$ attains asymptotic rate $1/H$. Exact Bellman checks through horizon 64 are recorded separately from this all-horizon asymptotic argument. The compiled family attains the bound; arbitrary exact code choices cannot improve it under the same law and reset contract. Active recovery is supplied after every repair attempt; neither reset counts nor source-pulse counts are physical energy. The complementary fair-coin mobile-memory construction has a different law and a mean edge excursion of 150 attempts. Its finite-horizon theorem is separate from the biased asymptotic result. Neither controlled memory is the interacting field pair or a native topological particle. The complete service and its bounds are retained with the executable model [10]. A separately certified variant omits 3,120 identity recoveries among 11,580 positive flags, saving about 48,391 supplied pulses per stationary native attempt. It preserves the admitted logical/reference channel and does not change the reset rate.

Smaller exact controls and a separate clock. At side four and zero total momentum, the contact-generated cyclic subspace has seven dimensions. Its free spectral weights are $(1,3,3,2,3,3,1)/16$ at energies $2c(-\sqrt{3}, -\sqrt{2}, -1, 0, 1, \sqrt{2}, \sqrt{3})$. The resulting rank-one resolvent gives exact rational pole enclosures and intertwines complete contact/complement histories with arbitrary references. Other spatial observations require the full field sector.

Complete Fock dynamics and the thermodynamic limit. The same local action acts on all 2^{2V} occupations of a box with V sites. For E spatial bonds, each of norm at most c, its full-Fock stability bound is $-cEI \leq H \leq (cE + UV)I$, with $E \leq 3V$. The completed two-mode hopping, contact and fermionic-route words apply to all number sectors and arbitrary references. With $B = cE + UV$, a symmetric n-step split has ideal operator error at most $B^3|t|^3/(3n^2)$. The corresponding complete-history error is summed over segments without conditioning on outcomes. This bound depends on volume; a bound restricted to two particles is not substituted for it.

Removing the onsite contact terms by a local interaction picture leaves even, time-dependent

two-site bonds of norm at most c. The commutator-chain estimate for disjoint, initially commuting observables is

$$\|[\tau_t(A_X), B_Y]\| \le \|A_X\|\|B_Y\|\,|X|\,2^{-d(X,Y)}(e^{48c|t|}-1).$$

At least one of the observables is even; odd–odd locality uses the graded CAR relation. For a normalized observable in a radius-r cube and strictly padded radius $R > r$, the comparison to any larger box is bounded by

$$\min\{2,\, 6c|X|(2R+1)^2|t|\,2^{-(R-r)}(e^{48c|t|}-1)\}.$$

The limit gives a strongly continuous automorphism group on the quasi-local CAR algebra at fixed spacing. The contact contribution $[H_U - H_0, a_0^* a_2] = U a_0^* a_2(n_1 - n_3)$ remains nonzero in that limit. Thus the construction is an interacting lattice field with three spatial coordinates and continuous Hamiltonian model time. Neither an infinite-volume physical state nor a Lorentzian chiral continuum is selected. The executable full-Fock control and rational locality bounds are retained with the executable model [10].

A corrected memory branch and a clock instrument on a separate factor compose exactly, including correlated logical/clock/reference inputs. The clock requires its preparation, control, readout and event association; the tensor identity does not establish a stationary product reference, physical duration, dispersion relation or memory mass.

Extension and obstruction for the full band. The same native record-rate estimator drives counted local source words on the full 8,128-state field sector, with all 64 pair momenta and all four terminal occupation-pointer outputs. Its uniform complete average half-diamond bound is below 0.037991 for $|s| \le 1$. This requires a supplied globally normalized field Gibbs reference and a rate-transfer actuator; it is not an inherited native encoding. Every finite independent product of the canonical native reference has integer modular gaps in units $\log(25/8)$, whereas the same zero-momentum pair has $13/12 < E/U < 3/2$ relative to an eta dark state. The inherited intertwiner therefore fails, with generator error greater than $1/24$ in the stated units. A separate native coherent state has an exactly reference-matched exterior through every native record history and a computed finite contrast lifetime, but its coherent core is immobile and its modular-energy excess is zero.

6.9 One ordered spin state, continuum clock and spectral readout

A separate source trajectory gives a canonical field in three continuous spatial dimensions and time. Its starting point is the declared two-flavor field extension on the cubic lattice, with the complete colored repair instrument. On the single-occupancy subspace every repair acts as identity on spin and retains its fair coin record. For a box with E bonds, the centered source with $c_E = JE^4$ and $U_E = JE^8$ converges, including its complete finite histories, to

$$H_J = J\sum_{\langle xy\rangle}(\mathbf{S}_x \cdot \mathbf{S}_y - 1/4), \qquad J > 0.$$

This is an interacting fixed-spacing spin dynamics. Its cubic lattice and coherent field encoding are specified source inputs, distinct from the one-bit occupation process on the spherical federation.

To select the ordered reference, prepare the unique finite ground of $H_B = H_J - B\sum_x \epsilon_x S_x^1$, where $B > 0$ and $\epsilon_x = (-1)^{x_1+x_2+x_3}$. The finite ground has a terminating source-control preparation algorithm. Taking volume to infinity before $B \downarrow 0$ gives the ordered state used throughout this construction. The matched infrared inequalities give staggered order $m > 1/1000$ and filtered

excitation energies between positive multiples of momentum [123]. They do not assert a sharp spectral pole.

Let K implement zero-field evolution in this ground representation. With $X = px$, $T = Jpt$, the fields

$$Q_p(F) = p^2 \sum_x \epsilon_x F(px) S_x^3, \qquad P_p(G) = -p \sum_x G(px) S_x^2$$

have a common bounded-history subsequential limit. The resulting regular Weyl field obeys the canonical relation with coefficient $m \int FG$, admits continuous spatial translations and positive time translations, and has a finite propagation cone. Both Stone fields create nonzero positive-energy weight at arbitrarily small momentum. These statements concern the same ordered state and clock $K/(Jp)$; they do not identify its limiting dynamics with the separate free Weyl branch.

Selection-field and zero-field clocks. For the same finite input on N sites and m interventions, let T_{abs} bound the sum of absolute controlled evolution durations on every branch. Replacing H_B by H_J changes the complete output state by at most

$$D_{\mathrm{output}} \le \min\{1, BNT_{\mathrm{abs}}/2\},$$

where D is half trace norm and all records and untouched references are retained. This follows from $\|H_B - H_J\| \le BN/2$, Duhamel's formula and complete-channel contraction. A local version combines this estimate on a finite buffer with the spin locality bound, so its error does not grow with an unrelated exterior volume. It controls dynamics on the same input; it does not supply the missing effective preparation modulus for the chosen infinite ordered state. Centering either clock at its own ground also requires its own energy-origin certificate.

For $\varepsilon_x = \frac{1}{2}\sum_{y\sim x} h_{xy} - B\epsilon_x S_x^1$ and $j_{xy} = i[\varepsilon_x, \varepsilon_y]$, exact finite evolution gives

$$\dot{\varepsilon}_x + \sum_y j_{xy} = 0, \qquad \frac{d}{dt}\sum_x f_x \varepsilon_x = \sum_{x<y} (f_y - f_x) j_{xy}.$$

The Hermitian currents are antisymmetric and vanish beyond graph distance two. Adjacent bond currents contain $i[h_{xy}, h_{yz}] = J^2 \mathbf{S}_x \cdot (\mathbf{S}_y \times \mathbf{S}_z)$. For the bipartite staggered field the site-pair currents agree with their zero-field values, although the energy densities and evolution differ. An intervention has a separately counted source-energy jump. These identities supply energy continuity, not a symmetric Lorentz stress tensor or a responsive geometry.

Relaxed response in the same continuum. Let L denote the continuum positive generator obtained from $K/(Jp)$. For either actual continuum field vector $a = A\Omega$, its inverse-energy susceptibility $\chi_A = \|L^{-1/2}a\|^2$ is positive and finite on the specified momentum annulus. The exact variational identity is

$$\inf_{2\,\mathrm{Re}\langle a,v\rangle = b} \|L^{1/2}v\|^2 = \frac{b^2}{4\chi_A}.$$

The vectors are perpendicular to $\ker L$. For $\eta > 0$, writing $\chi_\eta = \langle a, (L+\eta)^{-1}a\rangle$, $v_\eta = b(L+\eta)^{-1}a/(2\chi_\eta)$ gives a feasible sequence whose energy tends to this infimum. Attainment at $\eta = 0$ requires $L^{-1}a$ to belong to the Hilbert space. Positive Fourier averages of the same Heisenberg evolution implement the regularized filter, retaining complementary outcomes. Finite spectral certificates bound its response without treating bare twist energy as the relaxed stiffness. The bounded source probes $\sin(sA_p)/s$ give its regularized susceptibility through the same bounded-history limit at fixed s and η. Taking $s \to 0$ and then $\eta \downarrow 0$ recovers the actual continuum coefficient, without a finite-state convergence rate. The quantity above is the cost of an infinitesimal response, not an excitation mass.

Testing a particle branch. On an even periodic box, set $K_B = H_B - E_0(H_B)$. The commuting projectors Q_k for the combined translation and internal rotation, and the complete energy-clock operators

$$B_j(K_B) = \frac{1}{M} \sum_{r=0}^{M-1} e^{2\pi i j r/M} e^{-ir\tau K_B}$$

give joint effects $Q_k B_j^* B_j$ summing to the identity. Every unsuccessful preparation and source-control export retains its own outcome. For a phase margin δ measured in turns, the summed clock leakage is bounded by $\min\{1, [2(M\delta)^2]^{-1} + (2M\delta)^{-1}\}$. Inner and outer bin sets therefore bound the original, unnormalized spectral weight in a prescribed energy–momentum tube, with source, input and sampling errors added. The clock step must also resolve aliases across the full finite spectrum.

The unchanged positive-field eight-site cube supplies a finite control: its complete 256-vector certificate and 128-bin clock bound the weight in one reflection character and excitation interval $[4/5, 7/5]$ below by $1/5$. This interval contains the first excitation cluster near 1.0922846777 in the specified source units. The cube's characters are reflections, not small translation momenta; the larger native 128-bin tape is specified rather than reported executed. A positive continuum particle component additionally needs a uniform nonzero fraction of the original response in shrinking tubes about one common dispersion, with readout error small relative to that response. For width $Jp\delta_p$, the clock requires $M\tau Jp\delta_p \to \infty$. Fixed-history convergence alone supplies no such growing-time estimate. Lorentz covariance, surviving connected interactions, massive species and the corresponding gravitational stress are separate conclusions requiring proof. The complete constructions and finite certificates are in the executable model [10].

6.10 Compiled instruments on interacting sources

Three executed components act on specified interacting sources. This subsection gives their resources, retained records and error bounds; the owning papers give the mathematics.

The composite probe. On the authenticated hidden-frame ground preparation of the open eight-site Weyl/contact source, whose inherited 262,137 pulses are not reexecuted, a selector word of 1,008 source pulses in 56 contraction blocks with five work bits implements the success Kraus operator $C_0/3$ of the composite charged probe of the particle paper [116]. It retains all 7,776 raw selector records, native export failures, inherited preparation aborts and 2,097,152 coherent joint amplitudes over 256 reference columns. Its complete operator error is below 1.6×10^{-13} on every field input and untouched reference, and its success probability is approximately 0.08261623248916 [37]. It detects the total response, with no executed clock resolving the 138 spectral lines.

Intervening observations. On the compact 180-mode core the complete-history comparison of the particle paper [116] retains every native record, detector result, feedback choice and the final field and reference state, with the memory continuing by its actual transition law. Its rational schedule for three intervals of duration $1/3$ and complete accuracy $1/100$ requires 46,370,150,349 native steps, a finite resource bound. An independent execution of the 128-record law with binary filters of probability $3/8$ retains all 12 old memory inputs and 512 fine labels on a four-mode probe, reaching 52,488 class-history rows at four steps per interval, and obeys the bound [36].

The regional modular actuator. On the prepared open plaquette, a palindromic source word for the regional modular polynomial at dimensionless parameter $1/2$ executes all 39 Pauli terms on all 16 regional input columns in 37,568 pulses, with the same number of proper-code loads and

exports. It retains the eight preparation record bits, every earlier abort, every first-failure output, the exterior field bits and any untouched reference, and its complete half-diamond error is below 6.4×10^{-9}, with no postselection on source success [38]. The Einstein paper states the theorem [8]; this operational clock identifies no geometric flow or physical duration.

7 One repair history: geometry, continuum, clocks and fields

This section derives what one native repair history supplies on the carrier and on the refining spherical source. Its statements are dimensionless carrier displacements, attempt counts and event counts; physical position, length scale and propagation speed need their own operational identification.

7.1 Native geometry, causal cones and field relations

Proposition 7.1 (Native geometric readout). *(i) Exact enumeration of all 4,680 state, seam and coin branches gives the one-attempt mean $T = I - L/60$ under uniform seam selection.*

(ii) With Q removing the constant mode and $C_n = QT^{2n}Q$, the normalized response $12C_n/\operatorname{Tr} C_n$ tends to $G = 4P_{\mathrm{slow}}$, where P_{slow} projects onto the rank-three band $5 - \sqrt{5}$; all eleven nonconstant modes are present at every finite attempt count.

(iii) For the readout $x(n) = \sum_p n_p u_p$ with frame directions u_p, a committed current j on the seam (a, b) gives the relative displacement $x(n') - x(n) = j(u_b - u_a)$, computable from the record without hidden loads.

(iv) The two-event controller spans two displacement directions. A declared continuation that sends the final scalar bit to $(0, 5)$ or $(0, 7)$ after the first event re-executes all 1,056 second events, and its three displacement directions $(2, 0, 0)$, $(1, 1-\varphi, \varphi)$, $(1, 1-\varphi, -\varphi)$, with $\varphi = (1+\sqrt{5})/2$, have determinant 4.

Proposition 7.2 (Native causal cones). *For a fixed seam schedule and shared coins, the load at a port v after the schedule is a function of the initial loads and coins in its backward dependency set. An initial difference supported on D influences v only along a time-ordered path, so $\operatorname{dist}(D, v) \leq$ edge crossings on that path $\leq$ elapsed seam events, and at most one edge is crossed per causal layer of disjoint simultaneous events.*

Exhaustive checks cover all 78 states for 27 schedule and coin pairs, and two executed examples show influence at graph distance two arriving only after two properly ordered seams [83]. One edge per causal layer is a native signal speed of at most one seam per ordered repair event; remote repairs can carry a difference to a receiver before its first local tick [14]. Units are observer conventions [85]: if a common operational propagation law fixed a dimensionless speed c_{model}, choosing the SI speed of light would impose only $L/T = c_{\mathrm{SI}}/c_{\mathrm{model}}$ on the model's length and time units, an operational action constant would add $ET = \hbar_{\mathrm{SI}}/\hbar_{\mathrm{model}}$, and one overall scale would stay free. The construction supplies no c_{model} relating its native distance readout to its modular clock.

Proposition 7.3 (Native field relations). *(i) Every event satisfies the discrete continuity law $x' - x = -b_e j$ with $b_e = \mathbf{e}_a - \mathbf{e}_b$.*

(ii) Conditional on the past and the selected seam, $\mathbb{E}[j] = d/2$ and $\operatorname{Var}(j) = (|d| \bmod 2)/4$.

(iii) Under the uniform calibration the mean load obeys $\mu_{k+1} - \mu_k = -L\mu_k/60$, exactly, by $\mathcal{K}X = XT$ for the 78-state kernel $\mathcal{K}$ and load matrix X; the deviation $V = \frac{1}{2}\|\mu - \frac{1}{6}\mathbf{1}\|^2$ decreases strictly unless the mean is constant, while every binary trajectory keeps $\|x - \frac{1}{6}\mathbf{1}\|^2 = 5/3$.

(iv) The slow band has mean multiplier $q = (55+\sqrt{5})/60 \approx 0.953934466291663$*, every nonconstant multiplier lies in* $(0,1)$*, and the second-difference coefficient* $q + 1/q - 2 > 0$ *has the opposite sign to the coefficient* $-4\sin^2(\omega\Delta t/2) \le 0$ *of an undamped oscillator; no positive rescaling of the attempt count turns this closed mean law into a wave law.*

Proof sketch of Propositions 7.1 to 7.3. The fair coin averages a seam to its endpoint mean, which gives the mean operator and the conditional current moments; the other two positive response eigenvalues have strictly smaller magnitude than $1 - (5-\sqrt{5})/60$, which gives the limit in Proposition 7.1(ii). Conservation gives the displacement identity. For the causal cones, scanning the schedule backwards and adding both endpoints of every seam that touches the dependency set proves the dependency claim by induction [82, 83, 84]. □

The rank-three readout is a local load-weighted coordinate. An absolute value needs an initial reference, while every current fixes its displacement. On the native quantum carrier, each complete fine branch K_h is a scaled partial injection. Applying the classical increment identity to each supported basis column proves, for $d(h)$ the recorded net displacement,

$$[X_i, K_h] = d_i(h)K_h, \qquad e^{iq\cdot X}K_h e^{-iq\cdot X} = e^{iq\cdot d(h)}K_h.$$

These are operator identities and retain arbitrary untouched references. The branch acceptance depends on the initial load. A unitary on the whole finite carrier cannot instead satisfy $U^*X_iU = X_i + a_iI$ for nonzero a_i: taking the trace gives a contradiction and the operator-norm residual is at least $|a_i|$.

The protected spin source makes the distinction concrete. Its inclusion P has one fermion at each site, so $N_xP = P$. For any finite real profile f, set $X_f = \sum_x f_xN_x$ and $c_f = \sum_x f_x$. Then $(X_f - c_f)P = 0$. The low-band isometry W with $\|W - P\| \le 2\tau/U$ consequently obeys

$$\|W^*X_fW - c_fI\| \le 4\Big(\sum_x |f_x|\Big)(\tau/U)^2.$$

Indeed only $(W-P)^*(X_f - c_f)(W-P)$ survives after expansion. Thus fixed local total-load coordinates become scalar on the stated strong-coupling trajectory. A spatial detector of a moving spin excitation must read its internal field response.

There is a conjugate coordinate constructed from that same continuum field. Write $q(F) = Q(F)\Omega$ and use Fourier measure $d\nu = d^3k/(2\pi)^3$. The Weyl relation and the second-moment bounds imply $2\,\mathrm{Im}\langle q(F), P(G)\Omega\rangle = m\int FG$. Cauchy–Schwarz, optimized with $\widehat{G} = \widehat{F}/|k|$ on Fourier supports away from zero, gives

$$m^2\int \frac{|\widehat{F}|^2}{|k|}\,d\nu \le \|q(F)\|^2 \le 3\int \frac{|\widehat{F}|^2}{|k|}\,d\nu.$$

Strong commutativity of the equal-time Q fields makes their Gram form real on real tests, so these bounds extend complex linearly. The map $T(\widehat{F}/\sqrt{|k|}) = q(F)$ is bounded below onto the closed subspace $\mathcal{H}_Q$ of the original field representation. It intertwines the actual spatial translations with multiplication by $e^{-ik\cdot a}$. Its positive Gram operator commutes with these multipliers, and the polar factor $V = T(T^*T)^{-1/2}$ is a unitary onto $\mathcal{H}_Q$. Define

$$X_i = Vi\partial_{k_i}V^*, \qquad P_j^{\mathrm{space}} = V\hbar k_jV^*.$$

The second operator is the same field translation generator in the chosen action units. On the common invariant core $V\mathcal{S}(\mathbb{R}^3)$,

$$U(a)^*X_iU(a) = X_i + a_iI, \qquad [X_i, P_j^{\mathrm{space}}] = i\hbar\delta_{ij}I, \qquad \Delta X_i\Delta P_i^{\mathrm{space}} \ge \hbar/2.$$

The coordinate spectral measure extends to the full field space with the no-position effect $I - \Pi_Q$. A normalized, translation-covariant measure on all of space without this outcome would give the invariant vacuum a translation-invariant probability measure on $\mathbb{R}^3$, which is impossible. This source-defined coordinate is generally nonlocal and differs from the carrier load coordinate and the field quadrature $Q(F)$.

Complete finite detectors use the declared coherent load, pointer and refinable six-level control services. For q spins and b fine Kraus branches, a computable Stinespring completion has dimension $D = 2^{q+\lceil \log_2 b \rceil}$. Complex rank-one elimination and rational angle synthesis give at most $20D^2+4D+10q$ native pulses. Their precision and complete load/export failures enter the channel bound. The eight-site coordinate detector executes a 49,228-pulse decoded core with an untouched reference; its 49,324-pulse encoded completion has analytic half-diamond error below 1.53×10^{-10}. The full encoded tape is not claimed executed. This construction consumes the coherent service specification; bare occupation repair does not supply it.

Let R_n exhaust a computable Fourier basis and write $C = T^*T$. The compressed polar maps

$$V_n = TR_n(R_nCR_n)^{-1/2}, \qquad m^2 I \le C \le 3I,$$

use the inverse on $\operatorname{Ran} R_n$, extended by zero. They converge strongly to V, with their adjoints converging on the field representation. For a fixed computable spatial partition, compress its ordinary coordinate effects by R_n, conjugate by V_n, and retain $I - V_nV_n^*$. These complete effects and their positive square roots converge strongly. The associated Lueders instruments therefore converge in trace norm on each fixed state and reference. Source vectors $\sin(sQ_p(F))\Omega/s$ realize each finite Gram through bounded histories, first taking the source limit at fixed s, n, then $s \downarrow 0$, then $n \to \infty$. Positive finite Gram gaps and the compiler control each fixed stage. No effective box-size modulus, uniform infinite-dimensional channel norm or second-moment tail bound follows from this ordered limit.

A localization space invariant under the actual time generator is

$$\mathcal{H}_{\mathrm{resp}} = \overline{\operatorname{span}}\{e^{-itL}Vf : t \in \mathbb{R},\ f \in L^2(\mathbb{R}^3)\}.$$

Its spatial spectral measure is absolutely continuous: a momentum-null projector kills $\mathcal{H}_Q$ and commutes with every e^{-itL}. The countable rational-time history Gram consequently constructs an isometry $J : \mathcal{H}_{\mathrm{resp}} \to L^2(\mathbb{R}^3; \ell^2)$ that intertwines the same spatial translations and satisfies $JVf = f \otimes e_0$. Compressing ordinary position gives $E(B) = J^*E_0(B)J$, with the invariant dark effect $I - \Pi_{\mathrm{resp}}$. For normalized states in $\mathcal{H}_{\mathrm{resp}}$, its position-outcome and momentum spreads obey $\Delta_E x_i \Delta P_i^{\mathrm{space}} \ge \hbar/2$ whenever both moments are finite. The measure restricts to the preceding sharp coordinate on $\mathcal{H}_Q$. Its general outcomes need not be projections, and its effects alone do not determine measurement back-action. No time invariance of $\mathcal{H}_Q$, preservation of finite moments, sharp particle shell or operational continuum realization of the entire response measure is inferred. Proofs and finite certificates are in `model/spin_dynamical_localization`, `model/spin_operational_position` and `model/spin_position_limit` [2].

The recovered frame also supplies an unbounded three-component *axial circulation*. With oriented incidence matrix B, seam current J_n and columns $W_{(a,b)} = \hat{u}_a \times \hat{u}_b$ of the unit frame vectors, exact source identities give $WB^T = 0$, $WW^T = 8I_3$, $\mathbb{E}[J_n \mid x_n] = -B^T x_n/60$ and $\mathbb{E}[J_n J_n^T] = I_{30}/198$ in the uniform binary equilibrium. Hence $Z_N = \sum_{n<N} WJ_n$ is a record-readable martingale with $\mathrm{Cov}(Z_N) = (4N/99)I_3$, and $N^{-1/2}Z_{\lfloor Nt \rfloor}$ converges to three-dimensional Brownian motion with covariance $(4/99)\min(t,s)I_3$ and Cameron–Martin form $(99/8)\int |\dot{z}(t)|^2\,dt$ in a rescaled attempt count. A three-event face loop restores the carrier state while adding a nonzero counter increment. The counter's memory grows with the observation horizon; it supplies neither a radial field nor a physical trajectory [11].

7.2 The common continuum operator

On the declared spherical source family, the native cell Green operators converge in norm along one common subsequence to a positive compact injective operator G on the mean-zero observation space. Its inverse $A = G^{-1}$ is densely defined and self-adjoint, and restoring the constant zero mode gives a conservative strongly local regular Dirichlet form with domain H^1 on the supplied flat-polyhedron comparison surface, energy bounded above and below by positive multiples of the gradient energy, and heat-trace spectral dimension two. A boundary-action test excludes a boundedly invertible L^2 energy identification of A with the static boundary action of a smooth compact three-dimensional Dirichlet bulk with smooth boundary, whose Steklov eigenvalues grow like $\sqrt{n}$; bounded-energy internal rank-three modes vanish at this scaling [11]. An exact terminal reduction gives local dual-edge conductances in $[5/8, 15/19]$ with Green-operator error at most $182\,2^{-l}$ relative to native cell means, so this weighted graph has the same continuum subsequences.

A positive native lag readout realizes the Poisson extension. For $K_l = I - L_l/(2M_l)$, Sibuya jump probabilities $p_n = \binom{2n}{n}/[4^n(2n-1)]$ and a Poisson number of jumps with mean R give $\mathbb{E}K_l^N = e^{-R\sqrt{I-K_l}}$. Restricting jumps to b and their count to j, with overflow retained, has maximum native lag bj and operator error at most $\min(1, R\binom{2b}{b}4^{-b} + R^{j+1}/(j+1)!)$; at $R = 1/2$, $b = 64$, $j = 8$ the full 66-state source gives 0.034581 within the bound 0.036. Taking $R_l = r\sqrt{2M_l 4^l}$ yields the compressed limit $e^{-r\sqrt{A}}$; here r labels a positive mixture of native lags, not a physical radius.

Full native density correlations converge to $e^{-\tau A}$, and the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. The full-space extension $\bar{A} = A \oplus 0$ satisfies $\|\mathbf{1}_F e^{-\tau\bar{A}}\mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)}$, and its cosine propagator vanishes between these supports when $|t| < \sqrt{2}\,d(E,F)$ in the declared flat comparison distance. Two consecutive long observation blocks of one history supply a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement [11]. A declared static filter $A_l^{1/4}$ changes it to $A_l^{-1/2}\cos((t-u)\sqrt{A_l})$, twice the real part of the free-field vacuum covariance; the stationary increments obey $2A_l^{-1/2}(I - \cos(\Delta t\sqrt{A_l})) \le (\Delta t)^2\sqrt{A_l}$ and, with $\sum_m \langle h_{jm,l}, \sqrt{A_l}h_{jm,l}\rangle \le 12j^2$ for centered cell averages of spherical harmonics, the geometric H^{-s} increment moment is at most $12[1 + 1/(2s-3)](\Delta t)^2$ for $s > 3/2$, giving convergence in $C([0,T], H^{-s}(S^2))$. The wave readout is declared classical processing, distinct from passive density waiting and from a quantum vacuum.

7.3 Native density histories and their limits

At level l, let $V_l = 240\,4^l$, $q_l = V_l/3$ and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$. In the uniform canonical equilibrium, with independent uniform seams and fair repair, the actual snapshot readout of a mean-zero test function repeated on its twelve native ports,

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\,16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l\chi_l}},$$

converges in joint distributions at finitely many times as space and the native attempt count are refined together. The limiting family is stationary and Gaussian, with covariance $\langle f, e^{-|\tau-u|A}g\rangle$. Representing each unread repair as a fair identity-or-swap, the conditional profile overlaps concentrate with variance at most $4\|g\|_\infty^2\|f\|_{\ell^2}^2/V_l^2$ uniformly in attempt count, and a conditional finite-population central limit theorem applies. The declared filter $G_l^{1/2}$ gives covariance $Ge^{-|\tau-u|A}$. The law is reflection positive with one-particle semigroup $e^{-\tau A}$, generator A and not $\sqrt{A}$. The interpolated density converges in $C([0,T], H^{-3}(S^2))$ and its Green-filtered readout in $C([0,T], H^{-2}(S^2))$,

by a forward–backward martingale decomposition with $g = 2(I + P_l)^{-1} f$ and harmonic tail bounds; the step paths have the same Skorohod limits [11]. For $\phi = \Phi/\sqrt{2}$ a mode of eigenvalue λ has covariance $e^{-\lambda|\tau-u|}/(2\lambda)$, so the Gaussian precision is the Euclidean form

$$S_E(\phi) = \frac{1}{2} \int_{\mathbb{R}} \left(\|\partial_\tau \phi\|^2 + \|A\phi\|^2 \right) d\tau,$$

with reconstructed Hamiltonian $d\Gamma(A)$ and spatial precision A^2. The action follows from the native readout law and has diffusive dispersion.

Two restrictions bar identifying this sector with a relativistic field. On the shorter attempt scale $b_l = 1260\,8^l$ both the density and Green-filtered paths freeze to their initial field, with expected squared supremum increments at most $50(T2^{-l} + a_l^{-1})$, $a_l = 1260\,16^l$. For a complete linear readout with bounded injective dense-range composite map $B = TC^{1/2}$, $C = I$ or G, the left polar decomposition $B = D_0^{1/2} U$ gives

$$D_0^{-1/2} D_\tau D_0^{-1/2} = U e^{-\tau A} U^*,$$

so normalization preserves the ordered decay rates $\lambda_n(A) \asymp n$. No common positive clock factor turns these into the $n^{1/3}$ Euclidean decay rates of a complete stationary free scalar on a smooth compact three-dimensional manifold without boundary, and the same argument excludes every spatial dimension $d \geq 2$, whose rates grow like $n^{1/d}$; a continuous nonconstant time change preserving the semigroup is a scalar factor. The excluded object is the complete linear Gaussian identification; restricted boundary observations, nonlinear dictionaries and interacting fields need their own analysis, and a causal microscopic theory may have this diffusive sector as an effective description [11].

7.4 Incident clocks and active waves

Let $a_l = 1260\,16^l$, $M_l = 630\,4^l$ and let a fixed event set contain $d_l > 0$ seams. Its incident counter $N_l(n)$ includes zero-current repairs, with $p_l = d_l/M_l$; write $T_l(k)$ for the attempt of its kth tick. For $n_* = \lceil a_l(T + \delta) \rceil$,

$$\Pr\left[\max_{n \leq n_*} |N_l(n) - n p_l| > a_l p_l \delta \right] \leq \frac{n_*(1 - p_l)}{a_l^2 p_l \delta^2},$$

and on the complementary event the normalized count $C_l(t) = N_l(\lfloor a_l t \rfloor)/(a_l p_l)$ and the inverse clock $U_l(t) = T_l(\lfloor a_l p_l t \rfloor)/a_l$ differ from t by at most $\delta + 1/a_l$ and $\delta + 1/(a_l p_l)$ on $[0, T]$. A primitive port has $d_l = 5$ or 6 and a whole carrier $d_l = 33$. The density path theorem transfers to these local count parametrizations without independence between clock marks and fields [13]. Passive port access chosen before each seam draw from earlier records also works: integer credits $R_n = \sum_{j \leq n} (30/d_j) I_j$ have conditional mean $30/M_l$ per attempt, deviation probability at most $n_*(M_l - 5)/(5 a_l^2 \delta^2)$, and inverse clock error at most $\delta + M_l/(5a_l)$.

For fixed incidence counts d_i and shared-seam counts c_{ij}, centered clock counts divided by $\sqrt{a_l/M_l}$ converge to Gaussian increments with covariance $\min(t, u) c_{ij}$, asymptotically independent of the density snapshots. Adjacent degree-five port clocks have limiting correlation $1/5$, the endpoints of an external seam $1/6$, and nonadjacent clocks zero. These are native model predictions under the stated schedule and preparation, without an atomic-clock or proper-time dictionary [13].

The admitted coherent walk can use the same tick. A protected wave register receives one fixed step W_l per incident event at a degree-d port, including zero current, while the timing source

pauses during coherent services. With $Q_l = 630\,16^l$, $D_l = 4^l L_l$, $\beta = 2d/\sqrt{630}$, $\eta = \delta + 1/a_l$ and $e_l = \Lambda/[6(4Q_l - \Lambda)]$, on the band $0 \le D_l \le \Lambda < 4Q_l$ and the count good event,

$$\sup_{\tau \le T} \left\| \cos(N_l(\lfloor a_l \tau \rfloor) \arccos K_l) - \cos(\beta\tau\sqrt{D_l}) \right\| \le \beta\sqrt{\Lambda}\,[\eta + (T+\eta)e_l],$$

so the compressed wave converges to $\cos(\beta\tau\sqrt{A})$. Step error ϵ_l accumulates at most $N_{\max}\epsilon_l$ with $N_{\max} = O(\sqrt{Q_l})$, so $\sqrt{Q_l}\epsilon_l \to 0$ suffices. An exact control retains 237,600 two-attempt clock histories; the continuing Ramsey-plus probability is 1297/1320 against 7901/7920 when zero-current triggers are dropped. The wave step is compiled apparatus; passive repair does not supply it [13].

7.5 An encoded quantum field and its modular clock

The classical covariance does not determine commutators. N independent proper-code registers in $|0\rangle$ give $Q_N = \sum_j X_j/\sqrt{2N}$ and $P_N = \sum_j Y_j/\sqrt{2N}$, with commutator $i\sum_j Z_j/N$, $\mathbb{E}Q_N P_N = i/2$ and $\mathbb{E}Q_N^4 = 3/4 - 1/(2N)$. For Weyl factors $W_N(v) = e^{i(sQ_N + tP_N)}$ the quantum central limit theorem [1] gives

$$\lim_{N\to\infty} \mathbb{E} \prod_j W_N(v_j) = \exp\left[-\frac{1}{4} \left| \sum_j v_j \right|^2 - \frac{i}{2} \sum_{j<k} \sigma(v_j, v_k) \right], \qquad \sigma((s,t),(s',t')) = st' - ts',$$

the CCR vacuum with its ordered-product phase. Source pulses execute four codes and one Ramsey pointer with all 32 outcomes retained; the pointer quadratures give $0.7452257133 - 0.3013609949i$ for the declared three-factor word, reversal conjugates the phase and the tracial preparation removes it. The native density bands fix the mode rotation $e^{i\theta_l Z/2}$ per incident trigger, $\theta_l = \arccos(1 - \lambda/(2Q_l))$, with encoded frequency $\Omega_l = \sqrt{Q_l}\theta_l \to \sqrt{\lambda}$, ordered covariance $e^{-i\Omega_l(t-u)}/(2\Omega_l)$ and kick response $\sin(\Omega_l(t-u))/\Omega_l$, executed for both kick signs at three native bands [11]. Polarized preparation, mode encoding, storage, pulses and readout are specified services. The sine kernel $A^{-1/2}\sin(t\sqrt{A})$ vanishes outside the comparison cone for mean-zero tests; unrestricted smearings add tP_{const}, and restoring the compact massless zero mode gives a free particle with no normalizable ground state, so the mean-zero field is no complete local net.

For $\rho_r = (I + rZ)/2$, $0 < r < 1$, and $H = \Omega(I - Z)/2$, the finite Gibbs and Kubo–Martin–Schwinger identity gives $\sigma_s = \alpha_{-bs}$ for every real s exactly when $\log\frac{1+r}{1-r} = b\Omega$, that is $r = \tanh(b\Omega/2)$; equal polarization at unequal frequencies fails a common b, the pure endpoint is not faithful and the tracial endpoint has trivial flow. With $Q_N = \sum X_j/\sqrt{2Nr}$ and $P_N = \sum Y_j/\sqrt{2Nr}$ the ordered copy limit has thermal covariance $C(\delta) = [\cos(\Omega\delta) - ir\sin(\Omega\delta)]/(2r\Omega)$ with $C(t - ib) = C(-t)$, executed on two codes with retained purifying references. The native bound $\lambda_n(A) \ge n/49920$ gives $\operatorname{Tr} e^{-b\sqrt{A}} \le 99840/b^2$, so the mean-zero Fock space has the faithful normal state $Z_b^{-1} e^{-b\,d\Gamma(\sqrt{A})}$ with the same modular flow, and under $t = \kappa\tau$, $\kappa = 2d/\sqrt{630}$, its modular parameter is $s = -\kappa\tau/b$. The common b is a measurable preparation condition, not a repair-selected temperature.

7.6 The product field, its clock and its Lorentzian density

Independent long blocks of one native history supply radial innovations. Their mean-zero covariance is $B_l = G_l - P/(2520\,16^l)$, and a second block adds a normalized constant-direction readout, giving $B_l + P_{\text{const}}$. On a declared interval $(0, L)$ with Dirichlet ends, a mode with $h = \sqrt{\lambda}$ has covariance

$$C_\lambda(r,s) = \frac{\sinh(h\min(r,s))\sinh(h(L - \max(r,s)))}{h\sinh(hL)}, \qquad C_0(r,s) = \frac{\min(r,s)(L - \max(r,s))}{L},$$

realized by applying $Q_{rs}^{1/2}(B_l+P_{\mathrm{const}})^{-1/2}$ to successive blocks, with Q_{rs} the conditional innovation. The whitening means the radial direction carries no native information beyond the spectrum of A. Complete 96-event current tapes check the finite decoder. The spatial subsequence gives the product operator $\mathcal{B}=-\partial_r^2+\bar{A}$ with action $\frac{1}{2}\int_0^L(\|\partial_r\phi\|^2+\mathcal{E}(\phi))\,dr$; for $L=\pi$ its count satisfies $\Lambda^{3/2}/(12\sqrt{2})\le N_{\mathcal{B}}(\Lambda)\le 49921\Lambda^{3/2}$ for $\Lambda\ge 3$, and its tensor form gives comparison wave speed one in $d^2_{\mathrm{product}}=|r-s|^2+2d^2_{\mathrm{screen}}$. On a radial grid the sampled bridge has an explicit sine-mode covariance, and a spectral filter realizes the thermal covariance $\coth(b\sqrt{\mathcal{B}}/2)/(2\sqrt{\mathcal{B}})$ from the same records; the full-port generator and the cell Schur generator have compressed resolvents within $1/(8\,4^l)$ at each positive shift, so both give the same product target. Radial labels, filters, boundaries and block access are specified operations.

For a full-port angular mode set $a_l=\sqrt{Q_l}\arccos(1-\lambda/(2Q_l))$ and $c=\sqrt{\rho_{k,h}}$. Each incident trigger applies the source Z pulse of angle $a_l/(2\sqrt{Q_l})$ and the radial X pulse of angle $c/(2\sqrt{Q_l})$; anticommutation gives $\omega_{l,k}=\sqrt{a_l^2+c^2}$ with no fitted product frequency, and the split word differs from the joint Hamiltonian by at most $na_lc/(4Q_l)$ after n triggers. The faithful reference is polarized along $(cX+a_lZ)/\omega_{l,k}$ with $r=\tanh(b\omega_{l,k}/2)$; ordered two-time measurements retain all 32 outputs. A deterministic packet $v\propto(12I-L)^{1024}(\mathbf{1}_{\mathrm{cell}\,0}-\mathbf{1}_{\mathrm{cell}\,1})$ on the 240-port carrier has Rayleigh quotient $\lambda\simeq 0.044442251696$, residual $\|(L-\lambda)v\|<0.000304$ and squared cell compression above 0.996458773594; Householder encoders and a Bell reference read it after incident-triggered phase, with 25-trigger Chebyshev discrepancy about 0.000149026 at $M=630$, at a cost of 1,747,068 source pulses. Copy and spectral limits followed by radial refinement give

$$C_b(t)=\frac{\coth(b\sqrt{\mathcal{B}}/2)\cos(t\sqrt{\mathcal{B}})-i\sin(t\sqrt{\mathcal{B}})}{2\sqrt{\mathcal{B}}},\qquad D(t)=\frac{\sin(t\sqrt{\mathcal{B}})}{\sqrt{\mathcal{B}}},\qquad \sigma_s=\alpha_{-bs},$$

and with $C_L=L(49920+L^2/9)/3$, $\operatorname{Tr}e^{-b\sqrt{\mathcal{B}}}\le 6C_L/b^3$, so the encoded Fock algebra has the faithful normal state $Z_b^{-1}e^{-b\,d\Gamma(\sqrt{\mathcal{B}})}$ with $s=-\kappa\tau/b$. Vanishing count tolerances require $4^l\eta_l^2\to\infty$ and complete-step errors $\sqrt{Q_l}\epsilon_l\to 0$. Radial design, common b, preparation, mode access and source precision are declared services [11].

Lorentzian density. The regular strongly local angular form with its H^1 domain admits a measurable uniformly elliptic tensor in each chart [118, Theorem 7.2 and Corollary 7.3], recoverable from complete native records: with the disclosed total-one kernel $K_l=I-L_l/(2M_l)$, including every zero-current hold, the ordered limits

$$\lim_{t\downarrow 0}\lim_{l}\int\phi\,\frac{\bar{P}_l(t)(fg)-f\bar{P}_l(t)g-g\bar{P}_l(t)f+fg}{2t}\,d\mu=\int\phi\,a^{ij}\partial_i f\partial_j g\,d\mu$$

recover weighted tensor averages. In barycentric face coordinates the one-attempt increment tensor obeys $\frac{3}{7}a_{\mathrm{bare}}\le a_{\mathrm{screen}}\le\frac{15}{19}a_{\mathrm{bare}}$ with $a_{\mathrm{bare}}=\begin{pmatrix}1/36 & -1/72\\ -1/72 & 1/36\end{pmatrix}$, so the one-attempt-first limit reads a different tensor. The radial product adds $a^{rr}=1$. With coordinate measure $\mu(x)\,d^3x$ the chosen wave action has coefficient density $C=\mu\,\mathrm{diag}(-1,a)$, and

$$g=\sqrt{-\det C}\,C^{-1}=\mu\sqrt{\det a}\;\mathrm{diag}(-1,a^{-1}),\qquad \sqrt{-\det g}\,g^{-1}=C$$

has signature $(-,+,+,+)$ almost everywhere, with weak wave equation $\partial_\alpha(C^{\alpha\beta}\partial_\beta\phi)=0$ [121, Equations (23)–(26)]. The signature is that of the chosen action; the derived content is the measurable tensor a. The native measure is the radial pushforward of flat face area, with density

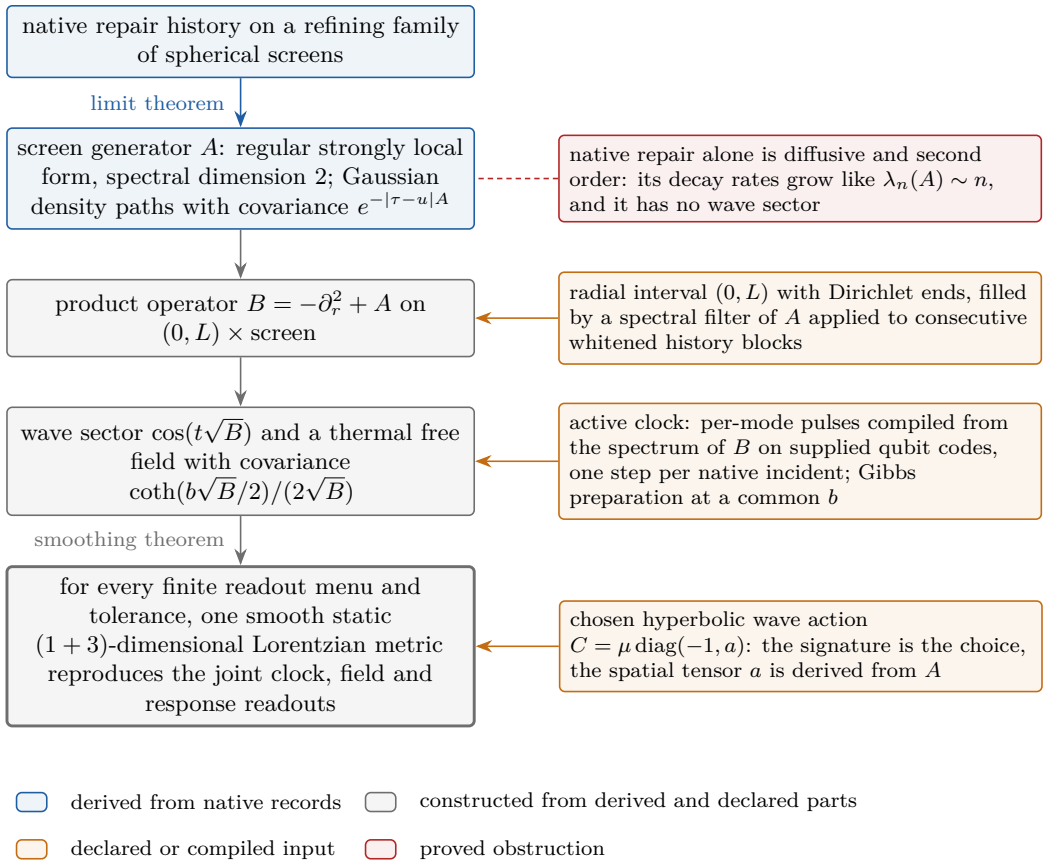


Figure 3: What the effective $(1+3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

$\rho_F(u) = c/(n_F \cdot u)^3$, $c = \pi r^2/(5a_0)$, on a face of unit normal n_F and plane distance r. Adjacent faces agree on their common edge but their spherical gradients differ by $-3c(n_F - n_G)/(n_F \cdot u)^4 \neq 0$, so the density is not C^1, and since $C^{tt} = -\mu$ the exact dictionary in this atlas fails the smooth-density premise. An alternative smooth dictionary would need a separate argument.

The density is globally Lipschitz relative to smooth sphere volume. Smoothing only the angular variables and the tensor, with the unitaries $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\, f$ between the changing L^2 spaces, gives transported Mosco convergence and then norm-resolvent convergence with gap $(\pi/L)^2$. Thermal covariance and sine response converge in operator norm, and ordered Weyl laws, finite adaptive Ramsey histories and finite detector outputs converge. Hence for any fixed finite admissible menu on bounded wave times and any positive tolerance one smooth static $1+3$ metric approximates the joint readouts. Both sides are free fields on the same supplied codes; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state, and gives no computable rate. In the smooth comparison $g = N^2\,\mathrm{diag}(-1, a^{-1})$, $N^2 = \mu\sqrt{\det a}$, the canonical momentum is $\mu\partial_t\phi$ and the thermal covariance and response have the standard static CCR/KMS normalization with a reflecting radial boundary.

Angular observations do not select the radial law. For the 240-port L, three 720-port prod-

ucts $B_R = I_3 \otimes L + R \otimes I$ with the unit path, fourfold path and unit triangle Laplacians satisfy $J^* F(B_R) J = F(L)$ for the radial average J and every spectral function, while a radial-contrast probe has cosine second derivatives $-5/2, -10, -3$.

For positive finite-measure regions E, F of the product, let $d(E,F) = \sup_u(\operatorname{ess\,inf}_F u - \operatorname{ess\,sup}_E u)$ over bounded local-form functions with $\Gamma_{\mathcal{B}}(u) \le 1$. Davies–Gaffney estimates and the short-time theorem of Ariyoshi and Hino [117, Theorem 2.7], $\lim_{\tau\downarrow 0} -4\tau \log\langle \mathbf{1}_F, e^{-\tau\mathcal{B}}\mathbf{1}_E\rangle = d(E,F)^2$, combined with heat-cosine transmutation give

$$\inf\{t>0 : \|\mathbf{1}_F \cos(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = \inf\{t>0 : \|\mathbf{1}_F \mathcal{B}^{-1/2}\sin(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = d(E,F).$$

This is the sharp finite-propagation front of the chosen wave equation, fixed by its intrinsic metric; it supplies no independent propagation speed, and same-field modular time has onset $|s| = d/b$.

7.7 Interacting readouts on the product

Retain the angular constant and radial profiles $\sqrt{2}\sin r$ and $\sqrt{2}\sin(2r)$ on $(0,\pi)$. The projected local quartic polynomial is

$$W = \tfrac{3}{2}(q_1^4 + q_2^4) + 6q_1^2 q_2^2, \qquad H = \tfrac{1}{2}(p_1^2 + q_1^2 + p_2^2 + 4q_2^2) + \frac{g_{\mathrm{eff}}}{24} W,$$

with $g_{\mathrm{eff}} = (\lambda/\pi)\int \mu_{\mathrm{screen}}^2 \sqrt{\det a_{\mathrm{screen}}}\, dx$ for a constant covariant coupling; $g_{\mathrm{eff}} = 1$ is an input. At occupation cutoff 4×4, native source pulses prepare the purification of $e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$ and execute two reads with feedback, retaining all work, reference and record outputs. For $\beta = 1/4$, intervals $1/16$ and 64 split steps, the complete-output distance from an independent exponential is below 1.97×10^{-8}; deleting the interaction or feedback changes the record law by about 7.31×10^{-4} or 1.95×10^{-3}. The worst branch uses 217,110 source pulses, 434,220 load/export events and two native repairs. A separate protocol advances this Hamiltonian on every incident event of two native timing attempts, retaining all 3,600 tapes and 4,900 field-prefix outputs, with timeout probability $35/36$ and complete error below 3.44×10^{-9}. On the refined clock family, incident rate $R_l = 2d4^l$ gives a controlled inverse-count comparison; each metric readout uses its own clock pullback, and no universal physical conversion is inferred.

For $g \ge 0$ the positive quartic form has a Friedrichs Hamiltonian $H \ge H_{\mathrm{harm}}$, Hermite compressions form a core, and

$$Z_d \longrightarrow Z, \qquad 0 < Z \le \frac{e^{-3\beta/2}}{(1-e^{-\beta})(1-e^{-2\beta})}, \qquad \rho_d \longrightarrow \rho \quad \text{in trace norm},$$

so every fixed finite complete adaptive history converges and the limiting Gibbs state has $\sigma_s = \alpha_{-\beta s}$. This proves occupation existence for the two-mode theory, without a small-code certificate: the complete $d=4$ versus $d=8$ output distance is about 0.56045, and $d=8$ versus $d=12$ about 0.24501. An inverse control recovers $(g_{\mathrm{eff}}, \beta, c) = (1, 1/4, 1)$ within 10^{-4} from six noiseless source probabilities and predicts a withheld interval within 10^{-8}, with local Jacobian singular values about $0.264713, 0.001489, 0.000110$. For fixed covariant λ the smoothed metrics give convergent quartic projections. The quartic excites omitted modes; no invariant two-mode band, interacting ultraviolet limit or finite-cutoff microcausality follows [39].

The reference-state boundary. Increasing native-return caps match modular phase magnitudes to the cosine limit, and a fixed cap freezes the low-band phase. On the full configuration algebra the uncapped reference approaches normalized trace with half-trace error at most $3/(8\,2^l)$,

its density-wave sector has vanishing weight, and fixed-parameter modular evolution becomes trivial in normalized-trace L^2. If the relevant fluctuation moments converge, any limiting scalar canonical commutator in this bounded-overlap regime is zero. Refinement-rescaled time, other states and other fluctuation regimes are not excluded [11].

8 A common matter metric and its gravitational response

The conditional scalar description supplies a concrete dictionary for comparing matter and geometry. With coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$ and metric $g = \sqrt{-\det C}\,C^{-1}$, the scalar wave symbol, both Weyl principal determinants and the Maxwell symbol share one characteristic cone, and the product has induced measures $dA = \mu\,d^2x$ and $dV = \mu N\,dr\,d^2x$ with $N^2 = \mu\sqrt{\det a}$. The same dictionary yields a control. Both radial null Einstein contractions of a smooth static product vanish, while a coherent scalar excitation has positive relative null stress, so that geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed. The spacetime paper [8] proves these statements, gives the finite field operators and entropy variations, and derives the necessary area-response equations.

9 Prediction protocols

A response without a fitted coefficient. In the stationary binary number sector, disclose local occupations and apply one extra fair native repair on seam (a, b) exactly when $n_a = 1$. Detailed balance gives, for any fixed future record event F whose continuation does not consult the earlier pulse mark,

$$\Pr_{\text{feedback}}(F) - \Pr_{\text{passive}}(F) = \tfrac{1}{2}\mathbb{E}_{\text{passive}}[(n_b - n_a)\mathbf{1}_F].$$

Complete capped clock and field forecasts and the native spatial Green response satisfy this identity exactly, with every trigger, coin and current label retained [11]. No matched public intervention dataset exists for these forecasts.

A controlled continuum response. For a disclosed port profile h, a uniform seam (a, b) executes its unchanged fair repair with probability $[1 + \epsilon(h_b - h_a)(n_a - n_b)]/2$, and the clock counts executed repairs, including zero currents. The exact linear occupation response is $2\chi K(I - K^r)h$, and on the refining source

$$\partial_\epsilon \mathbb{E}\eta_l(g)|_0 \longrightarrow 2\langle g, (I - e^{-\tau A})f\rangle.$$

At $\tau = 1$ with four selection slots per target repair, stopped likelihood bounds control the nonlinear mean error by $36\epsilon^2\|g\|$, and for a unit-axis flat coordinate at $\epsilon = 10^{-6}$, $\liminf_l \mathbb{E}_\epsilon\eta_l(f) \geq \epsilon\, 9/32792 - 36\epsilon^2 > 0$ [11]. In the selection-slot clock the derivative converges to $\langle g, R_\tau f\rangle$ with $R_\tau = 2(I - e^{-\tau\bar{A}})$, and positivity improvement of the heat semigroup gives, for nonzero nonnegative Lipschitz profiles at distance $r > 0$,

$$0 < -\langle g, R_\tau f\rangle \leq 2e^{-r^2/(2\tau)}\|f\|\|g\|, \qquad \tau > 0.$$

This derivative has no exact finite support cone. Reading its control, support and slot parameter as physical local interventions and time would contradict an exact relativistic cone; that identification is not derived, and a causal microscopic theory with diffusive effective observables is consistent with it.

9.1 The phase experiment

The classical repair law fixes the diagonal of every admissible record law and none of its coherences (Theorem 2.3). A fixed interference experiment reads the multiplier g that separates the members, and its count protocol, comparing the executed laws C and E, is fixed before any data [64, 65]. All members are ordinary quantum instruments, so the test separates record laws and makes no claim to separate OPH from quantum mechanics.

Write $x = \{2, 3\}$, $z = \{1, 2\}$ and $w = \{0, 2\}$. Prepare $|\phi_\theta\rangle = (|x\rangle + e^{i\theta}|z\rangle)/\sqrt{2}$ for $\theta \in \{0, \pi, \pi/2\}$, or the incoherent mixture $\frac{1}{2}(|x\rangle\langle x| + |z\rangle\langle z|)$, apply one repair on the seam $(0, 1)$, retain its record, and measure $P_+ = |\phi_0\rangle\langle\phi_0|$ and $P_- = I - P_+$ inside each committed record block.

Proposition 9.1 (Phase law of the two record instruments)**.** *For the preparations above:*

(i) the nonzero recorded source densities of C *are* $\frac{1}{2}|\phi_\theta\rangle\langle\phi_\theta|$ *for* $(c, j) = (0, 0)$, $\frac{1}{4}|x\rangle\langle x|$ *for* $(1, 0)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*; those of* E *are* $\frac{1}{4}|x\rangle\langle x|$ *for* $(c, d) = (0, 0)$ *and* $(1, 0)$, $\frac{1}{4}|z\rangle\langle z|$ *for* $(0, -1)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*;*

(ii) the plus probabilities are $p_{\mathsf{C}}(\theta) = \frac{3}{8} + \frac{1}{4}\cos\theta$ *and* $p_{\mathsf{E}}(\theta) = \frac{3}{8}$*, and the incoherent preparation gives* $3/8$ *for both; the phase-reversal contrast is* $1/2$ *for* C *and zero for* E*;*

(iii) all four preparations have identical classical load probabilities, the readout preserves the probability of every old record, and no record is erased and no outcome postselected.

Proof sketch. On the seam $(0, 1)$ the configuration x has contrast zero for both coins, and z has contrast -1, current 0 for coin zero and current -1 for coin one, which moves it to w. The coherent sum survives only in the C branch $(0, 0)$, where both currents vanish; E separates x from z by their contrasts [65]. □

Table 4: Plus probabilities of the four fixed settings [64, 40].

Preparation	C_0	C	E	T
Phase 0	$\frac{3}{8} + \frac{\sqrt{2}}{4}$	5/8	3/8	3/8
Phase π	$\frac{3}{8} - \frac{\sqrt{2}}{4}$	1/8	3/8	3/8
Phase $\pi/2$	3/8	3/8	3/8	3/8
Incoherent mixture of x and z	3/8	3/8	3/8	3/8

For every admissible record law the plus probability is $p_\Phi(\theta) = \frac{3}{8} + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$, since the empty seam of x stays in place and z waits with probability $\frac{1}{2}$ while keeping coherence g with x. The settings 0, π and $\pi/2$ therefore estimate $\mathrm{Re}\,g$ and $\mathrm{Im}\,g$ for whichever law is realized; laws with equal g, such as E and T, are separated by the same construction with the full seam $\{0, 1\}$ in place of z, which gives $\frac{1}{2} + \frac{1}{2}\,\mathrm{Re}(k\,e^{-i\theta})$ [40].

The count protocol fixes four settings with 8,192 fresh independent attempted trials each and no optional stopping. Each attempt falls in one of twelve common bins of coin, current and outcome, or aborts. At each setting it scores the twelve bin indicators, the plus indicator and the abort indicator, $m = 4(12 + 2) = 56$ statistics, with sampling half-width $b = 1/32$, independently established apparatus probability budget $\eta = 1/64$ and familywise sampling requirement $\delta = 0.001$.

Proposition 9.2 (Sampling certificate)**.** *If the attempted trials of each setting are independent, then with* $n = 8192$ *and* $b = 1/32$ *the probability that any of the 56 observation intervals* $[\hat{p}-b, \hat{p}+b]$ *misses its actual probability is at most* $112e^{-16} < 112/S_{32} < 0.000012606 < \delta$*, where* $S_{32} = \sum_{k=0}^{32} 16^k/k!$*.*

Proof. Hoeffding's inequality [106] gives $2e^{-2nb^2} = 2e^{-16}$ for each statistic, a union bound over 56 statistics gives $112e^{-16}$, and $S_{32} < e^{16}$; the final comparison is exact rational arithmetic [65]. □

A candidate is rejected when some observation interval is disjoint from its ideal probability widened by η. Under the apparatus contract, rejection occurs only when an observation interval fails, and both candidates use the same intervals; if the calibration guarantee fails with probability at most β, the combined bound is $112/S_{32} + \beta$. An abort has ideal probability zero, and a resolved excess returns the verdict that the apparatus budget is exceeded. Without supplied counts the verdict is "no data", and software demonstrations are labelled synthetic [64]. One survivor discriminates the conditional instruments, two survivors do not separate them, and two rejections mean neither specified apparatus model explains the counts; in every case the same counts estimate g.

9.2 The source-executed implementation

Both complete recorded instruments, the four preparations and the readout also run as emitted twelve-port control words [62, 63]. A configuration is written as the seam pair and the ten spectators: the six endpoint pairs use three bits and the spectators six, placing the 78 configurations injectively in a nine-qubit register with 434 unused addresses. A local permutation with blank record bits realizes each repair from phase-exact transpositions, and the fair coin is prepared by a source pulse and recorded. The encoding is a mathematical isometry, not a physical interface from occupation states.

Proposition 9.3 (Exact source realization of the two instruments)**.** *In the nine-qubit encoding J:*

(i) the encoded recorded channels intertwine the native ones,

$$\mathcal{E}_{\mathrm{code}}(J\rho J^*) = (I_{\mathrm{record}} \otimes J)\, \mathcal{E}_{\mathrm{native}}(\rho)\, (I_{\mathrm{record}} \otimes J^*),$$

on all 6,084 matrix units of the 78-state carrier and therefore with every untouched reference; the complete Gram identity has 156 nonzero Kraus entries;

(ii) E *is* C *followed by a three-pulse source controlled-NOT (CNOT) that copies the seam-total parity p, with contrast decoded as $d = 2j + (2c-1)p$; relative to the retained coin and current labels one additional binary record is necessary and sufficient, and forgetting it leaves parity dephasing without restoring the coherence of* C*;*

(iii) the readout is the Lüders instrument [110] of the rank-one plus projector and its complement, realized by coupling the plus projector to a blank pointer and measuring only the pointer; the minus outcome is one rank-77 projector whose internal coherences survive, and reading the full computational basis would implement a different instrument.

Proof sketch. For fixed current the endpoint map is an injective translation, and for fixed contrast the spectators and endpoint sum reconstruct the input, so every input column of the joint coin, record and source isometry has two entries $2^{-1/2}$ with disjoint supports; an independent enumeration checks all 6,084 matrix units. For (ii), $\{2,3\}$ and $\{1,2\}$ share the C record $(0,0)$ and have contrasts 0 and -1 [63]. □

The complete attempted trial costs 32,006 to 32,010 native pulse requests, with one proper-code load and one export per pulse; a readout costs 26,632 pulses, a local C repair 5,370 and a local E repair 5,373, with peak live width 11 qubits for C and 12 for E [62]. Executed terminal states

reproduce Table 4, and on synthetic counts drawn from executed branch norms the evaluator keeps the generating candidate and rejects the other. The compositional error contract

$$N_p u + N_l l + N_e e + p + z + c + a + r + s + m + g + n \leq \frac{1}{64},$$

with pulse, load and export counts and their channel errors and whole-trial allowances for preparation, initialization, coin, randomizer, records, storage, measurement, coefficient compilation and numerical execution, gives sufficient conditions for $\eta = 1/64$; an equal allocation at the largest schedule is $1/6146496$ per term. The same source alphabet, code services and classical transitions admit two quantum instruments whose phase-zero observable differs by $1/4$, so these ingredients alone do not select the record law.

9.3 Calibration and electromagnetic response

The detector in Section 3.4 evaluates its declared coupling map. The complete electromagnetic identification additionally requires the same matter coefficients, current normalization, thresholds and physical low-energy readout. The particle construction states its certified conditional feedback roots and the omitted response terms [116]. Numerical root closure alone supplies no Thomson-limit identification. The source response archive includes exact rational tests of the direct-endpoint hypothesis [64, 65]; this hypothesis must be tested separately from existence of a mathematical fixed point.

9.4 A sequential design on one carrier

The phase test above assumes fresh preparations. A sequential design runs it on one continuing carrier in its binary total-two sector without resets or independent trials. Each trial applies 379 unread native attempts to the state the carrier holds, performs the fixed-seam native current herald, whose useful yield is $5/33$, and on either useful branch executes the source mixer, three unread native attempts, a second mixer and the native coin and current pointer; a failed herald counts as a trial. After exactly 3,927 trials the two locked likelihoods are compared on all six counts, and an impossible record returns a verdict outside both models. Conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most $3/400$ under the exact-source model; a uniform service allowance of $1/1570800$ per complete trial would suffice for a one-percent bound, and that allowance is not authenticated [25]. Two consecutive trials per candidate and their reference controls are executed.

9.5 Native preparation and calibration of a complete readout

The clocked phase and work readout is a complete instrument at cap N: an independent work preparation, a fixed-seam native herald of useful yield $5/33$, stopping slots, a supplied balanced mixer, the native phase pointer, all sixteen work records and their feedback, the failed herald and two unmeasured timeouts, with $128N + 3$ joint suffix records [33]. Nineteen fixed-order sweeps of the thirty seams, 570 native attempts with the tape ignored and no conditioning, replace its exact input $I/66$ and initial binary support: every density on $\mathcal{H}_{78}$, with any finite untouched reference, reaches the stationary reference within half trace norm 3.7187×10^{-8}, and at cap 64 all 8,195 suffix records enter a complete comparison below 3.03719×10^{-6} [34]. Joint preparation-and-return tests on a trusted basis and on one uniform coherent vector per outcome block of rank above one determine a complete Lüders instrument, with a reference-uniform error bound [35]. The observers

paper proves both theorems [6]. The total-two sector, the quantum instrument semantics, the coherent mixer, the work action, the trusted tests and the model clock are inputs.

9.6 Native records against public data

Published records test precise attachments of native records without a laboratory realization of the carrier; each test is retrospective and states its readout hypotheses.

The binary endpoint map is an identity or a local exchange, so diagonal preparations stay diagonal under recorded operations and local classical feedback. Under independent settings and local factorized responses the diagonal branch obeys $P(\text{win}) \leq 3/4$ in the CHSH game, that is $S \leq 2$; a control that lets one side read both settings wins every trial, and the native example on the 66-state carrier, with all 1,056 histories, gives $S = 26/33$. Replaying the deposited loophole-free records of Hensen et al. [28] reproduces 245 eligible trials, 196 wins and $S = 16036094/6619647 \approx 2.42250$, with a published fixed-count tail bound near 0.03917 [26]. This rejects the local classical reading of the diagonal branch under that experimental contract; it does not test the quantum C/E instruments.

Two native ports, or two regions of one carrier, counting incident repairs under the uniform scheduler have independent interval increments with exact count laws. Attaching the published 300-second blocks of the Ar^{13+}/Yb^{+} frequency ratios of King et al. [29] to the normalized count ratio, through a gain of at least one, a declared bracket on the Yb^{+} frequency, an absolute error of 10^{-12} per block and direct phase-count readback, gives conditional probability bounds below 10^{-104} for two ports and below 10^{-22} over all ordered pairs of the 3,433 regional event sets, after charging every choice of subwindow, gain and retained blocks [27]. This rejects the direct attachment with its readout assumptions; the native event clocks are unaffected as clocks of the model, and a clock calibration needs an explicit model of the measuring apparatus.

Published trapped-ion records test a supplied experimental model executed through the same source interface: two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876427868 in two-point position-estimator units [10]. They test the supplied oscillator model, whose action and apparatus do not follow from native repair.

10 Finite premises and physical interpretation

A conclusion supplies a subsequent theorem's premise only on the same state, algebra, operation, clock and refinement domain, with compatible error bounds. The model–theorem crosswalk records these maps [43, 44], and Table 5 states the division of supply for the main results. No measured particle mass is derived anywhere in this construction, and its finite energies carry no physical unit. Finite realization does not establish its physical interpretation. The following entropy bounds and countermodels constrain that interpretation without excluding other specified designs.

Finite entropy and the gravitational premise. For positive sector counts d_i, normalized probabilities p_i and fixed $M = \sum_i d_i$, the counting entropy obeys

$$S_{\text{count}} = H(p) + \sum_i p_i \log d_i = \log M - D(p\|d/M) \leq \log M.$$

The maximizer is $p_i = d_i/M$. In the positive-real relaxation, joint fixed-M variations have zero first derivative there and second derivative $-\sum_i (p_i' - d_i'/M)^2/p_i$. This is no operation varying

Table 5: What native repair and its counts supply and what each result declares.

Result	**From native repair and counts**	**Declared input**
Seam rule and record family	Current law, coherence table, uniform stationary density, gap $(5-\sqrt{5})/60$	Tie-break and covariance premises, uniform schedule, fair coin
Binary repair [6]	Conditional resampling at constant energy, entropy identities, complete tape accounting	Total-two sector, fair coin, exogenous schedule
Record costs [6]	Prefix weights, Gibbs register reference, bounded refresh kernels	Fair coin, stopping rule, multiplier b, comparison controller
Closed-history functional [6]	Count moments, $H=-\log P$, modular correlations from path counts	Linear extension, complexification, census settings; ρ_L is unprepared
Measurement	Diagonal acceptance effects, complete classical record laws	Complex tomography: coherent services and quantum semantics
Preparation (Section 9.5)	Finite warm-up to $I/66$ with a reference-uniform bound	Total-two sector, fair coin, instrument semantics
Compiled instruments (Section 6.10)	Source pulses, raw records, aborts, complete-output bounds	Source generators, load and export services, target action, prepared reference
Strain and pressure [6]	Charge weights from a complete two-attempt E census	Volume control, energy meter, response class, calibrated sector forces
Interacting fields and mass law [116]	None	Weyl/contact action, texture, memory law, couplings, links, mass-law architecture
Maxwell form [9]	Incidence tables and their exact ranks	Locality, Faraday, conservation, reversal, field definitions, coefficients
Carrier	Rank-three response band of the given incidence	Incidence and response layer (premise A1); the readout selector takes a supplied three-dimensional quadratic readout
Physical units	None	Laboratory clock, energy scale, temperature, measured masses: supplied by no component

integer dimensions, and with bulk factors full saturation also requires maximally mixed bulk blocks. Uniform native preparation realizes counting weights. Fixed count is not identified with fixed physical volume, nor the logarithmic sector term with a geometric area term of universal coefficient [12]. For every cap of the wave reference's return-length law, an absolute bound holds on the full $D = \binom{V}{V/3}$ configuration space:

$$\Delta = \log D - S(\sigma) \le \frac{2025}{64V}, \qquad I_\sigma(R : R^c) \le \log J + \Delta,$$

where J counts the allowed regional particle numbers. A complete base-face patch has $V/20$ ports and $3\,2^l$ crossing seams. Its stationary reduced entropy is extensive in port count, while the return reference's mutual information divided by crossing-seam count vanishes. Neither observable therefore has a nonzero universal coefficient times that boundary count. This does not exclude interpreting a whole spherical screen as a bulk boundary; that physical region map is a premise [12].

Four countermodels fix boundaries of the construction [39]. Each rejects a named identification or an overly broad inference; none rejects the theorem premises.

Complete-carrier center. The centers of M_{66} and M_{78} consist of scalars, so the unextended working algebra cannot supply twelve nonzero primitive central ports. Binary occupations are noncentral, overlap and sum to $2I$, and total-two occupations also admit the eigenvalue two. The native matrix algebra is kept as a sector; a separate twelve-block central-address extension supplies a central algebra with port-controlled operations, and occupation measurement is a separate contract.

Marked triplets. The slow Gram matrix satisfies $G^2 = 4G$, $\operatorname{Tr} G = 12$ and $(5I - A)G = (5-\sqrt{5})G$. One fixed order-five symmetry has trace $(1-\sqrt{5})/2$ on the slow frame and $(1+\sqrt{5})/2$ on the lower response triplet, which excludes an invertible intertwiner for the same action, so spatial and internal response domains are kept distinct. The comparison separates these two specified triplets only; it neither shows every internal triplet inequivalent to the frame nor identifies carrier rechartings with physical rotations. The local rank-three carrier band and the refining global low band are distinct. The conditional Spin–matter construction has independent commuting factors, and its locked diagonal action fails to give a uniform fermionic central sign. A physical separation requires the spatial and internal actions to be identified on the same clocked interacting response and preserved by its dynamics.

Feedback class. With the two endpoint loads in the uniform mixture of their one-particle configurations, a rational number-preserving exchange gives endpoint probabilities 769/1250 and 481/1250 for the two private preparations, a difference of 144/625. Four Kraus operators certify that it is CPTP, and its private commutant has dimension one, which excludes every single-product exponential representation, alternate logarithm branches included. The no-feedback theorems for reduced loads and single products hold within their stated classes, and this channel marks the class boundary.

Configuration equilibrium. For $A = I + L_{\text{path3}}$ the controls $\varepsilon = \pm 1/16$ keep the normalized configuration Gibbs marginal and change the full reference and the vacuum. For a stiffness response $Y = \gamma(A + \varepsilon A^2)$ one has $Y_{13} = \gamma\varepsilon$ and a third-mode fit residual $6\gamma\varepsilon$, while the actual acceleration is $-Yq$. The theorem identifying inertia with the same capacity keeps that premise, and equal equilibrium configurations do not select the kinetic law.

10.1 Native relaxation and static-geometry obstructions

Three results exclude specified mechanisms of the tested law. Each is a fact about that law and its stated class.

The native double-load defect is transient. On the twelve-port carrier, the only configuration functions invariant under every admitted seam and coin are constants, and the binary shell is the unique closed sector. A collision state carries one double load above that shell. Under the uniform seam probe, the complete C and E instruments both satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$ for the projector onto the twelve collisions, so after n attempts the collision probability is $(5/6)^n \operatorname{Tr}[(P_{\mathrm{coll}} \otimes I)\rho]$ for arbitrary quantum inputs and untouched references. The mean number of attempts to split is six, and the double load never hops before splitting. Load differences telescope around all twenty faces for all 78 states, so this gradient map carries no holonomy. Holding one port fixed creates sectors only by disabling its five repairs, which changes the law. The record costs of the corrected memory code have expectations independent of the logical state, so they cannot split logical-state masses [31]. No persistent quantum subspace supported on the collisions exists under this evolution, and the particle paper classifies the compact amplitude subspaces of the binary transfer [116]; the theorem covers no other record-encoded sector.

The static radial product fails null balance. For the smooth static product metrics of the conditional scalar description, both radial null Einstein contractions vanish, while a coherent scalar excitation in the restored angular-constant Dirichlet sector has positive relative null stress. That geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed [8]. A responsive geometry is required.

Passive repair has no wave sector. The record-discarded native channels are Hilbert–Schmidt self-adjoint with spectrum in $[0, 1]$ and scalar fixed algebra (Theorem 3.4). The executed quantum channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier, and the half-step map of its principal logarithm is not positive, so it generates no completely positive semigroup [10]. The closed mean law is damped (Proposition 7.3(iv)), ballistic rescaling freezes the density paths, and the native decay rates grow like Laplacian eigenvalues, so no common clock factor gives a wave dispersion (Section 7.3). Every wave in this paper is declared record processing or a compiled apparatus step on supplied codes; other dilations or generators are not excluded.

11 Particles, interacting fields and gravity

The physical target is one continuing source whose compatible observer records admit an interacting chiral field description in $3 + 1$ dimensions, its particle spectrum and a common classical-gravity limit. Carrier, protocol and memory choices are permitted design variables. For each admitted protocol p, a controlled identification requires

$$\sup_p \frac{1}{2} \left\| D_{\mathrm{native}}(\rho_{\mathrm{native}}(p)) - D_{\mathrm{eff}}(\rho_{\mathrm{eff}}(p)) \right\|_1 \leq \epsilon,$$

with the same declared menu, all failure and wait outcomes, untouched references and an error that vanishes in the claimed limit. The decoding channels map to one common output algebra, preserve every claimed observation and intervention response, and are fixed independently of the comparison data. Matching separate marginals or isolated eigenvalues does not prove this statement.

The companion papers state the results on each side of that target. The gauge paper derives the compact Lie type and the conditional matter module from complete-response premises and constructs finite local Gauss constraints and a gauge-covariant band on classical links [7]. The particle paper joins the twelve native currents, localized fermion modes and a pixel-dependent charged mass operator on a specified design, proves the transfer and charged spectra, memory limits, compact-defect bounds and interacting poles, and compares twelve outputs with measured values [116]. The Einstein paper gives regional entropy on native and prepared references, null tomography and causal counts, and the conditional Einstein implication, which requires geometrically normalized modular flow, the same conserved matter stress, a universal positive area coefficient and fixed-proper-volume entropy stationarity with controlled small-ball remainders and sufficient independent directions [8]. These conditions must hold together for the same source, state, clocks and refinement maps. A chiral continuum also requires controlled spatial refinement, locality, positivity and a complete accounting of tastes and residues; the photon-containing theory must retain massless sectors, and massive poles, unstable resonances and confined observables require their appropriate spectral conventions.

For a coherent rest angular frequency ω_0, a physical clock calibration T seconds per model unit and the physical relation $E = \hbar\omega$, the conditional rest mass is $m = \hbar\omega_0/(Tc^2)$. Assigning the model signal speed v to c sets the length unit $L = cT/v$ and determines neither T nor ω_0 [30], and a necessary small-momentum test of a massive relativistic branch is $\omega_0\omega''(0)/v^2 = 1$. Host work, relaxation rates and maintenance counts acquire no mass interpretation from a unit choice alone. Independent physical comparisons then test the complete dictionary: a failed candidate rejects the stated realization and interpretation, and a whole-class exclusion requires coverage of every admitted realization.

12 Reproducibility

Every executable result has a replayable receipt. Package manifests bind source, document and receipt artifacts by SHA-256. The repository's verification entry point recomputes these hashes, replays retained receipts and compares canonical bytes. Exact certificates must match exactly, and floating diagnostics carry declared replay tolerances. Independent test suites check the executable statements against separately written oracles, including adversarial mutations. Analytic limit proofs state their domains, normalizations and order of limits; a finite receipt does not replace those proofs. Model sources and the premise crosswalk are available at the pinned revisions cited with their results [11, 43]; the flagship synthesis is cited on the main branch [3].

Statements and declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Data availability. The model documents, executable sources, retained receipts and claim records supporting this study are available in the public OPH repository at the baseline revision [2] and the revisions cited with individual results.

Corresponding author. Correspondence should be addressed to Bernhard Mueller, bernhard@floatingpragma.ai.

References

[1] Jakšić, V., Pautrat, Y., and Pillet, C.-A. (2009). *A quantum central limit theorem for sums of IID random variables*. Author manuscript, Sections 3–4. https://jaksic.xyz/papers_pdf/qcltiid.pdf

[2] FloatingPragma (2026). *Observer Patch Holography*: papers, Lean library, executable model, receipts and claim registries. Public repository; model documents, receipts, papers and the basis document are cited at the release tag of this paper, and frozen custody records at their registration revisions. https://github.com/FloatingPragma/observer-patch-holography/tree/r2061

[3] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., Kim, J., Matscheko, D., Friedman, D. A., Hill, J., Glynn, W. T., Visser, M. A., Anirudha, K. A., and de La Fournière, B. (2026). *Observers Who Read Their Universe into Existence: Finite Consensus and the Reconstruction of Standard Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/flagship/from_observer_consensus_to_standard_physics.pdf

[4] FloatingPragma (2026). *OPH premise reference*. Formal statements of the theorem-library premises A1, A2 and A3. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/claims/axiom_registry.yaml

[5] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/reality_as_consensus_protocol.pdf

[6] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., and Friedman, D. A. (2026). *Observers Are All You Need*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/observers_are_all_you_need.pdf

[7] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Matscheko, D., and Friedman, D. A. (2026). *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[8] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[9] Mueller, B., Osika, A., Xue, K., Cassie, B., de La Fournière, B., and Friedman, D. A. (2026). *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/screen_microphysics_and_observer_synchronization.pdf

[10] FloatingPragma (2026). *The runnable OPH model*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/README.md

[11] FloatingPragma (2026). *A common native continuum operator, wave observations, modular phases and repair response*. Executable model, analytic bounds and retained source receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/README.md

[12] FloatingPragma (2026). *Joint counting stationarity and its geometric boundary*. Exact entropy identity and finite source controls. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[13] FloatingPragma (2026). *Native incident clocks, predictable access and active waves*. Analytic count and inverse-clock bounds, correlated clock fluctuations and exact native controls. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/relative_observer_geometry/README.md

[14] FloatingPragma (2026). *Observer-relative time, fields and shared records*. Recovery assessment of the native process results. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/OBSERVER_RELATIVE_RECOVERY.md

[15] FloatingPragma (2026). *One process, overlapping observer records and an effective transfer*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer_correspondence/README.md

[16] FloatingPragma (2026). *Coherent quantum histories on overlapping native clocks*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/quantum_observer_correspondence/README.md

[17] FloatingPragma (2026). *Source-compiled apparatus in a complete observer-clock history*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/compiled_clock_history/README.md

[18] FloatingPragma (2026). *Joint local clocks and quantum fields with explicit memory*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/joint_clock_process/README.md

[19] FloatingPragma (2026). *An executable modular clock for the native record-transfer description*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/transfer_modular_probe/README.md

[20] FloatingPragma (2026). *Fixed observer predictions and complete-count comparison*. Modular clock recovery and native repair response. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/predictions/README.md

[21] FloatingPragma (2026). *Identifying the complete current-resolving repair law*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/repair_identification/README.md

[22] FloatingPragma (2026). *A local action for native quantum observations*. Executable model, locality bounds and screen comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/native_action/README.md

[23] FloatingPragma (2026). *Joint time and field laws on matched source screens*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/observer_time_fields/REFINEMENT.md

[24] FloatingPragma (2026). *Coherent observer outputs on matched source screens*. Executable model and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/coherent_screen_observer/README.md

[25] FloatingPragma (2026). *A finite sequential native-law prediction protocol*. Executable model and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/closed_prediction_protocol/README.md

[26] FloatingPragma (2026). *Classical record responses and published Bell measurements*. Replay of the deposited records. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/predictions/BELL_RECORDS.md

[27] FloatingPragma (2026). *OPH predictions and public-data comparisons*. Retrospective audit, including the direct optical-clock attachments. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md

[28] Hensen, B. et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686. https://doi.org/10.1038/nature15759

[29] King, S. A. et al. (2022). An optical atomic clock based on a highly charged ion. *Nature* 611, 43–47. https://doi.org/10.1038/s41586-022-05245-4

[30] BIPM. *SI defining constants*. https://www.bipm.org/en/measurement-units/si-defining-constants

[31] FloatingPragma (2026). *Native defect decision*. Exact sector, collision-survival, load-gradient and memory-cost controls on the twelve-port carrier. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/native_defect_decision_results.json

[32] FloatingPragma (2026). *Native interface and complex tomography*. Model document and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/native_interface/README.md

[33] FloatingPragma (2026). *A native phase measurement on the stopped interacting work history*. Model document and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/clocked_quantum_readout/README.md

[34] FloatingPragma (2026). *Native preparation followed by the complete phase/work instrument*. Proof and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/prepared_quantum_readout/PROOF.md

[35] FloatingPragma (2026). *Selecting a complete instrument with finitely many probes*. Proof, Lean core and exact controls. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/finite_instrument_calibration/PROOF.md

[36] FloatingPragma (2026). *A complete adaptive observation law for the compact interacting source*. Proof, execution and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/interacting_observation_process/PROOF.md

[37] FloatingPragma (2026). *Native coherent detection of the cubic charged response*. Proof, execution and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/native_composite_probe/PROOF.md

[38] FloatingPragma (2026). *Complete native action of the corrected regional modular generator*. Proof, execution and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/native_regional_modular/PROOF.md

[39] FloatingPragma (2026). *OPH model construction from local repair*. Construction guide with completion target, acceptance criteria and audit controls. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/CONSTRUCTION.md

[40] FloatingPragma (2026). *Admissible record laws for native repair*. Model document and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/record_laws/README.md

[41] FloatingPragma (2026). *The coherence table and the recorded instrument*. Exact control on the actual carrier. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/record_laws/INSTRUMENT_CONTRACT.md

[42] FloatingPragma (2026). *Concrete repair-law candidates*. Candidate register. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/CANDIDATES.md

[43] FloatingPragma (2026). *OPH reverse-engineering table: model and theorem premises*. Generated premise map and concrete theorem bindings. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/model/PREMISES.md

[44] FloatingPragma (2026). *Machine-readable premise crosswalk*. Source of the generated premise map. https://github.com/FloatingPragma/observer-patch-holography/blob/4a3f9a6bdfd51867dcda8786c418ffea34bfec8f/tracking/premise_register.json

[45] FloatingPragma (2026). *Runnable finite observers*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/README.md

[46] FloatingPragma (2026). *Finite current-record controller: mathematical account*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/DERIVATION.md

[47] FloatingPragma (2026). *Native observer records reuse the existing capacity construction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/CAPACITY.md

[48] FloatingPragma (2026). *Quantum mixing of the native C/E repairs*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/QUANTUM_MIXING.md

[49] FloatingPragma (2026). *Recovery with complete native records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/RECORD_RECOVERY.md

[50] FloatingPragma (2026). *Coherent continuation: native attachment and nonselection*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/observer/COHERENT_CONTINUATION.md

[51] FloatingPragma (2026). *Selecting one source, interface and repair law*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/SOURCE_SELECTION.md

[52] FloatingPragma (2026). *P-dependent electromagnetic observation and native feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/p_response/README.md

[53] FloatingPragma (2026). *Charged realization, binary instrument and feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/p_response/DERIVATION.md

[54] FloatingPragma (2026). *Detector access and the conditional capacity budget*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/p_response/ACCESS_CAPACITY.md

[55] FloatingPragma (2026). *Validation of the native-record electromagnetic adapter*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/p_response/VALIDATION.md

[56] FloatingPragma (2026). *A scalar detector inside the native observer protocol*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/scalar_response/README.md

[57] FloatingPragma (2026). *A native record coupled to a scalar field instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/scalar_response/DERIVATION.md

[58] FloatingPragma (2026). *Conditional detector prediction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/scalar_response/PREDICTION.md

[59] FloatingPragma (2026). *Two reads of the retained field*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/scalar_response/CONTINUATION.md

[60] FloatingPragma (2026). *Scalar preparation, evolution and detection through source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/source_scalar/README.md

[61] FloatingPragma (2026). *A finite source implementation of the two-site scalar instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/source_scalar/DERIVATION.md

[62] FloatingPragma (2026). *Execute the native C/E discrimination experiment from source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/source_discrimination/README.md

[63] FloatingPragma (2026). *Source implementation of the complete C/E discriminator*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/source_discrimination/DERIVATION.md

[64] FloatingPragma (2026). *Fixed observer predictions and complete-count comparison*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/predictions/README.md

[65] FloatingPragma (2026). *Exact record-instrument discrimination*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/predictions/DERIVATION.md

[66] FloatingPragma (2026). *N4 memory survives the observer's local tick clock*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/local_tick/README.md

[67] FloatingPragma (2026). *N4 occupation memory at observer-local ticks*. Protocol specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/local_tick/SPEC.md

[68] FloatingPragma (2026). *A common interacting observer process*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/README.md

[69] FloatingPragma (2026). *One finite interacting source: derivation and theorem attachments*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/DERIVATION.md

[70] FloatingPragma (2026). *The interacting feedback signal survives infinite occupation*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/INFINITE_HISTORY.md

[71] FloatingPragma (2026). *Occupation refinement of the interacting retained history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/OCCUPATION.md

[72] FloatingPragma (2026). *Occupation limit of the same interacting observer history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/OCCUPATION_REFINEMENT.md

[73] FloatingPragma (2026). *Infinite coordinate-sign readout*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/READOUT.md

[74] FloatingPragma (2026). *A smooth binary observer with a complete occupation error bound*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/SMOOTH_REFINEMENT.md

[75] FloatingPragma (2026). The finite executable observer specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/COMPOSITION.md

[76] FloatingPragma (2026). Complete-output certificate for the continued smooth history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/SMOOTH_HISTORY.md

[77] FloatingPragma (2026). Coherent native repairs around one continuing quantum history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/QUANTUM_NATIVE.md

[78] FloatingPragma (2026). Active encoded memory in the continuing observer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/ENCODED_MEMORY.md

[79] FloatingPragma (2026). *Hilbert space, modular clocks and operational regions*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/OBSERVER_READOUTS.md

[80] FloatingPragma (2026). *A fixed thermal reference for the changing common history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/THERMAL_CLOCK.md

[81] FloatingPragma (2026). *One complete history from local constraint repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/HISTORY_REPAIR.md

[82] FloatingPragma (2026). *The existing three-dimensional response carrier on actual observer records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/SLOW_GEOMETRY.md

[83] FloatingPragma (2026). *Exact native causal cones*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/CAUSALITY.md

[84] FloatingPragma (2026). *Field relations of the existing native repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/NATIVE_FIELDS.md

[85] FloatingPragma (2026). *Testing the readouts before assigning physical units*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/CALIBRATION.md

[86] FloatingPragma (2026). Retained receipt of the exact native observer histories and calculation checks. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/results/native.json

[87] FloatingPragma (2026). Retained receipt of the 20,000-trial sequential native execution. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/results/run.json

[88] FloatingPragma (2026). Retained receipt of the record capacity and compact controller. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/results/capacity.json

[89] FloatingPragma (2026). Retained receipt of the quantum mixing and memory certificates. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/results/quantum_mixing.json

[90] FloatingPragma (2026). Retained receipt of complete-record recovery and identifying schedules. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/results/record_recovery.json

[91] FloatingPragma (2026). Retained receipt of the P-dependent detector. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/p_response/results.json

[92] FloatingPragma (2026). Retained receipt of the local-tick memory certificate. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/local_tick/results.json

[93] FloatingPragma (2026). Retained receipt of the scalar detector on four grids. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/scalar_response/results.json

[94] FloatingPragma (2026). Retained receipt of the two-site source apparatus. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/source_scalar/results.json

[95] FloatingPragma (2026). Retained receipt of the common interacting source. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/results.json

[96] FloatingPragma (2026). Retained receipt of the complete sharp-output comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/sharp_output_results.json

[97] FloatingPragma (2026). Retained receipt of the observer readouts and the modular clock. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/bridge_results.json

[98] Aharonov, Y., and Bohm, D. (1959). "Significance of electromagnetic potentials in the quantum theory." *Physical Review* **115**(3), 485–491. https://doi.org/10.1103/PhysRev.115.485

[99] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C^*- and W^*-Algebras, Symmetry Groups, Decomposition of States*, 2nd ed. Springer, Berlin.

[100] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. Springer, Berlin.

[101] Caputo, P., Liggett, T. M., and Richthammer, T. (2010). "Proof of Aldous' spectral gap conjecture." *Journal of the American Mathematical Society* **23**(3), 831–851. https://arxiv.org/abs/0906.1238

[102] Choi, M.-D. (1974). "A Schwarz inequality for positive linear maps on C^*-algebras." *Illinois Journal of Mathematics* **18**(4), 565–574.

[103] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[104] Olver, F. W. J., et al., eds. *NIST Digital Library of Mathematical Functions*, Chapter 18, Orthogonal Polynomials. https://dlmf.nist.gov/18

[105] Gurvits, L., and Barnum, H. (2002). "Largest separable balls around the maximally mixed bipartite quantum state." *Physical Review A* **66**, 062311. https://doi.org/10.1103/PhysRevA.66.062311

[106] Hoeffding, W. (1963). "Probability inequalities for sums of bounded random variables." *Journal of the American Statistical Association* **58**(301), 13–30. https://doi.org/10.1080/01621459.1963.10500830

[107] Horodecki, M., Shor, P. W., and Ruskai, M. B. (2003). "Entanglement breaking channels." *Reviews in Mathematical Physics* **15**(6), 629–641. https://arxiv.org/abs/quant-ph/0302031

[108] Knill, E., and Laflamme, R. (1997). "Theory of quantum error-correcting codes." *Physical Review A* **55**, 900–911. https://doi.org/10.1103/PhysRevA.55.900

[109] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Lecture Notes in Physics 190, Springer, Berlin.

[110] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[111] Peres, A. (1996). "Separability criterion for density matrices." *Physical Review Letters* **77**, 1413–1415. https://doi.org/10.1103/PhysRevLett.77.1413

[112] Reed, M., and Simon, B. (1980). *Methods of Modern Mathematical Physics I: Functional Analysis*, revised and enlarged ed. Academic Press, New York.

[113] Reed, M., and Simon, B. (1975). *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*. Academic Press, New York.

[114] Strang, G. (1968). "On the construction and comparison of difference schemes." *SIAM Journal on Numerical Analysis* **5**(3), 506–517. https://doi.org/10.1137/0705041

[115] Teschl, G. (2014). *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, 2nd ed. Graduate Studies in Mathematics 157, American Mathematical Society, Providence.

[116] B. Müller et al., *Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf

[117] Ariyoshi, T., and Hino, M. (2005). "Small-time asymptotic estimates in local Dirichlet spaces." *Electronic Journal of Probability* **10**, 1236–1259. https://doi.org/10.1214/EJP.v10-286

[118] Cipriani, F. (2006). "Dirichlet forms as Banach algebras and applications." *Pacific Journal of Mathematics* **223**, 229–249. https://msp.org/pjm/2006/223-2/pjm-v223-n2-p03-p.pdf

[119] Higuchi, Y., Segawa, E., and Suzuki, A. (2016). "Spectral mapping theorem of an abstract quantum walk." arXiv preprint, version 2. arXiv:1506.06457. https://arxiv.org/abs/1506.06457

[120] Szegedy, M. (2004). "Spectra of Quantized Walks and a sqrt(delta epsilon) rule." arXiv preprint. arXiv:quant-ph/0401053. https://arxiv.org/abs/quant-ph/0401053

[121] Visser, M. (1998). "Acoustic black holes: horizons, ergospheres, and Hawking radiation." *Classical and Quantum Gravity* **15**, 1767–1791. https://doi.org/10.1088/0264-9381/15/6/024

[122] Welch, J., Greenbaum, D., Mostame, S., and Aspuru-Guzik, A. (2014). "Efficient quantum circuits for diagonal unitaries without ancillas." *New Journal of Physics* **16**, 033040. https://doi.org/10.1088/1367-2630/16/3/033040

[123] T. Koma, "Dispersion Relations of Nambu–Goldstone Modes," arXiv:2105.04970v2 (2021), Theorem 2.1 and equations (2.11)–(2.16). https://arxiv.org/abs/2105.04970.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.